



THE UNIVERSITY OF
MELBOURNE

Probability Reliability and Quality

SPC, Incoming Inspection, PPAP

Kevin Otto

Schedule and Learning Outcomes

Week	Recorded Lecture Topic
1	Introduction.
2	Inspection. Yield.
3	ANOVA. Measurement Systems Analysis.
4	Statistics. Confidence Intervals.
5	Hypothesis Testing. Sample Sizes.
6	Quality Management Processes.
7	Regression.
8	Design of Experiments.
9	System Reliability. Availability.
	<i>BREAK – Non Teaching Week</i>
10	Parametric Distribution Analysis
11	Statistical Process Control. Supplier Quality.
12	Review

After completion of the course the student should be able to:

- Explain different statistical methods
- Parametric and non-parametric hypothesis tests
- Design and plan for collecting experimental data
- Concepts of reliability engineering and the relationship between TTF, R, H
- **How to improve its processes, products and services**
- **Apply TQM, 6S, and benchmarking**



Overview

Statistical Process Control

- Xbar-R charts

- Out of Control Action Plans

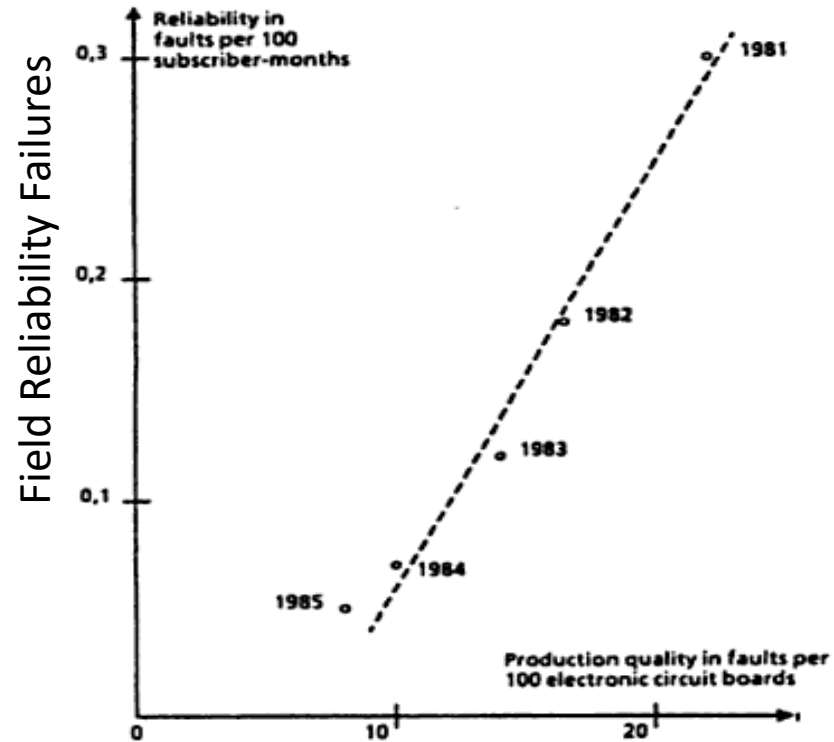
Incoming Inspection

- Statistical Sampling Plans

- Cpk Inspection Plans

PPAP

High Reliability Comes from High Quality

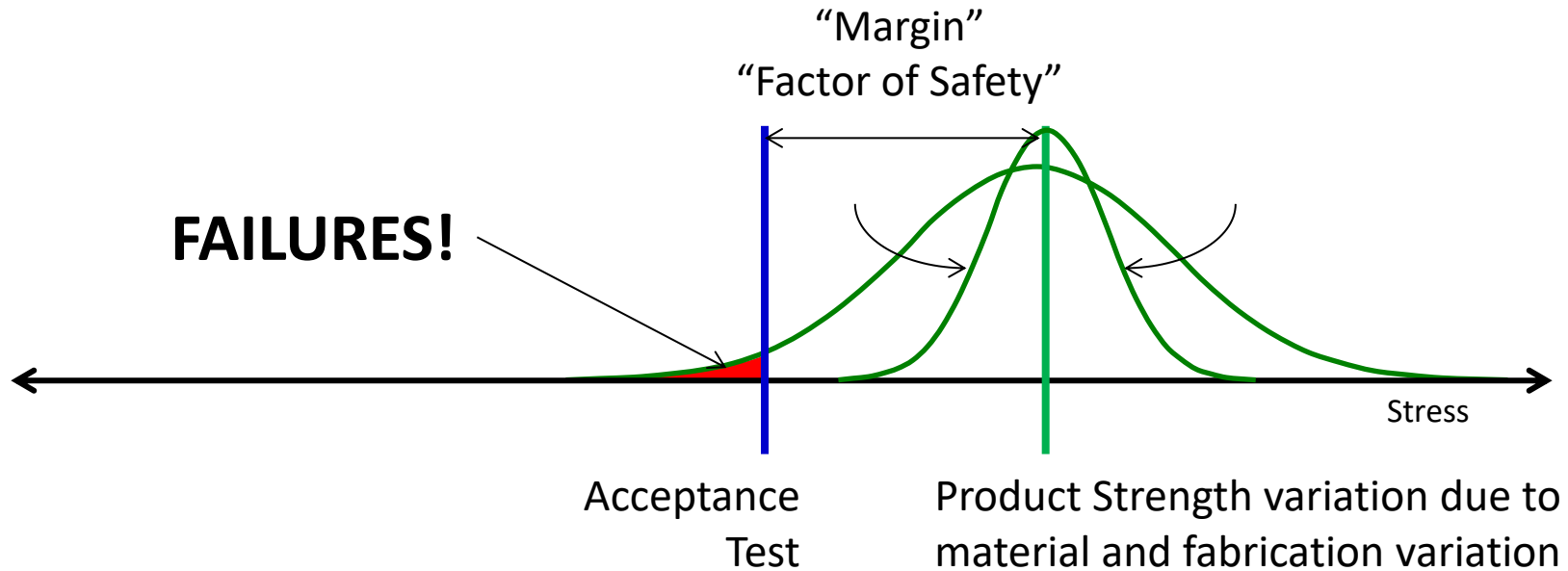


Factory Quality Turnbacks

5. Ender, U. and Gerling, W.: "Influence of Manufacture on Electronic Equipment and System Reliability". Quality and Reliability Engineering International, Vol.3, pp.195-201, 1987.

Quality and Consistency

High quality comes from reducing variability

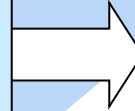
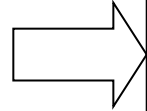


The Goal

The quality goal is to operate the process repeatably and consistently.



KPIV
KPIV
KPIV
KPIV

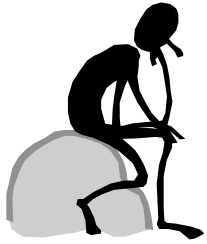


KPOV

MANAGE THESE ...

**... AND THIS WILL
BE IN SPEC**

You won't need to 100% inspect.



Keeping Processes in Control

The **control plan** is a list of KPOV / KPIVs at each process step, and what to when any process variable has become out of control.

“Inspect hole diameter. If 3 in a row oversized, replace drill bit”

“If vibration levels are exceeded, call Bob for maintenance”

“Inspect at the start of every shift. If out of calibration, adjust X until Y is on target”

Control Plan Number:				Key Contact/Phone:				Date:		Date (Rev):		
Part Number:				Core Team:				Customer Engineering Approval/Date (If Req'd):				
Part Name Description:				Supplier/Plant Approval/Date:				Customer Quality Approval/Date (If Req'd):				
Supplier Plant:		Supplier Code:		Other Approval/Date (If Req'd):				Other Approval/Date (If Req'd):				
Part / Process Number	Process Name / Operation Description	Machine, Device, Jig, Tools for Mfg.	Characteristics			Special Char. Class.	Methods				Reaction Plan	
			No	Product	Process		Spec Limits	Evaluation Measurement Technique	Sample			Control Method
									Size	Frequency		

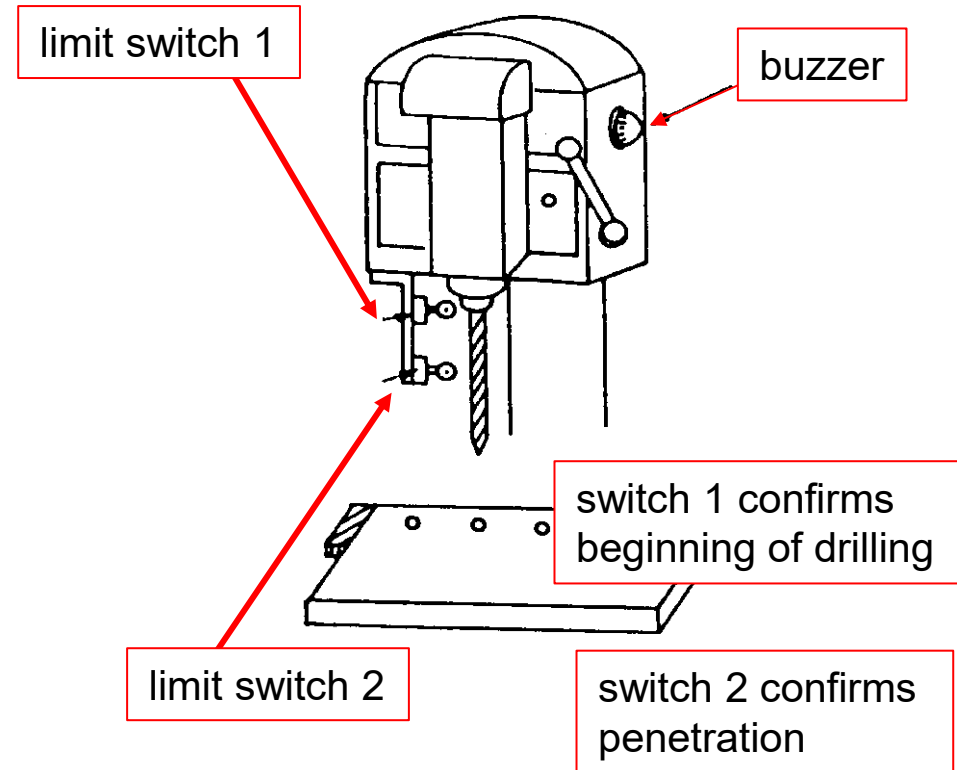
Keeping Processes in Control

The best approach is *mistake proofing*. Alter equipment so it cannot be done wrong.

Example: Add Go No-Go switches to equipment

Example:

- Holes were not being drilled to the appropriate depth. Two limit switches were mounted on the drill press. Faulty drilling is indicated if limit switch 1 is released before limit switch 2 has been tripped (indicating the start of drilling without penetration). A buzzer is sounded to alert the operator.





Keeping Processes in Control

Given an input cause, how to fix it so it stays fixed?

Change the design

Change the process equipment

Mistake proof the equipment

Mistake proof the standard work

Change the process standard work

Add measurement or inspection

Operator Training



Most reliable

Least reliable



Keeping Processes in Control

Given an input cause, how to fix it so it stays fixed?

Change the design

Change the process equipment

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Change the process standard work

Add measurement or inspection

SPC

Operator Training

Most reliable

Least reliable



What is SPC?

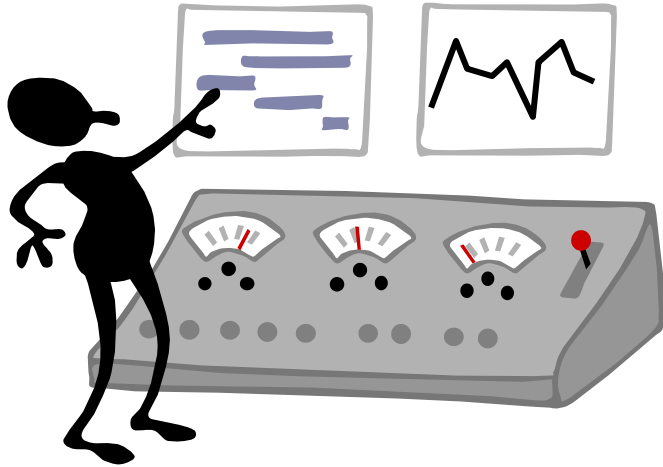
Statistical Process Control (SPC) is a statistically based method of

- Evaluating the stability of process and product variables
- Indicating when the process has changed
- Maintaining expected results despite normal process variation

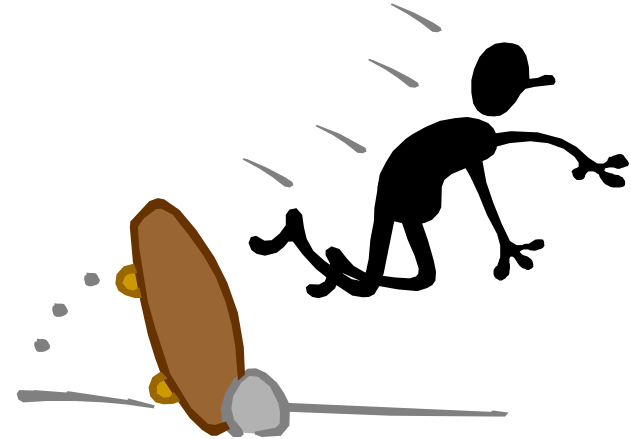
SPC is effective for monitoring inputs to detect a change. Nonetheless, prevention through mistake proofing is better.

Variation!

Unnatural Causes of Variation, sometimes called *Special Causes*, happen for identifiable (assignable cause) reasons.

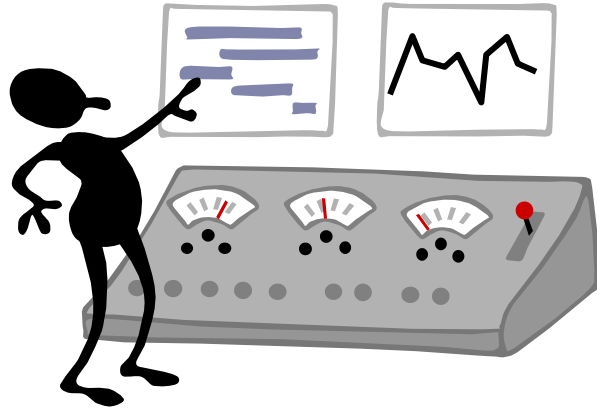


Natural Causes of Variation, sometimes called *Common Causes*, happen by chance.



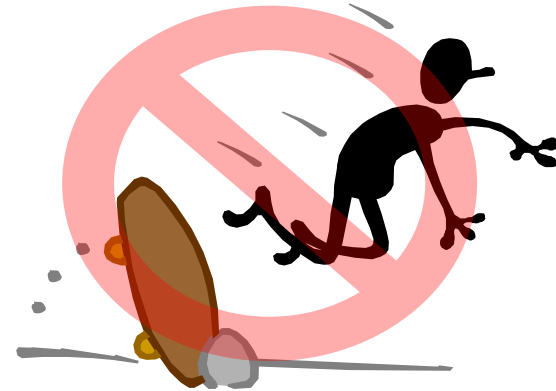
Natural & Unnatural Variation

SPC helps separate common from special causes of variation so that we can eliminate the special cause variation



Common Causes

- many in number
- have small effects



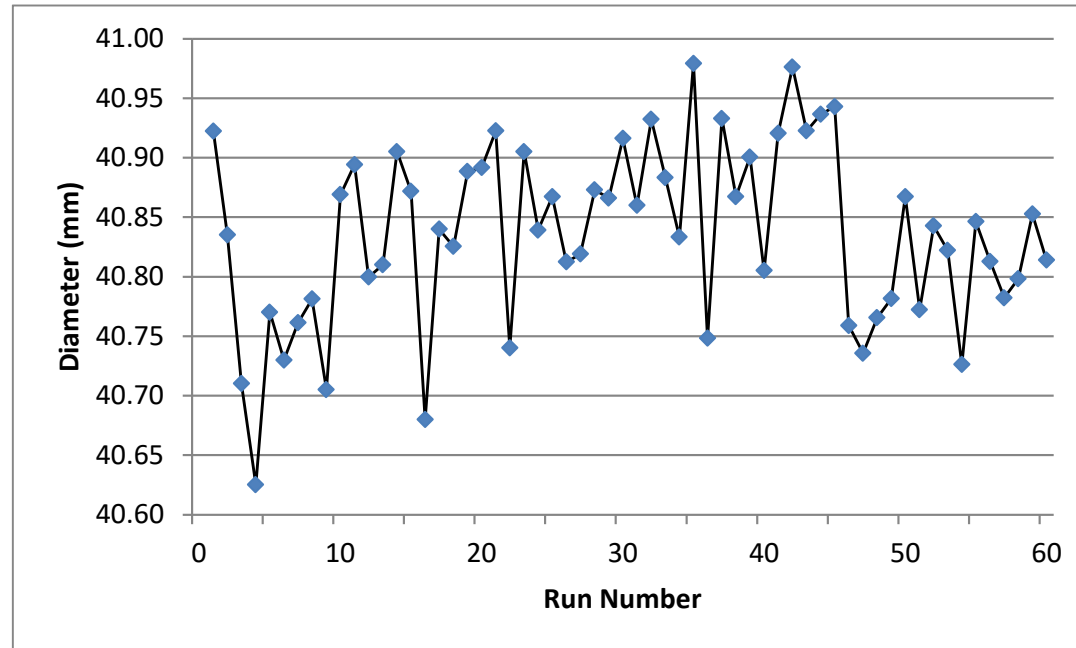
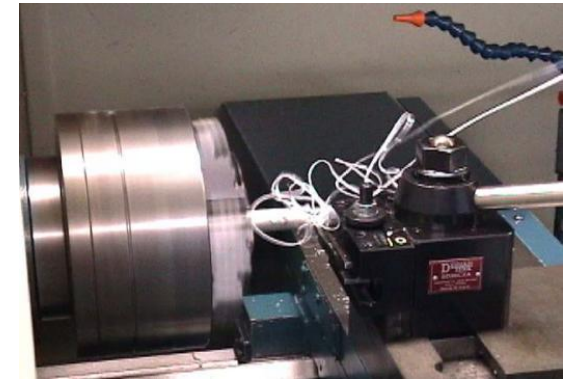
Special Causes

- few in number
- have large effects

The result is then stable, predictable processes!

Example: CNC Turning

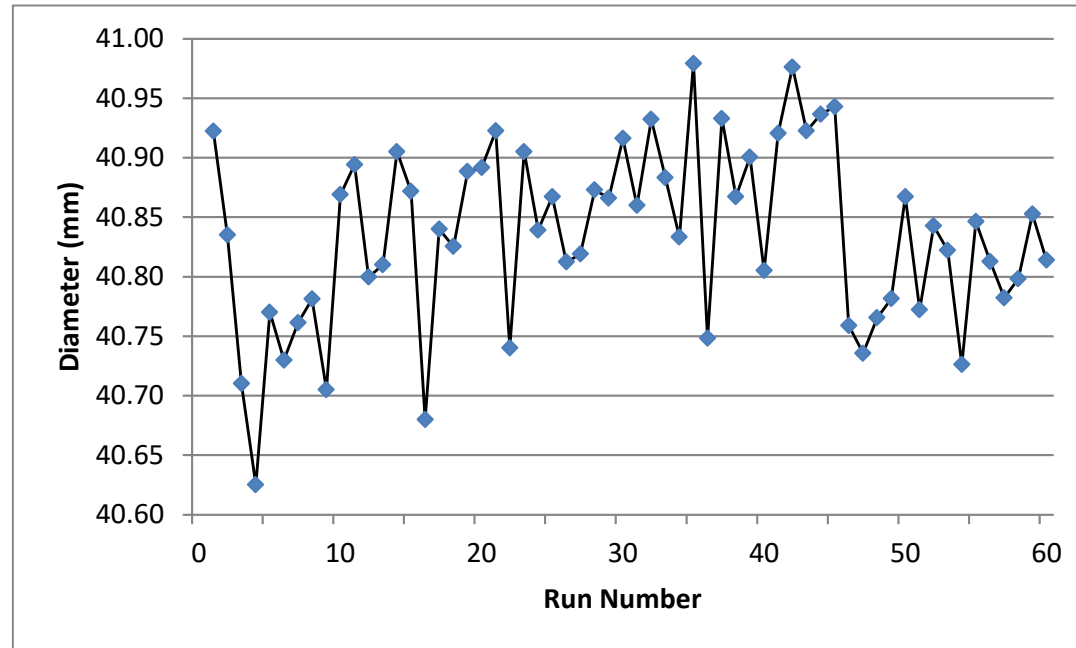
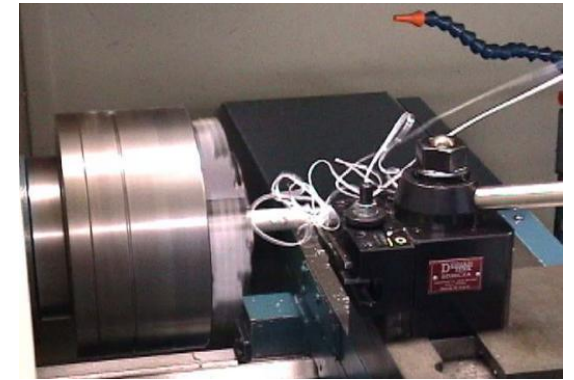
Is this all random variation?



Example: CNC Turning

Is this all random variation?

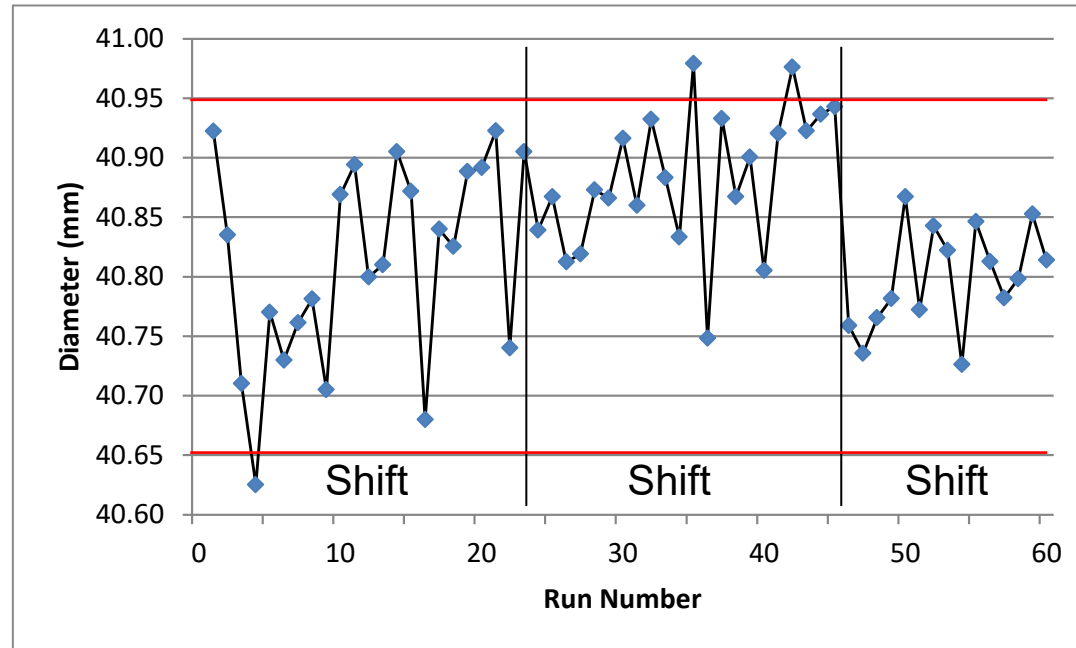
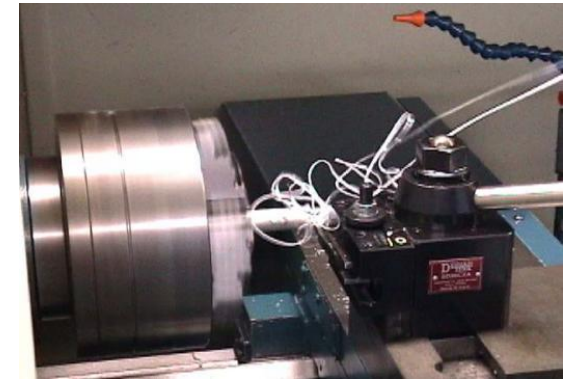
No! Process drifts and shifted at least once. Why?



Example: CNC Turning

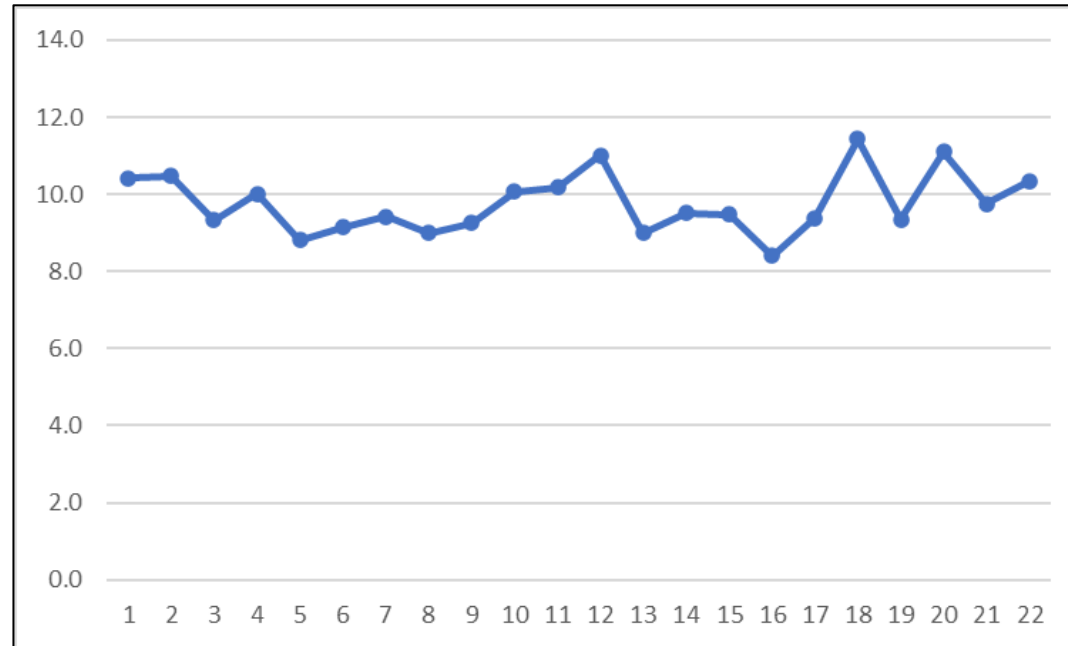
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No! Process drifts and shifted at least once. Why?



Example: Forming Pressure

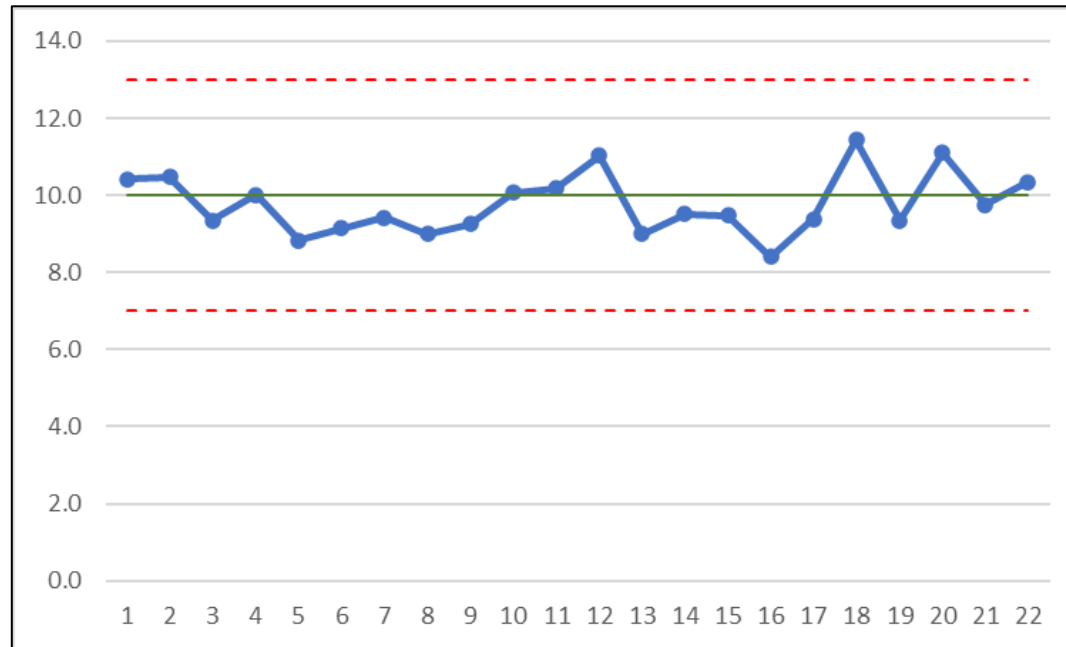
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Example: Forming Pressure

Is this all random variation?

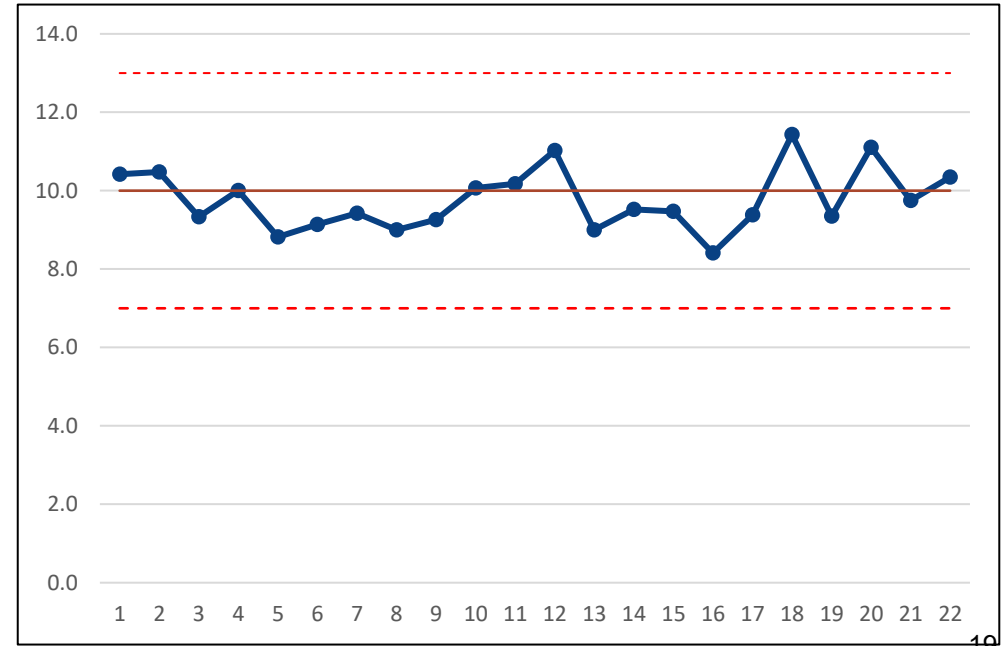
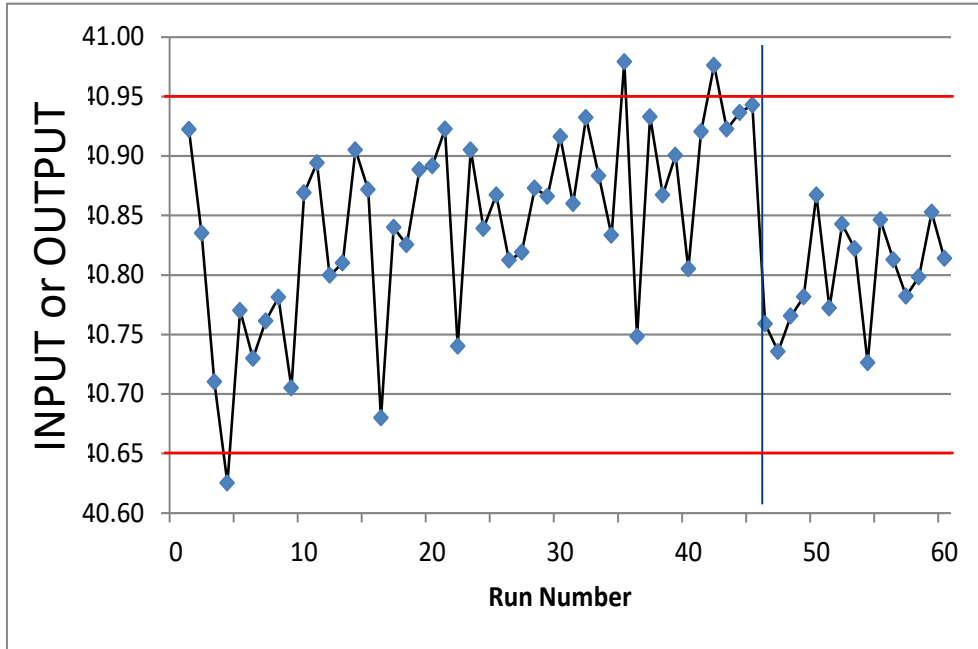
Looks like it...



Statistical Control

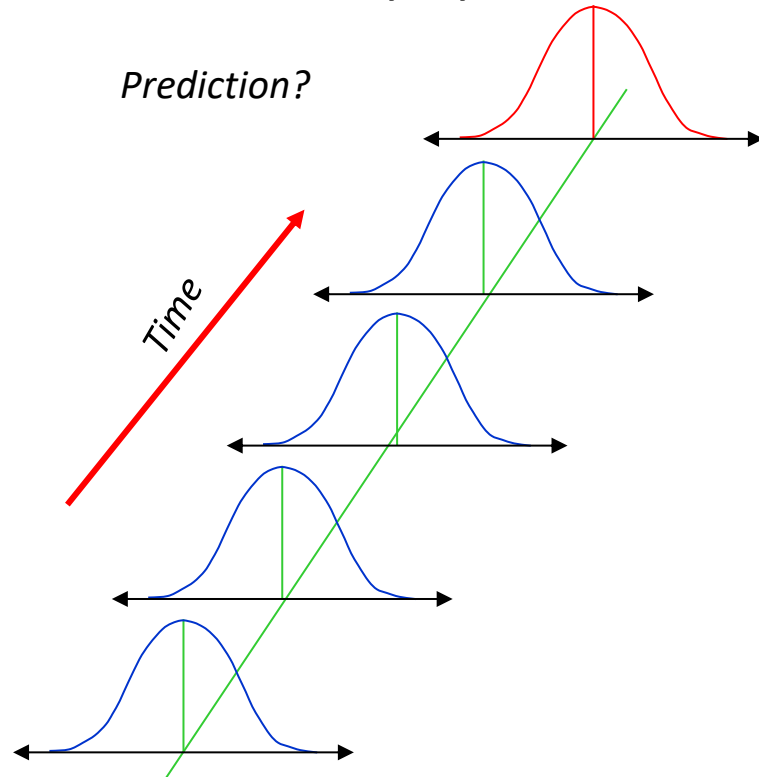
A process is in *statistical control* if

- the short term distribution parameters do not significantly change over time and
- there are no non-random patterns of variation.



Types of Variation: Common Cause

Processes affected only by common causes are said to be ***IN CONTROL***.

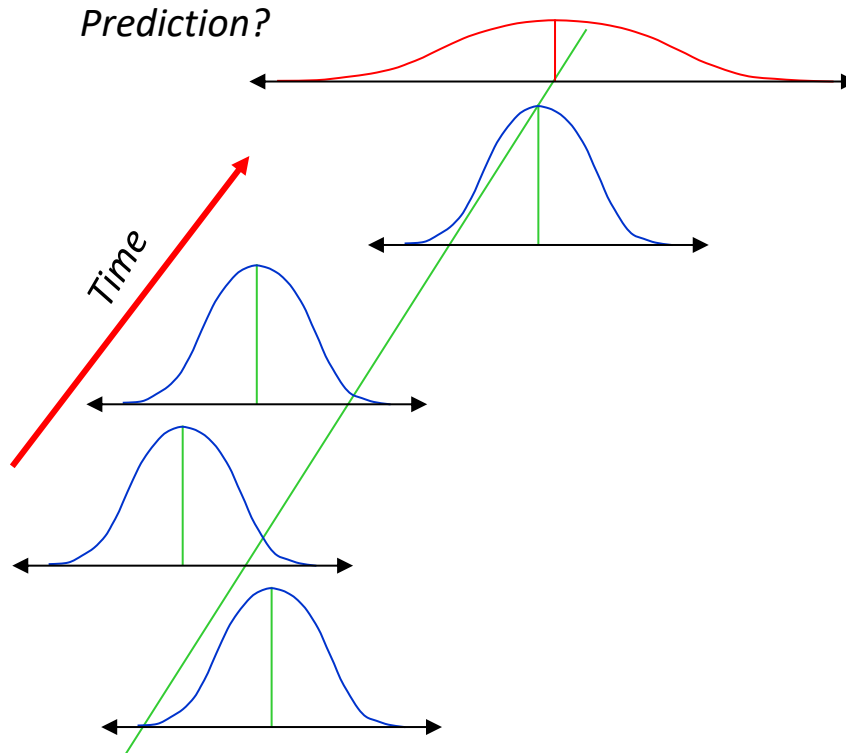


Characteristics of Common Causes:

- Affect the process uniformly.
- Usually consist of many small sources.
- Relatively stable with predictable output.

Types of Variation: Special Cause

Processes affected by special causes are said to be ***OUT OF CONTROL***.



Characteristics of Special Causes:

- Affect the process non-uniformly.
- Usually consist of a few large sources.
- Unstable and unpredictable output.

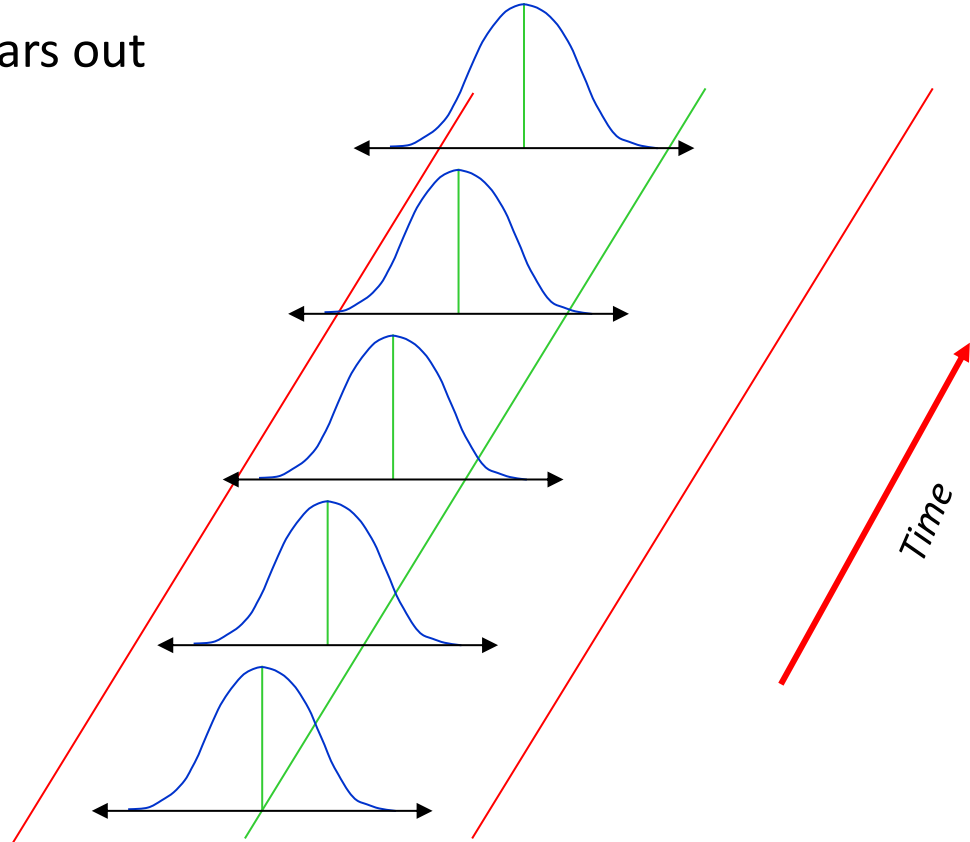
Should You Intervene?

Consider a process with a tool that wears out

A dimension grows correspondingly

- Eventually gets too large

When do you decide an intervention is needed to replace the tool?

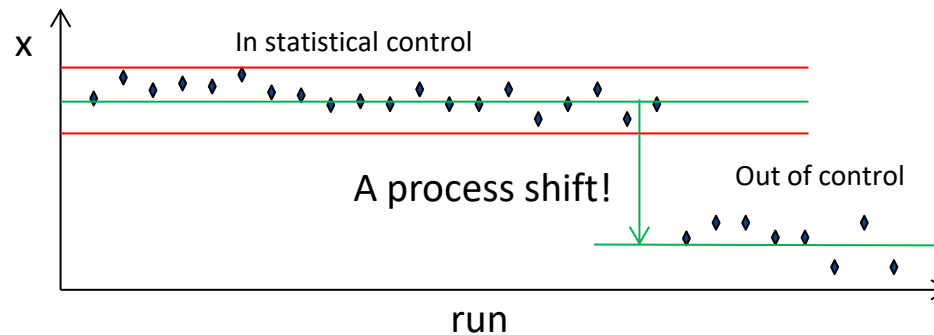


Out of Statistical Control

Only when the points change as a whole do we say the process has changed.

- We call this ***out of statistical control***.
- We do not see it by individual values.
- We see it by changes to the distribution.

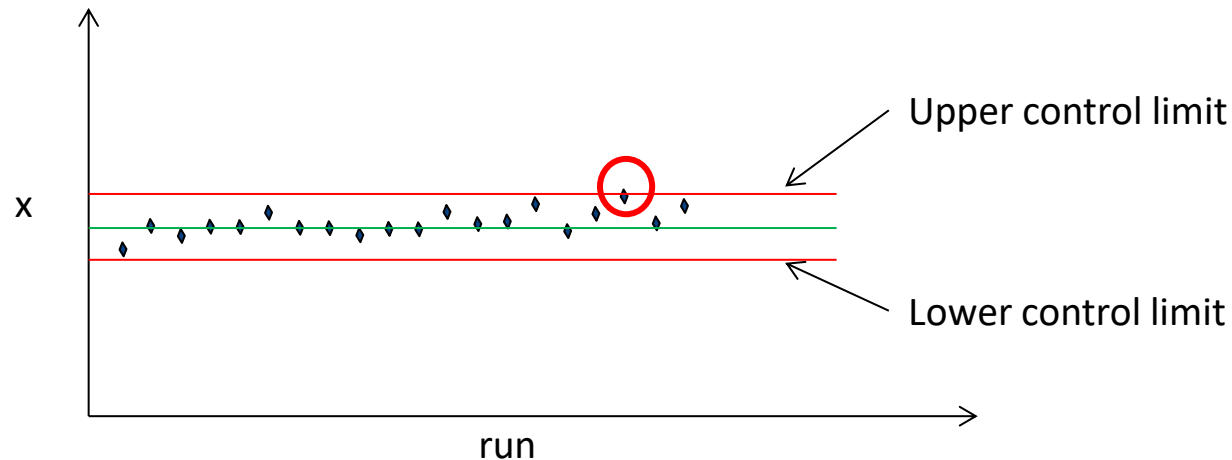
Directly, we look for ***out of control conditions***.



Out of Statistical Control

To ensure production is consistent, we measure and track a variable, such as the values on an acceptance test.

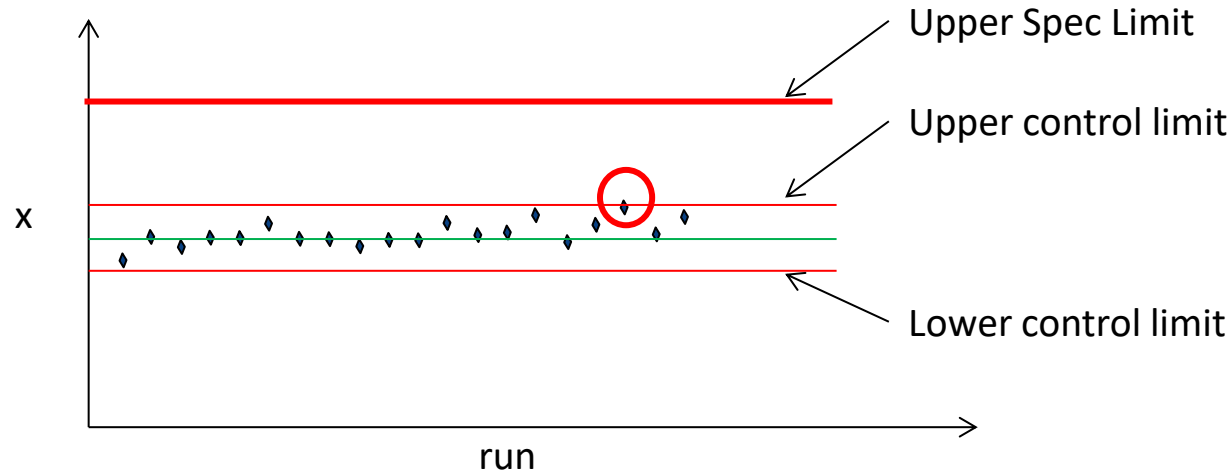
- We define upper and lower *control limits*.
- It is usually $\pm 3\sigma$. To detect when something has changed.



Statistical Control

To ensure production is consistent, we measure and track a variable, such as the values on an acceptance test.

- We define an upper and lower **control limits**.
- It is usually $\pm 3\sigma$. To detect when something has changed.
- A control limit is not a specification limit.

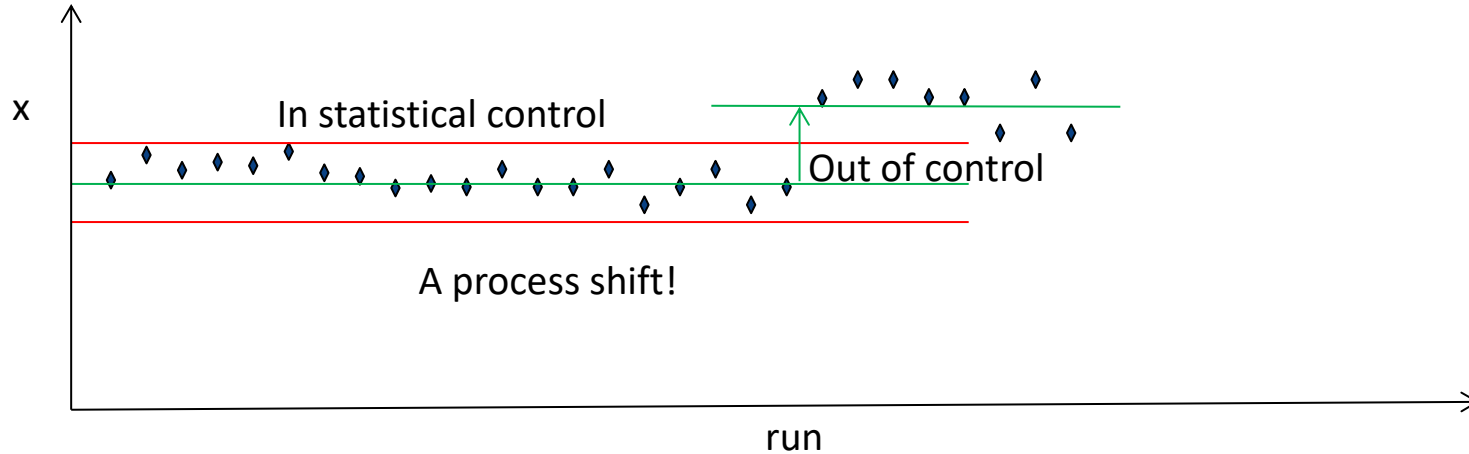


Manufacturing Uncertainty

Suppose you are a plant manager.

Your job is to produce parts. Adjust and replace equipment parts when needed. Ensure standard work.

You look for when a process operation has changed.

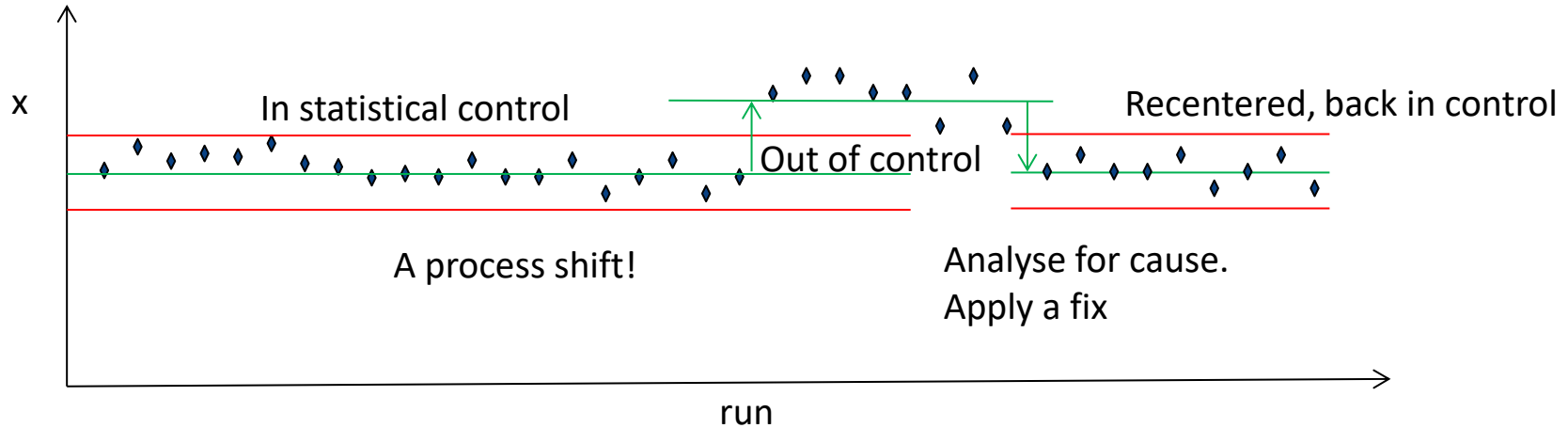


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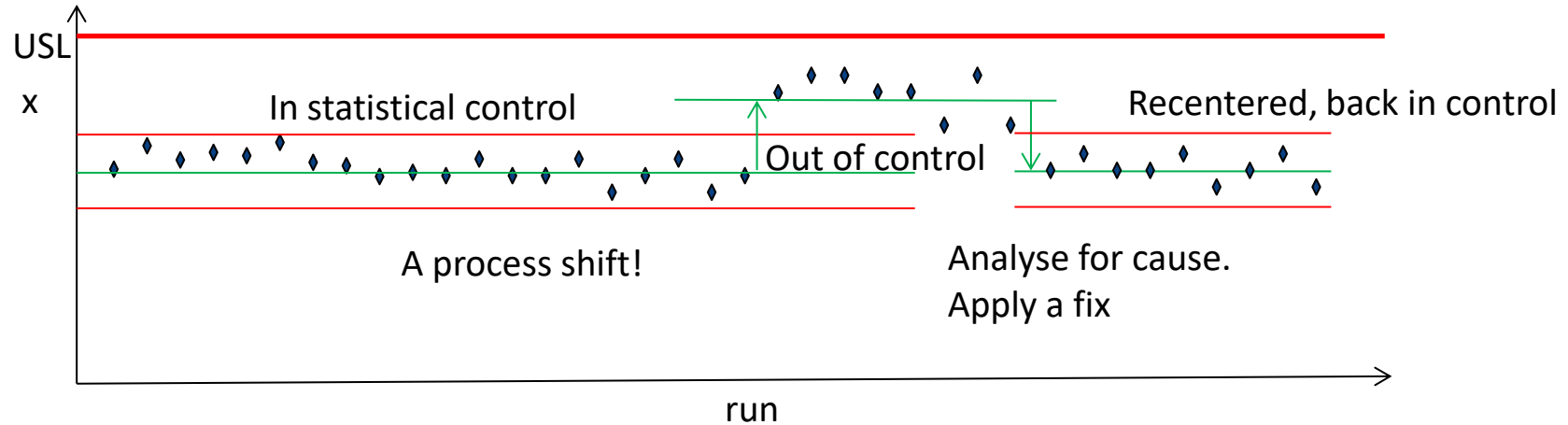


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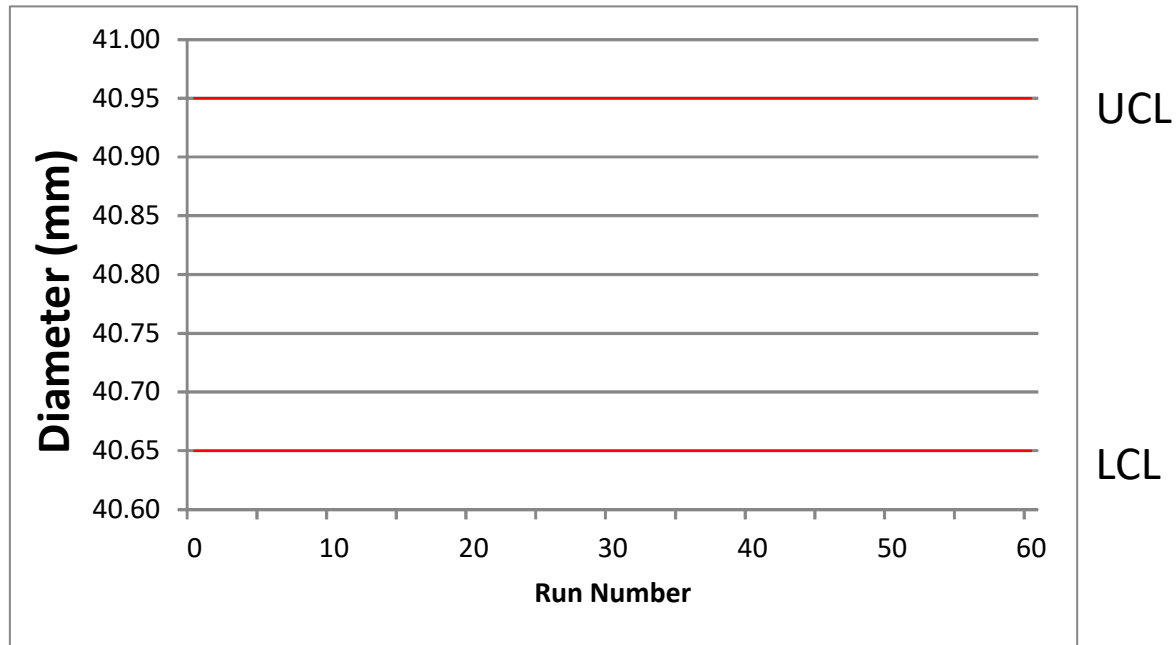
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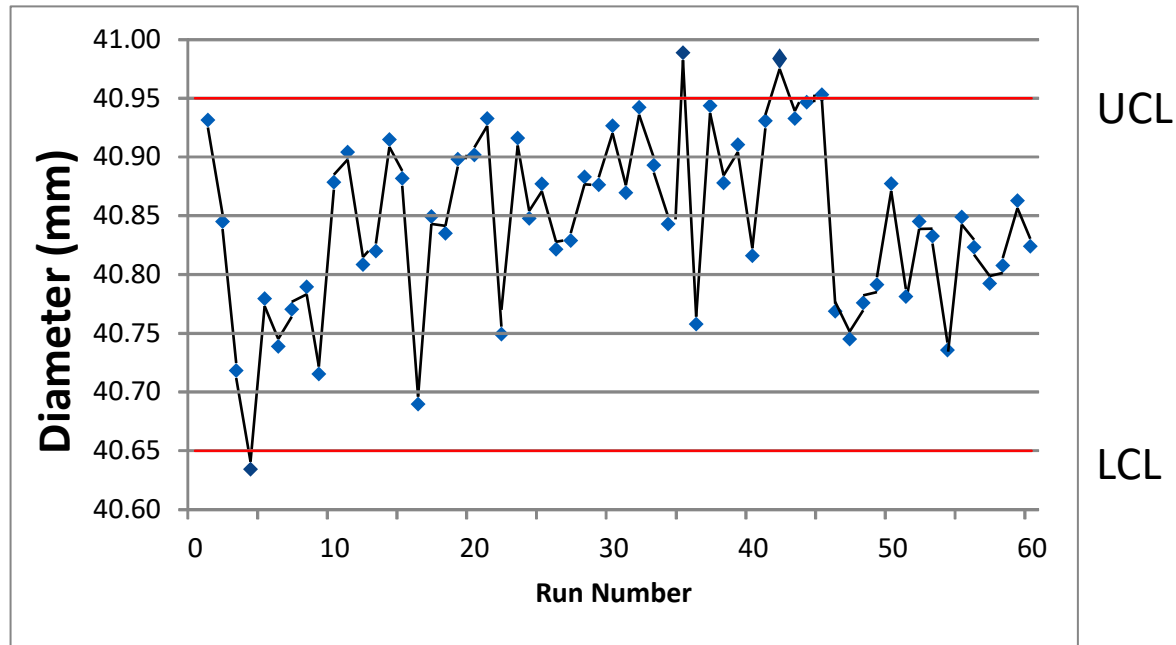
Should You Intervene?

When do you decide an intervention is needed to adjust or replace the tool?



Should You Intervene?

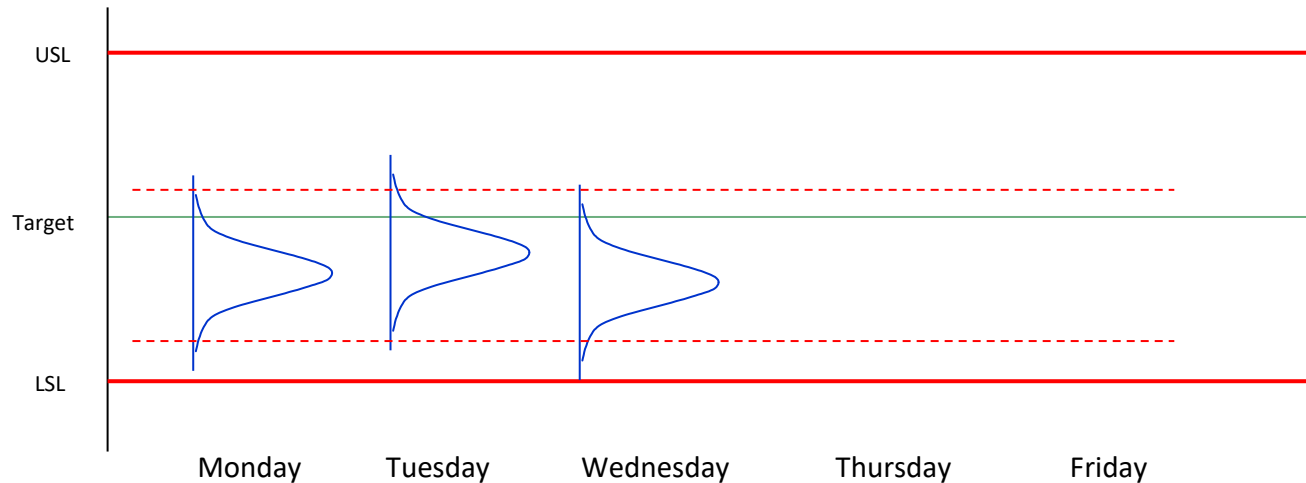
When do you decide an intervention is needed to adjust or replace the tool?



Manufacturing Process Control

The production process shifts day to day, batch to batch.

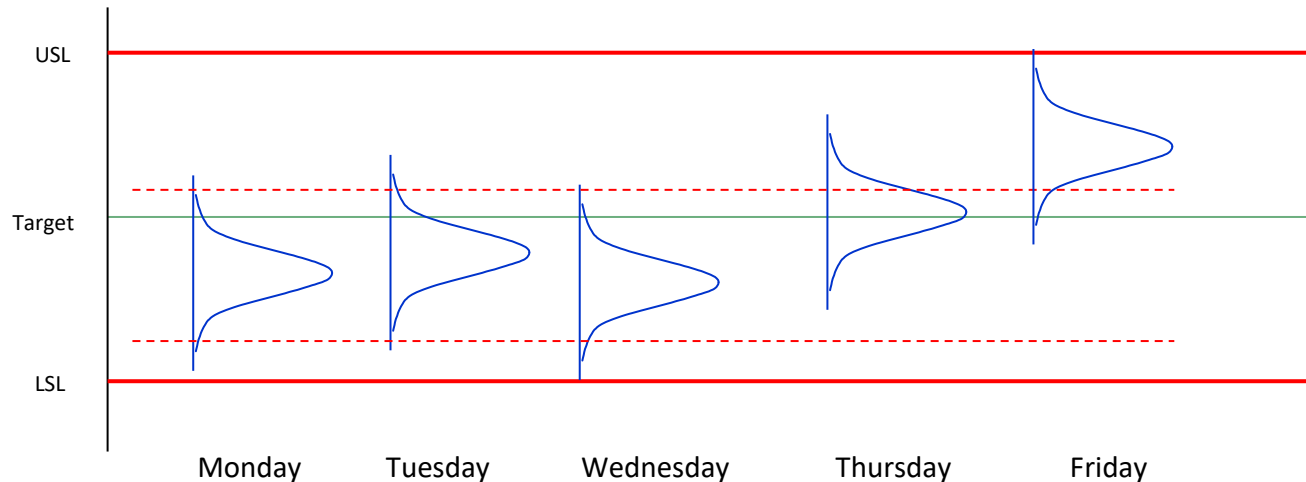
Plant engineers need allowance for this in the tolerance.



Manufacturing Process Control

The process is in control and requires no changes when in ± 3 sigma variability.

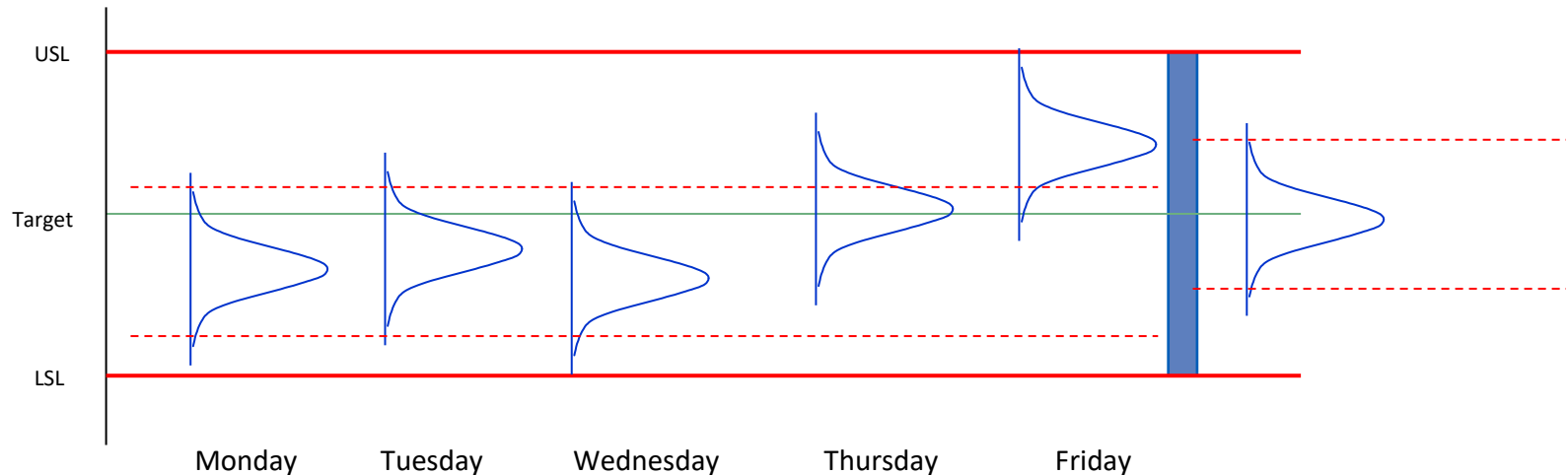
The engineers need at least another 1 sigma to see when the process has gone out of control and needs intervention.



Manufacturing Process Control

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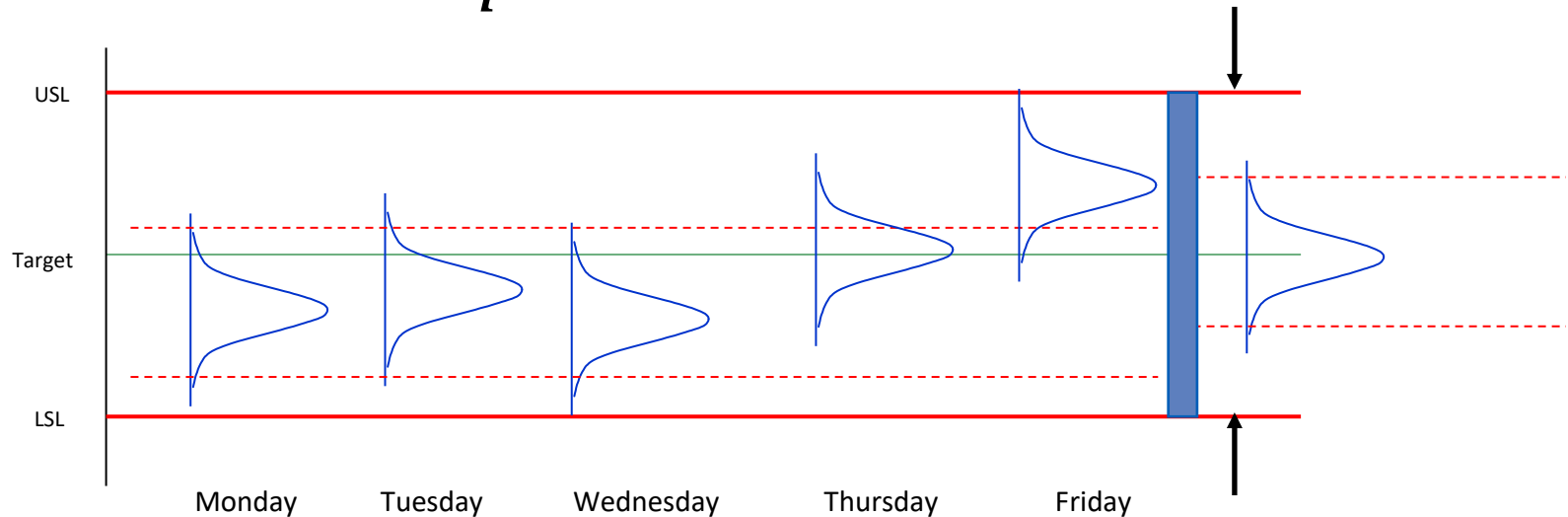
The engineers need at least another 1 sigma to see when the process has gone out of control and needs intervention.



Capability Requirement

Therefore design engineers should set tolerances to at least 4 standard deviations of process variation.

$$C_p \geq 1.33$$





Overview

Statistical Process Control

- Xbar-R charts

- Out of Control Action Plans

Incoming Inspection

- Statistical Sampling Plans

- Cpk Inspection Plans

PPAP



The X-bar, R Chart

SPC is mean to discover if the process statistics have changed.

Therefore we chart and track sample statistics.

Not individual values.

The ***X-bar, R chart*** is useful for many processes for which you have a continuous variable measurement

This chart plots sample averages and ranges

X-bar, R charts enable one to evaluate stability in nominal value and in variation

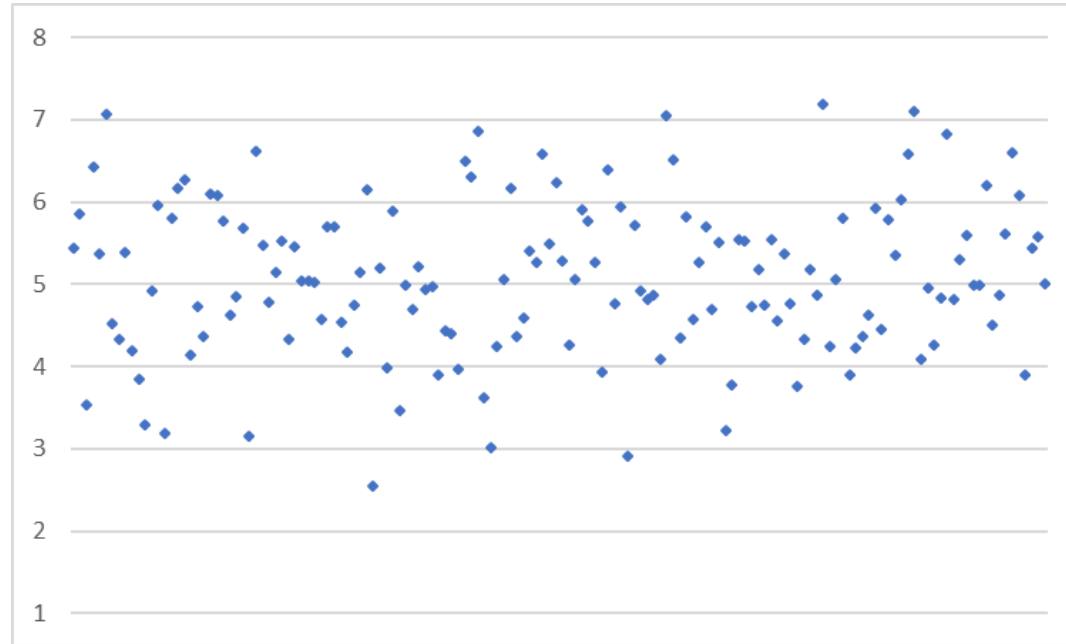
SPC Chart Statistics

SPC is mean to discover if the process statistics have changed.

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Not individual values.

**Here are the process
KPI values. Suppose
we don't know the
values, we have not
100% measured them.**



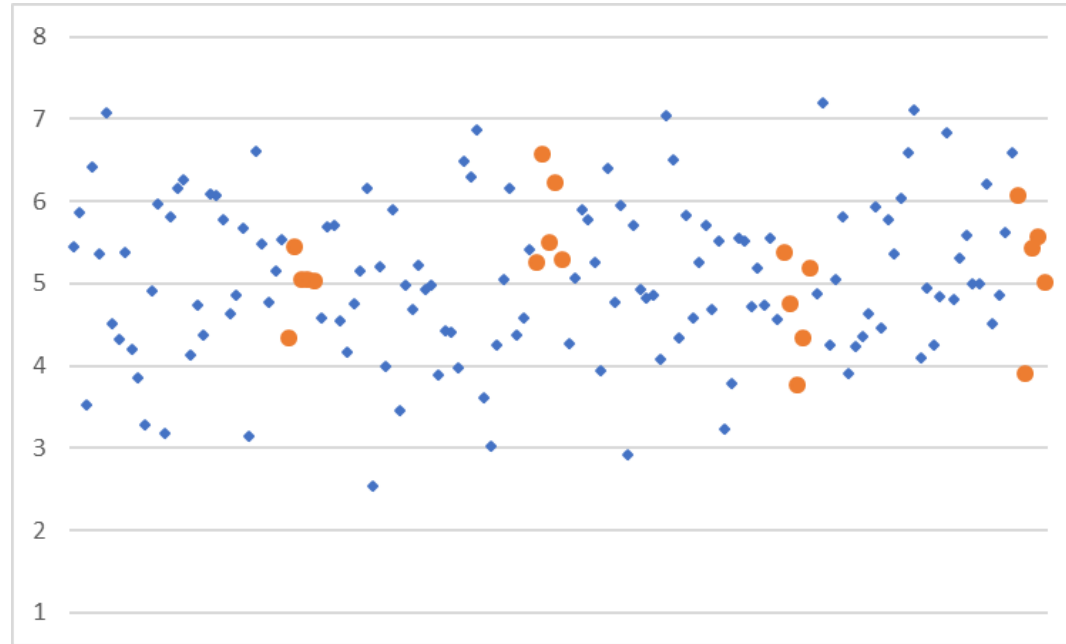
SPC Chart Statistics

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**Periodically take small
samples and measure
only these.**



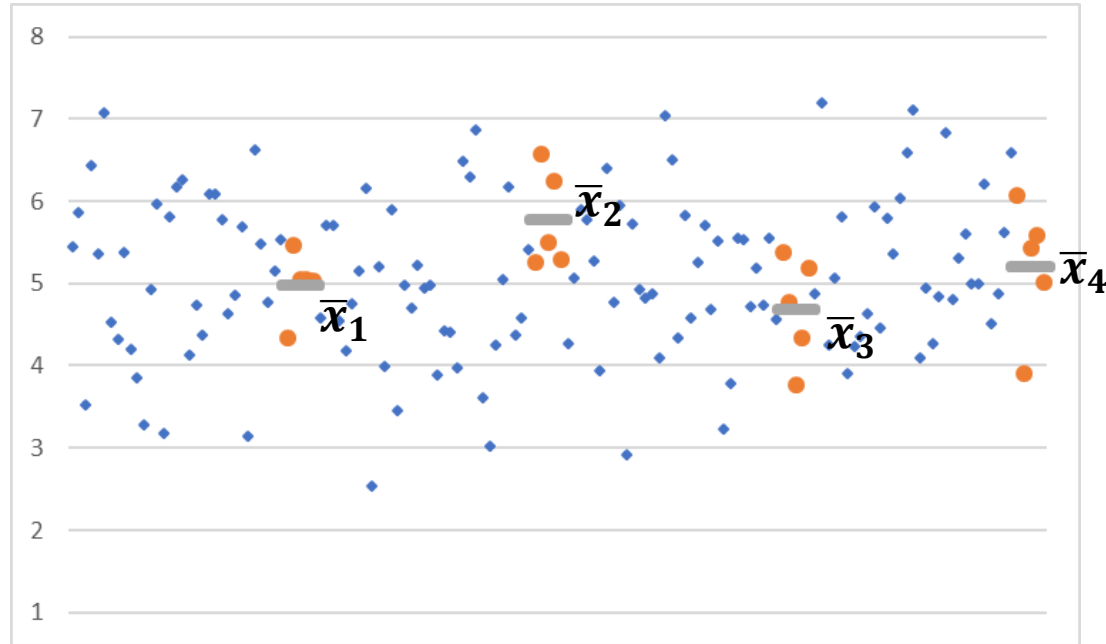
SPC Chart Statistics

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Compute the sample
statistic.
Here $\bar{x}$ of the 5 units in
each subgroup.



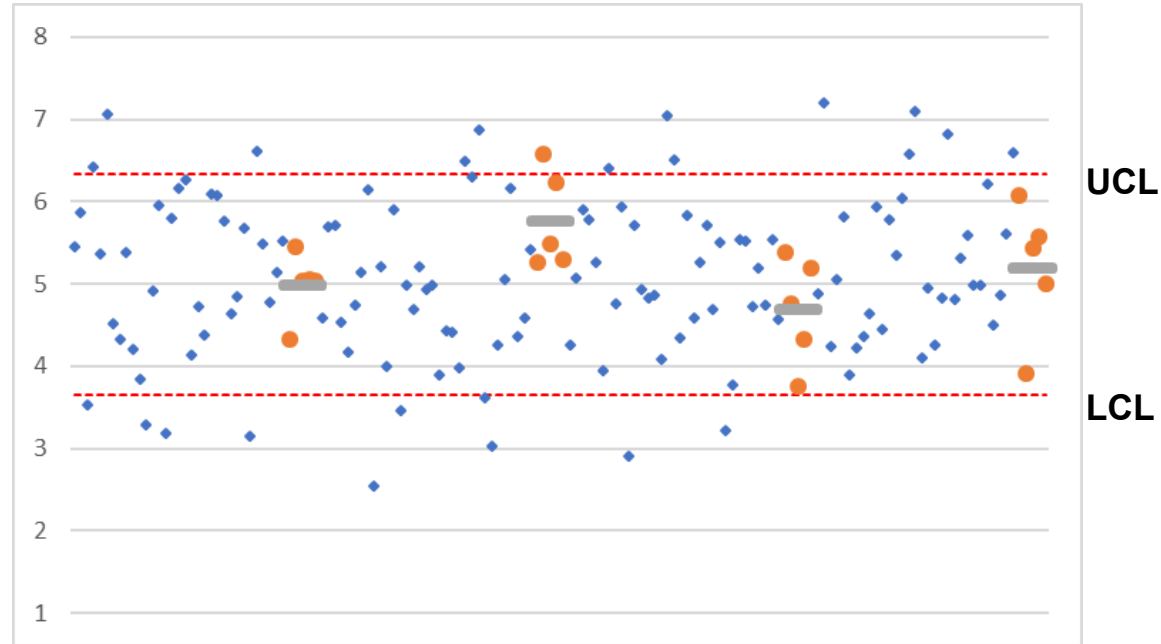
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**Has the sample's
statistic changed
unexpectedly?**



SPC Chart Statistics

SPC is mean to discover if the process statistics have changed.

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**The SPC chart.
Has the sample's
statistic changed
unexpectedly?**



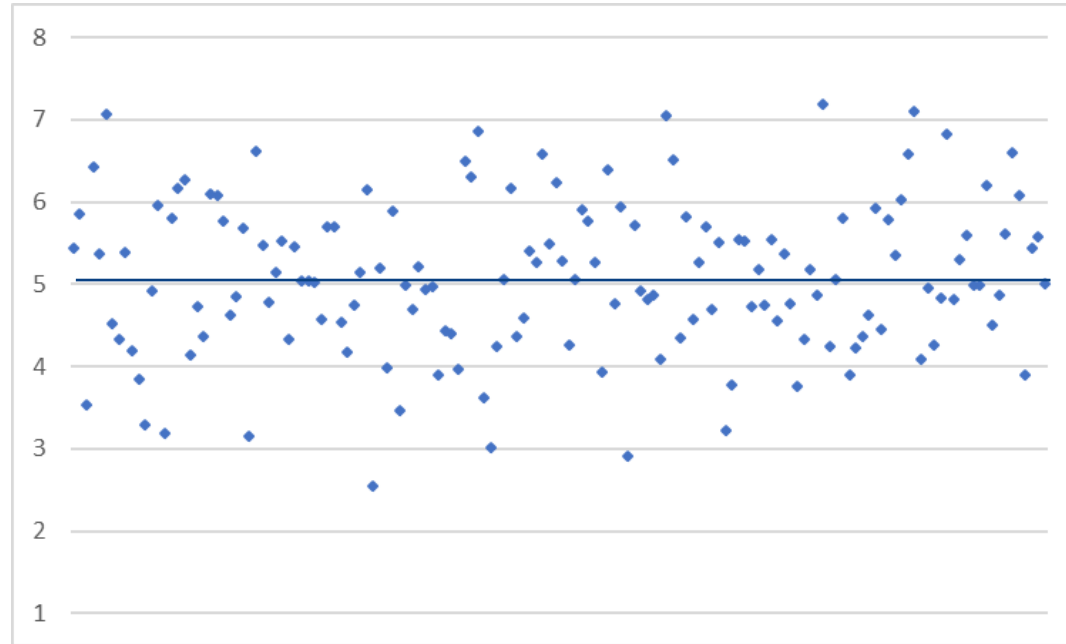
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**No, so you can be
confident the actual
unmeasured data looks
something like this.
Random about a constant
steady average.**



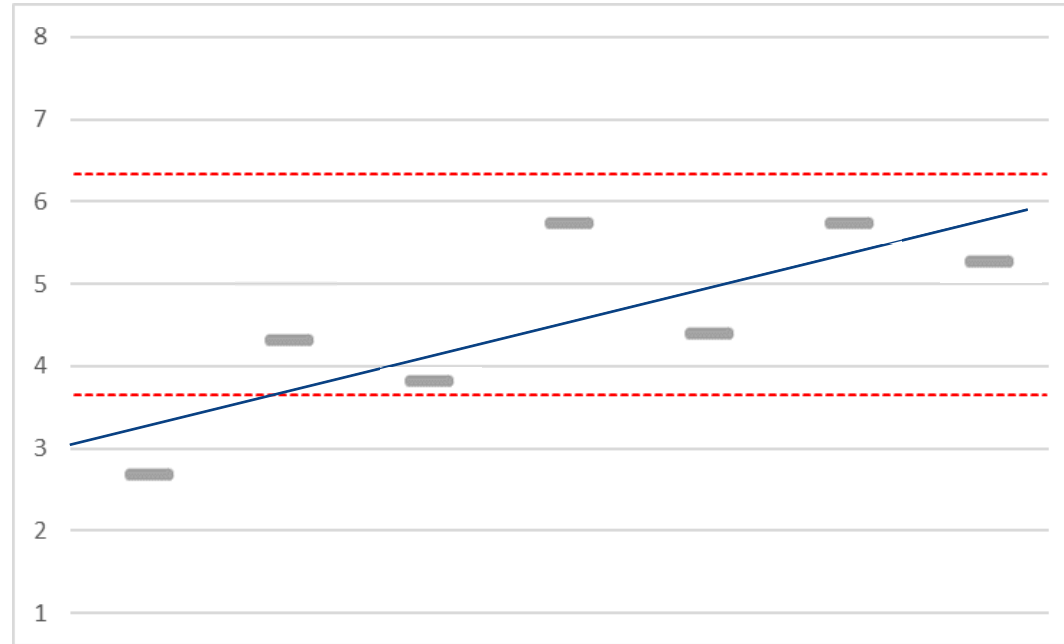
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SPC is mean to discover if the process statistics have changed.

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Not individual values.

**If the SPC chart would
have show the sample
averages increasing...**



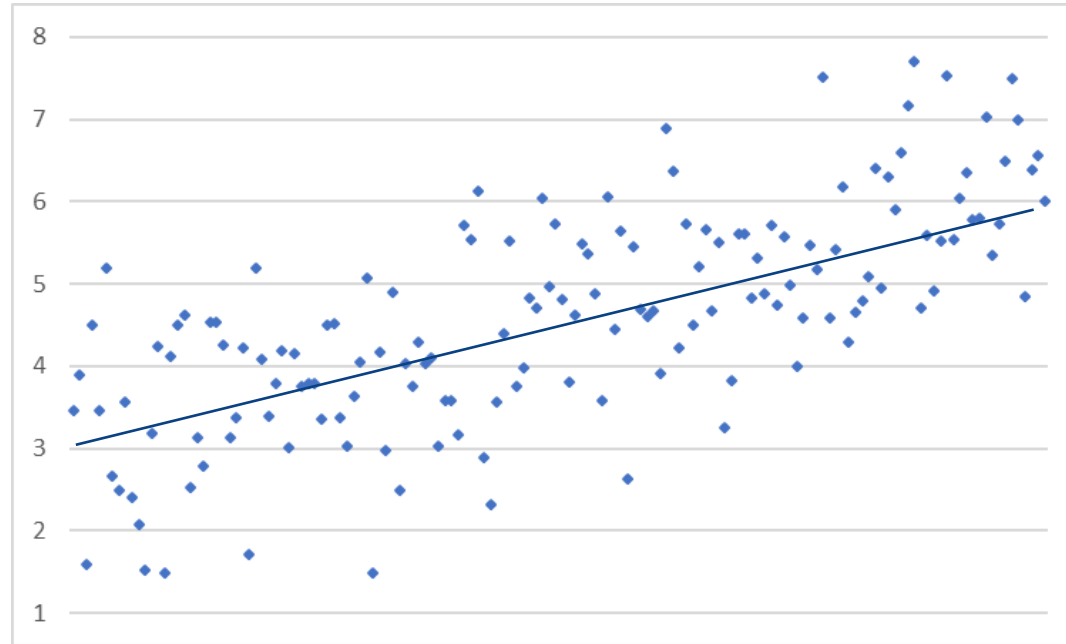
SPC Chart Statistics

SPC is mean to discover if the process statistics have changed.

Therefore we chart and track sample statistics.

Not individual values.

**The data itself is also
increasing.**



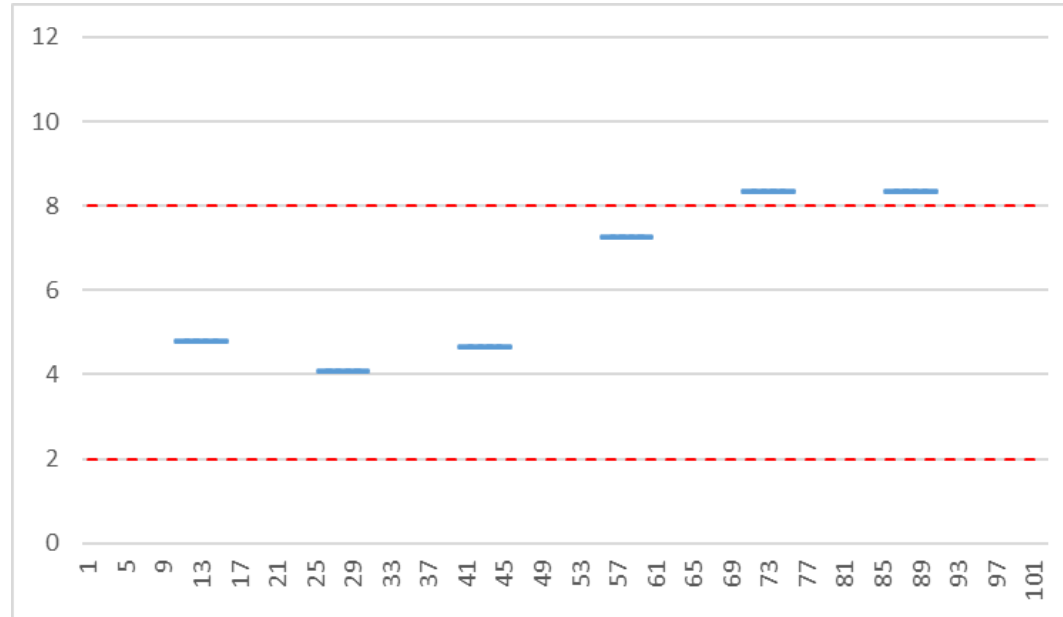
SPC Chart Statistics

SPC is mean to discover if the process statistics have changed.

Therefore we chart and track sample statistics.

Not individual values.

**If the SPC chart would
have show the sample
average shift...**



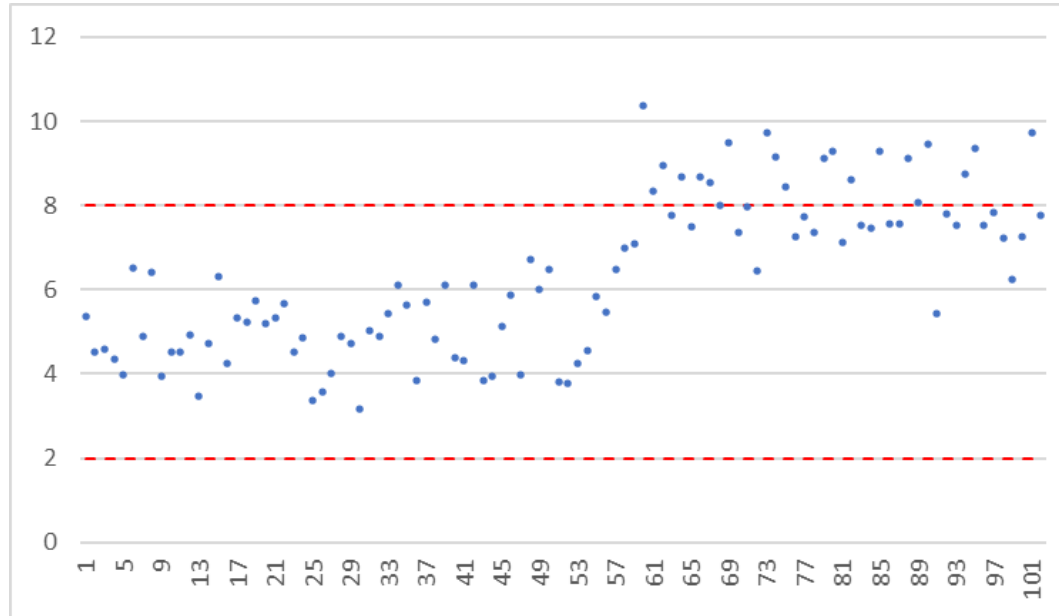
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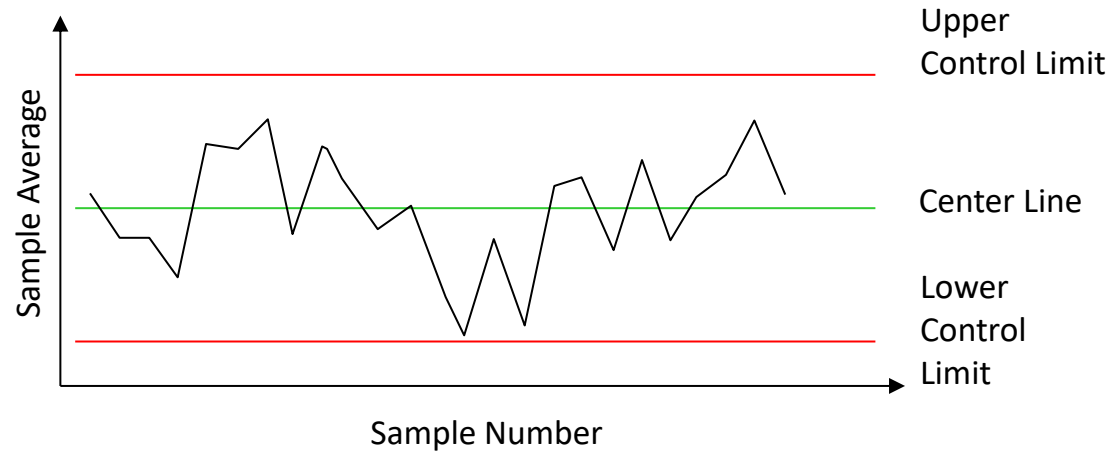
Not individual values.

**The data itself has
also shifted.**



Components of an X-bar Control Chart

Model for Center line, UCL & LCL for an X-bar chart



$$UCL = \mu + k\sigma$$

$$\text{Center Line} = \mu$$

$$LCL = \mu - k\sigma$$

But we must use estimates from samples:

$\mu \cong \bar{\bar{x}}$ = the mean of the sample means

$\sigma \cong s_{\bar{x}}$ = the standard deviation of the sample means

k = constants implying unusual observations (often ~ 3)

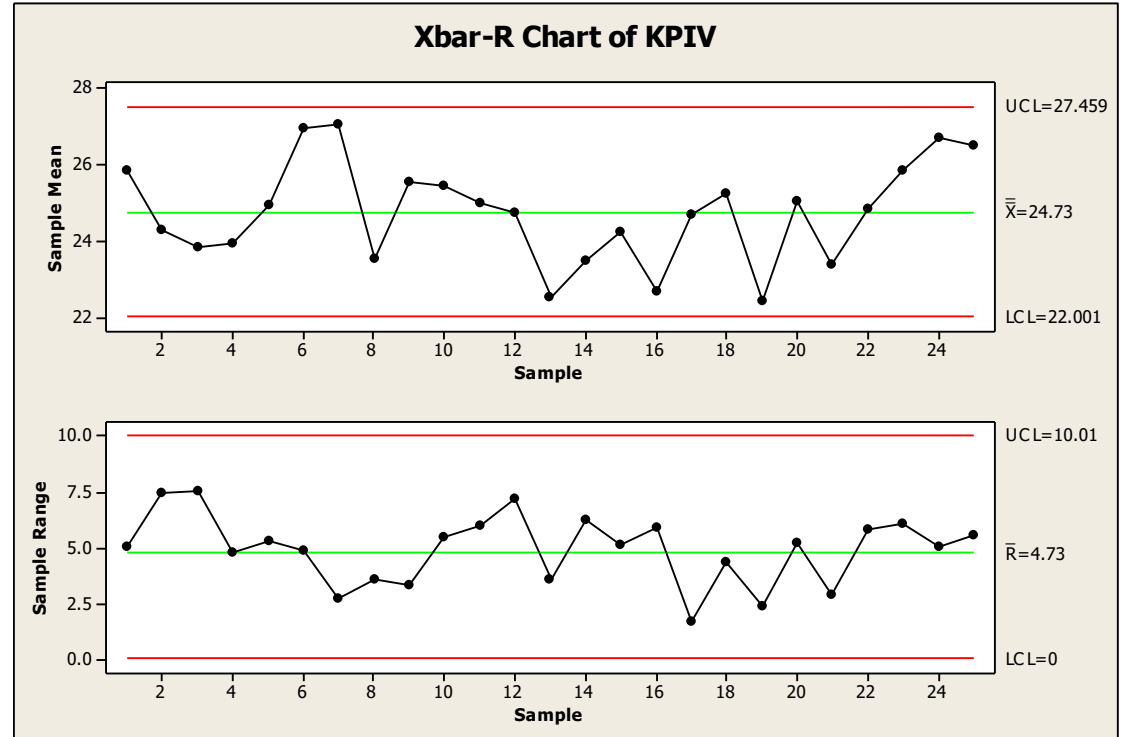
X-bar R chart

The **X-bar Chart** measures the central tendency of the variable over time. It plots $\bar{x}$ of samples.

$$CL_{\bar{x}} = \bar{\bar{X}} \pm A_2 \bar{R}$$

The Range, or **R Chart** measures the variability within subgroup over time. It plots the range R of samples.

$$CL_R = \bar{R} \pm D_{3,4} \bar{R}$$



SPC Chart Parameters

Xbar chart control limits

$$CL = \pm A_2 \bar{R}$$

$$CL = \pm A_3 \bar{s}$$

R chart control limits:

$$UCL = D_4 \bar{R}$$

$$LCL = D_3 \bar{R}$$

S chart control limit:

$$LCL = B_3 \bar{s}$$

$$UCL = B_4 \bar{s}$$

Estimating $\sigma_{\text{historical}}$:

$$\sigma_{\text{historical}} = \frac{\bar{R}}{d_2} = \frac{\bar{s}}{c_4}$$

n	d2	d4	c4	A2	A3	D3	D4	B3	B4
2	1.128	0.954	0.798	1.881	2.659	0	3.267	0	3.267
3	1.693	1.588	0.886	1.023	1.954	0	2.574	0	2.568
4	2.059	1.978	0.921	0.729	1.628	0	2.282	0	2.266
5	2.326	2.257	0.940	0.577	1.427	0	2.114	0	2.089
6	2.534	2.472	0.952	0.483	1.287	0	2.004	0.030	1.970
7	2.704	2.645	0.959	0.419	1.182	0.076	1.924	0.118	1.882
8	2.847	2.791	0.965	0.373	1.099	0.136	1.864	0.185	1.815
9	2.970	2.915	0.969	0.337	1.032	0.184	1.816	0.239	1.761
10	3.078	3.024	0.973	0.308	0.975	0.223	1.777	0.284	1.716
11	3.173	3.121	0.975	0.285	0.927	0.256	1.744	0.321	1.679
12	3.258	3.207	0.978	0.266	0.886	0.283	1.717	0.354	1.646
13	3.336	3.285	0.979	0.249	0.850	0.307	1.693	0.382	1.618
14	3.407	3.356	0.981	0.235	0.817	0.328	1.672	0.406	1.594
15	3.472	3.422	0.982	0.223	0.789	0.347	1.653	0.428	1.572
16	3.532	3.482	0.983	0.212	0.763	0.363	1.637	0.448	1.552
17	3.588	3.538	0.985	0.203	0.739	0.378	1.622	0.466	1.534
18	3.640	3.591	0.985	0.194	0.718	0.391	1.609	0.482	1.518
19	3.689	3.640	0.986	0.187	0.698	0.403	1.597	0.497	1.503
20	3.735	3.686	0.987	0.180	0.680	0.415	1.585	0.510	1.490
21	3.778	3.730	0.988	0.173	0.663	0.425	1.575	0.523	1.477
22	3.819	3.771	0.988	0.167	0.647	0.434	1.566	0.534	1.466
23	3.858	3.811	0.989	0.162	0.633	0.443	1.557	0.545	1.455
24	3.895	3.847	0.989	0.157	0.619	0.452	1.548	0.555	1.445
25	3.931	3.883	0.990	0.153	0.606	0.459	1.541	0.565	1.435

SPC Chart Constants

The fundamental constants are

$$d_2 = \int_{-\infty}^{\infty} \left(1 - (1 - \Phi(x))^n - \Phi(x)^n\right) dx$$

$$d_3 = \sqrt{2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(1 - \left(\Phi(y)^n - (1 - \Phi(x))^n\right) + (\Phi(y) - \Phi(x))^n\right) dx dy} - d_2^2$$

$$c_4 = \frac{\Gamma\left(\frac{n}{2}\right)\sqrt{2/(n-1)}}{\Gamma\left(\frac{n-1}{2}\right)}$$

Then for a confidence of 3 sigma, $k = 3$. Also n is the sample size. Then:

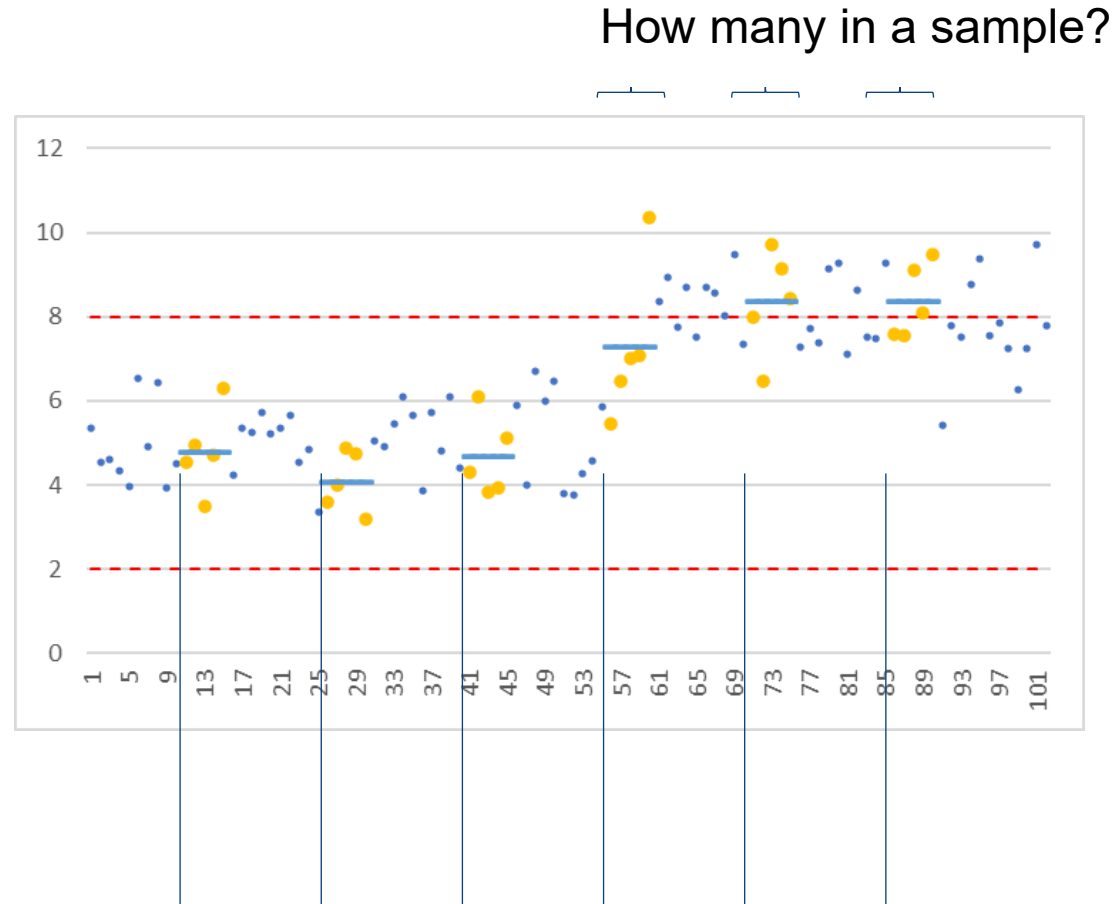
$$A_2 = k/(d_2\sqrt{n}) \quad B_3 = \max\left(0, 1 - \frac{k}{c_4}\sqrt{1 - c_4^2}\right) \quad D_3 = \max\left(0, 1 - kd_3/d_2\right)$$

$$A_3 = k/(c_4\sqrt{n}) \quad B_4 = 1 + \frac{k}{c_4}\sqrt{1 - c_4^2} \quad D_4 = 1 + kd_3/d_2$$

Choices on SPC

What sample size?

How often?



How frequently?



How often? Sampling Frequency

The frequency with which to sample subgroups and update the SPC chart is usually based on a *rational breakpoint*.

- Daily, hourly, etc.
- Shift, lot, etc.

This depends on production rate.

Determining the frequency is a trade off judgement between cost of sampling versus the cost of not detecting a special cause and so making defective product.

How many? Choosing the Subgroup Size

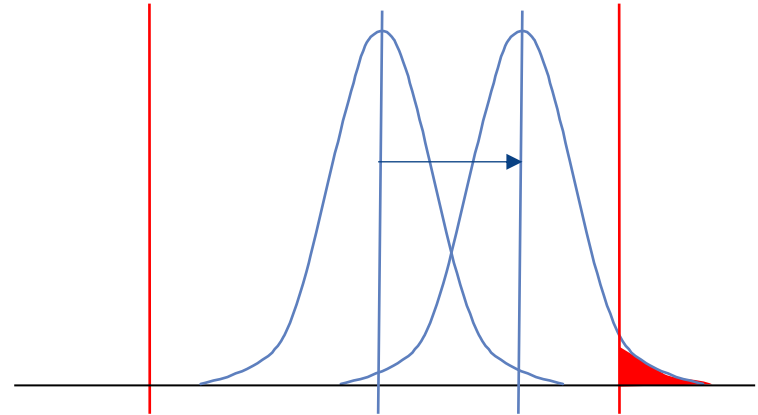
The subgroup must be large enough to be representative

- The sample adequately reflects the process performance during the time period which it represents
- The sample can be taken with a consistent methodology and accurately measured

The subgroup must be small enough to detect a change

- The subgroup size must also not be too large, or else signals for process shifts considered insignificant will occur

We want to know “Has it shifted”?

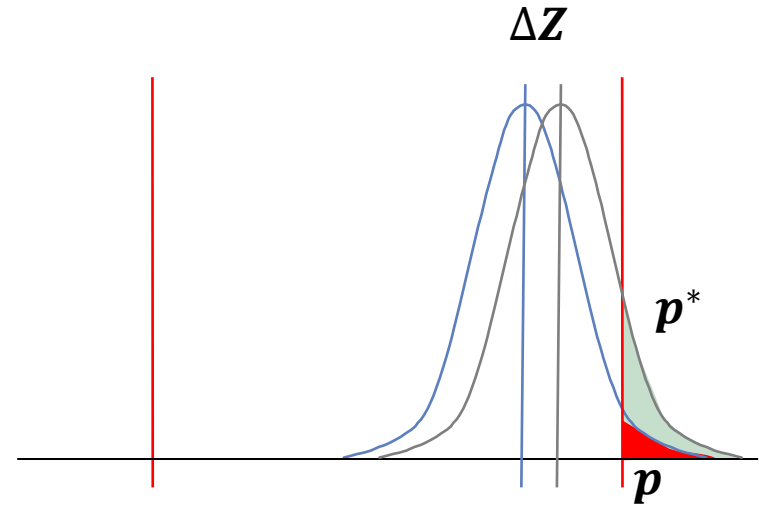


Computing a Sample Size

Has it shifted?

Define an acceptable increase in risk to a max p^* .

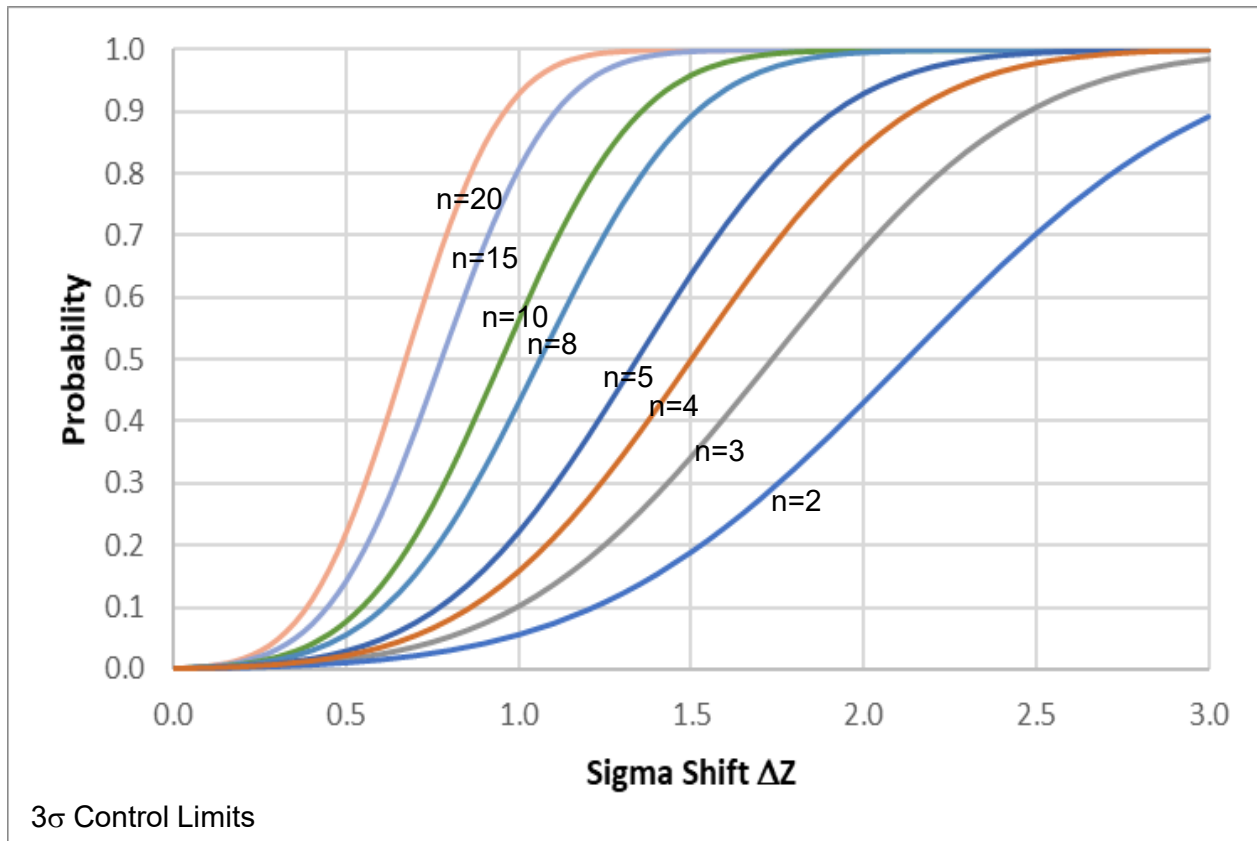
This defines the critical shift ΔZ_{mean} that must be detected.



$$\Delta Z_{mean} = \text{NORMSINV}(1 - p) - \text{NORMSINV}(1 - p^*)$$

Computing a Sample Size

Probability of Detecting a Shift in Mean



$$p = \text{normsdist}(\Delta Z \sqrt{n} - L)$$

where

p = probability of detecting

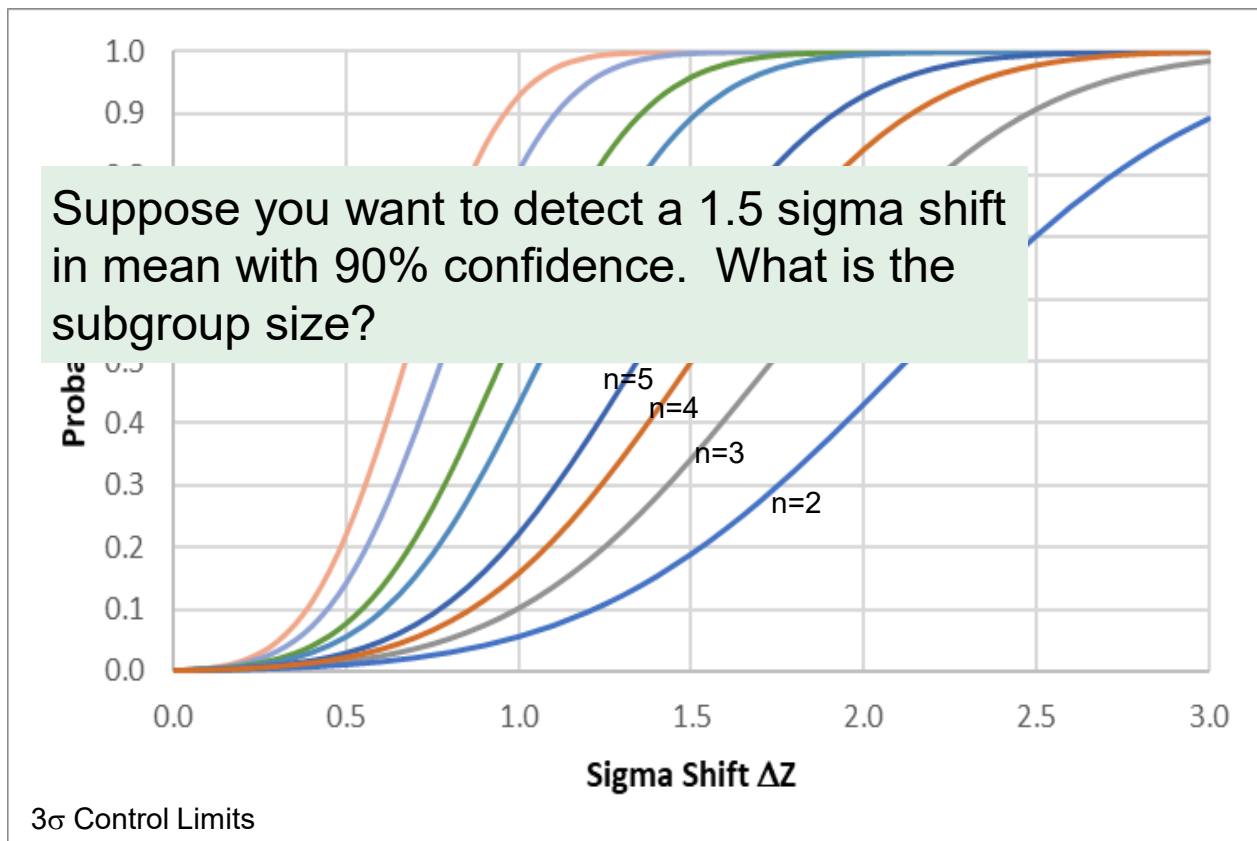
ΔZ = Z-score shift to detect

n = subgroup sample size

L = Z-score of control limit

Computing a Sample Size

Probability of Detecting a Shift in Mean



$$p = \text{normsdist}(\Delta Z \sqrt{n} - L)$$

where

p = probability of detecting

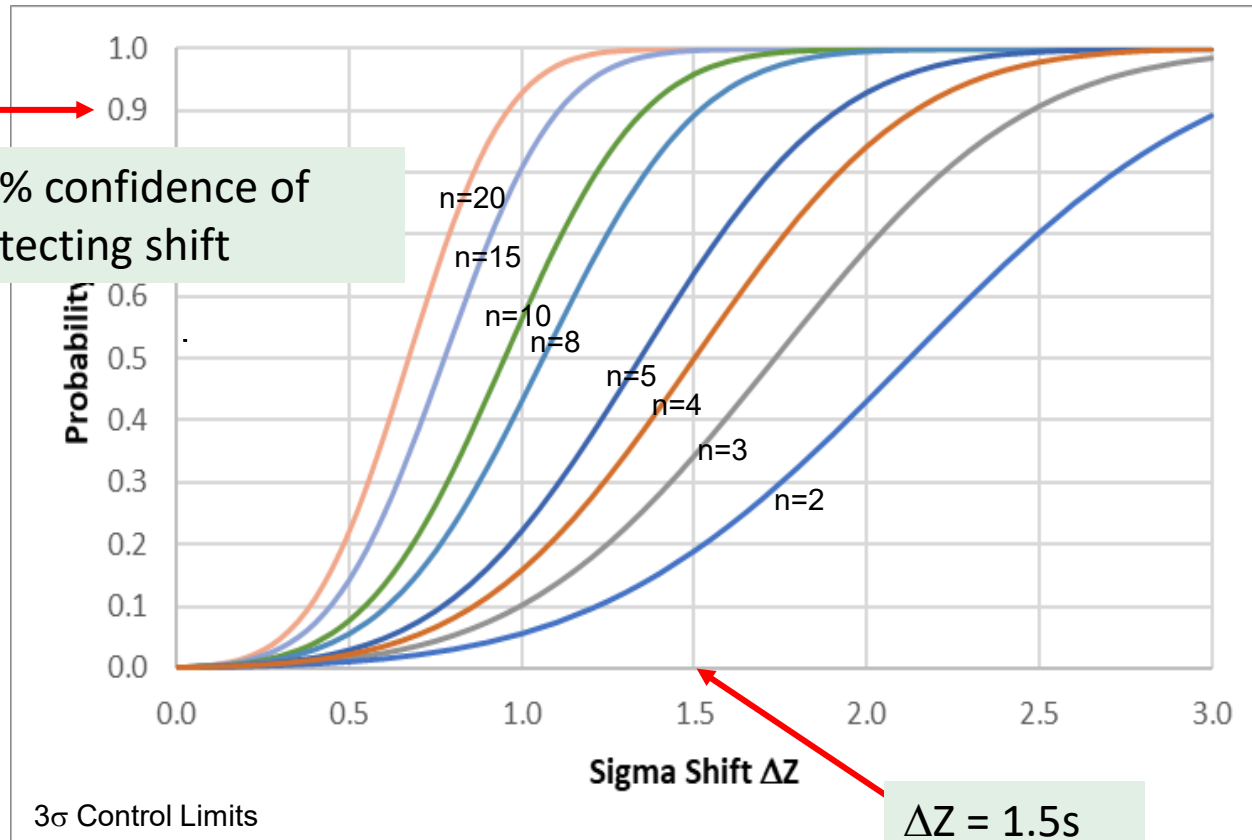
ΔZ = Z-score shift to detect

n = subgroup sample size

L = Z-score of control limit

Computing a Sample Size

Probability of Detecting a Shift in Mean



$$p = \text{normsdist}(\Delta Z \sqrt{n} - L)$$

where

p = probability of detecting

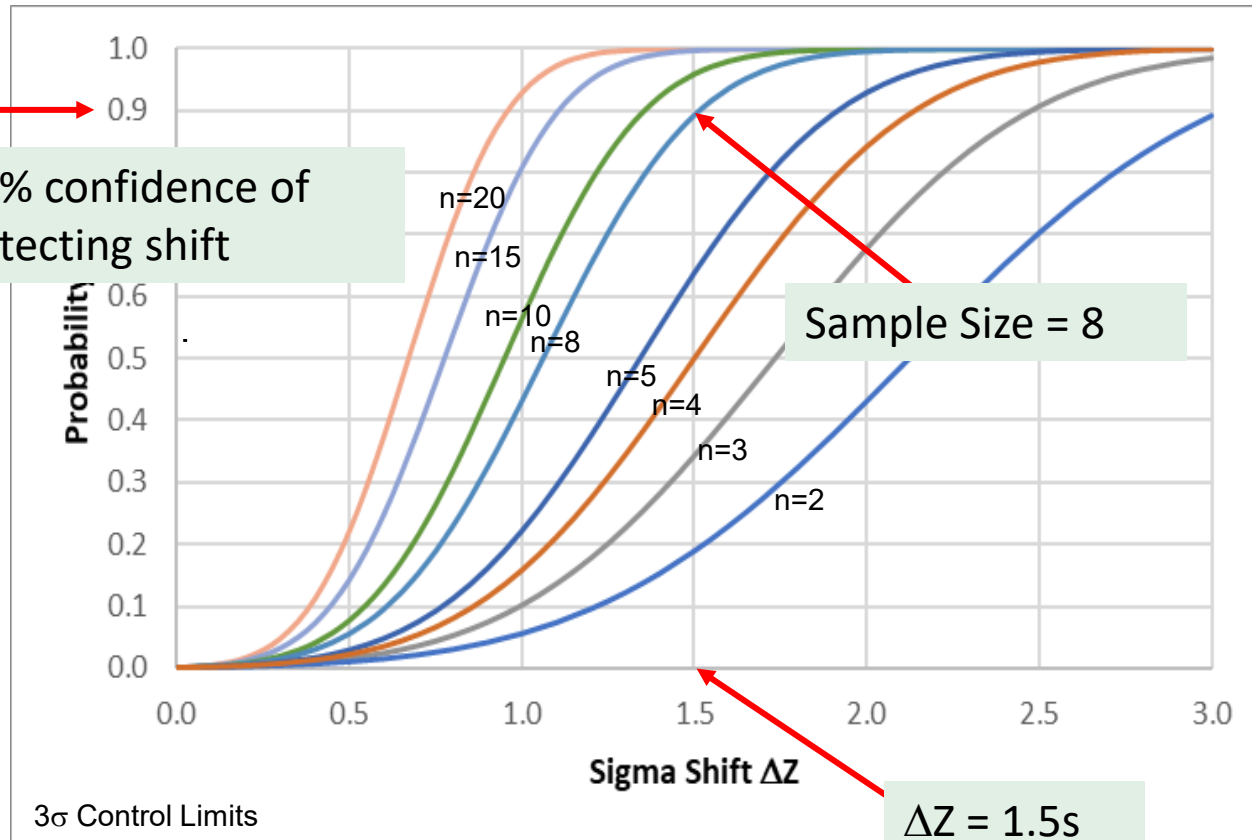
ΔZ = Z-score shift to detect

n = subgroup sample size

L = Z-score of control limit

Computing a Sample Size

Probability of Detecting a Shift in Mean



$$p = \text{normsdist}(\Delta Z \sqrt{n} - L)$$

where

p = probability of detecting

ΔZ = Z-score shift to detect

n = subgroup sample size

L = Z-score of control limit

Xbar-R Chart in Minitab

Stat > Control Charts > Charts for Subgroups > Xbar-R

Xbar-R Chart

Observations for a subgroup are in one row of columns:

x1-x5

Scale... Labels... Multiple Graphs... Data Options... Xbar-R Options...

Select

Help OK Cancel

Xbar-R Chart: Options

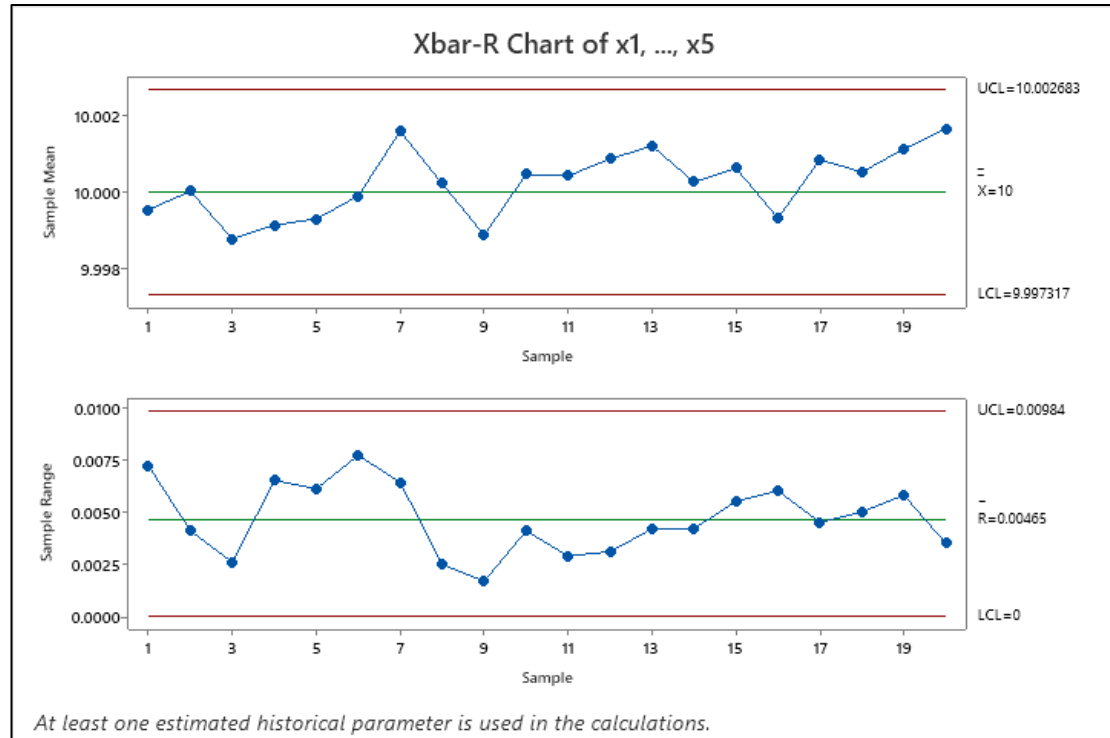
Parameters | Estimate | Limits | Tests | Stages | Box-Cox | Display | Storage

To specify the values for one or both parameters, enter them here. Minitab uses these values instead of estimating them from the data.

Mean: 10

Standard deviation: 0.002

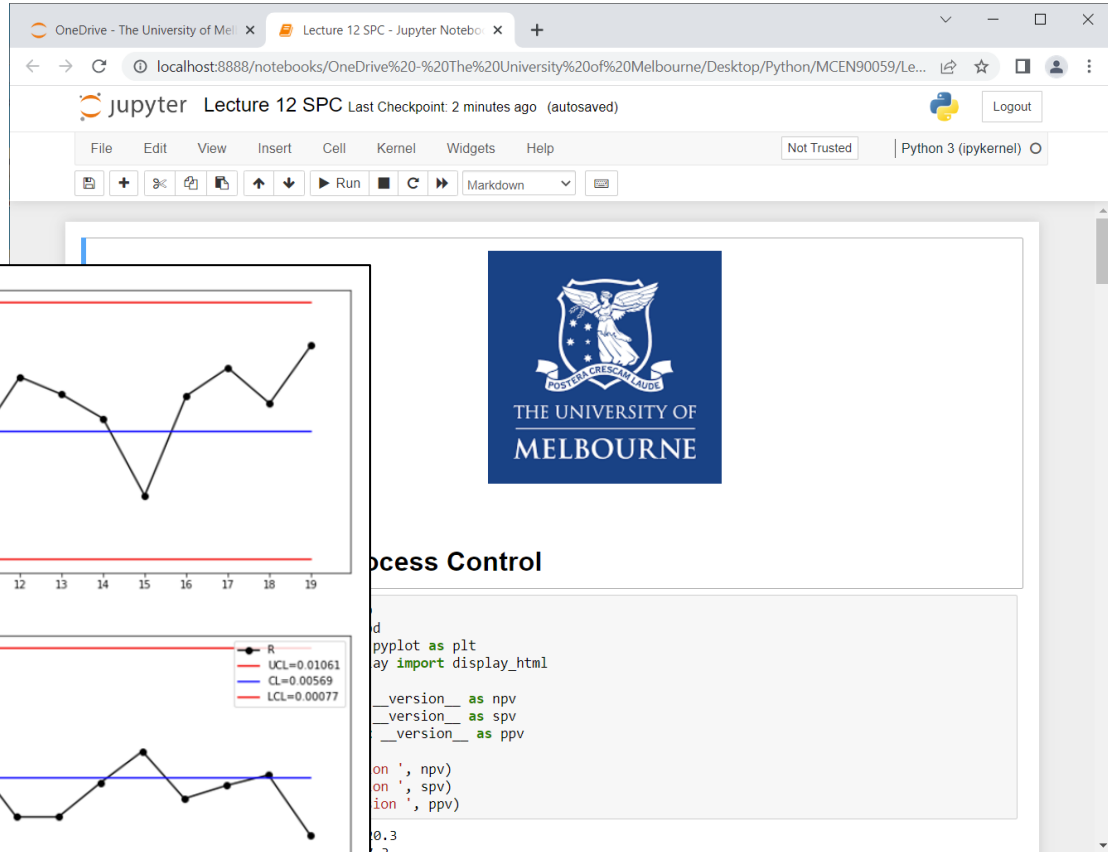
Help OK Cancel



Xbar-R Chart in Python

No good library.

A notebook “SPC Charts.ipynb”
can generate an Xbar-R chart.



EXERCISE

A firm produces precision ball valve bodies whose diameters are under SPC control. File “W21 Data.xlsx” has SPC sample data from the past weeks of production. If the historical mean has been 10.00 and the historical sigma has been 0.70,

a. Create an Xbar-R chart. Is the process in control?

b. A new sample has just been taken. It is (10.0019, 10.0027, 10.0038, 10.0036, 9.9973, 10.0037, 10.0028, 10.0017). Is the process now in control?

Xbar-R Chart

Observations for a subgroup are in one row of columns: ▼

x1-x8

Scale... Labels... Multiple Graphs... Data Options... Xbar-R Options...

Select

Help OK Cancel

Xbar-R Chart: Options

Parameters Estimate Limits Tests Stages Box-Cox Display Storage

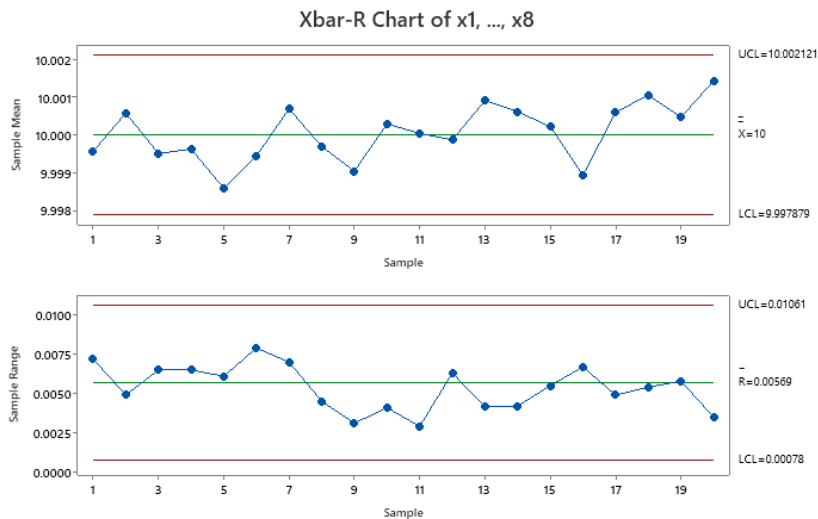
To specify the values for one or both parameters, enter them here. Minitab uses these values instead of estimating them from the data.

Mean:

10

Standard deviation:

0.002



At least one estimated historical parameter is used in the calculations.

x1	x2	x3	x4	x5	x6	x7	x8
9.9976	10.0021	9.9961	9.9986	10.0033	9.9979	10.0026	9.9983
0.0017	10.0021	10.0025	9.9999	9.999	9.9996	9.9988	10.0037
0.0028	10.0021	9.9999	9.9993	0.0008	9.9977	9.9968	9.9977
9.9968	10.0014	9.9966	9.998	0.0033	10.001	9.9976	9.999
0.0005	9.9973	9.9995	9.9996	9.9992	10.0007	9.9996	9.9976
9.9998	9.999	10.0011	9.9999	9.9989	9.9997	9.999	9.9995
9.9997	9.998	10.0002	9.9964	9.9998	10.001	10.0005	9.9997
0.0005	10.0019	10.0025	9.9992	9.9988	10.0005	9.9983	9.9998
0.001	9.9983						
9.989	10.0021						
9.995	10.0023						
9.996	9.9992						
0.017	10.0006						

Now we get an xbar outlier. This is out of control according to our SPC rules.

9.999	9.9995
10.0002	9.9964
10.0005	9.9997
10.0025	9.9992
9.9983	9.9998
10.0001	9.9983
9.9989	10.0021
9.9995	10.0023
9.9996	9.9992
10.0017	10.0006
10.0028	10.0017

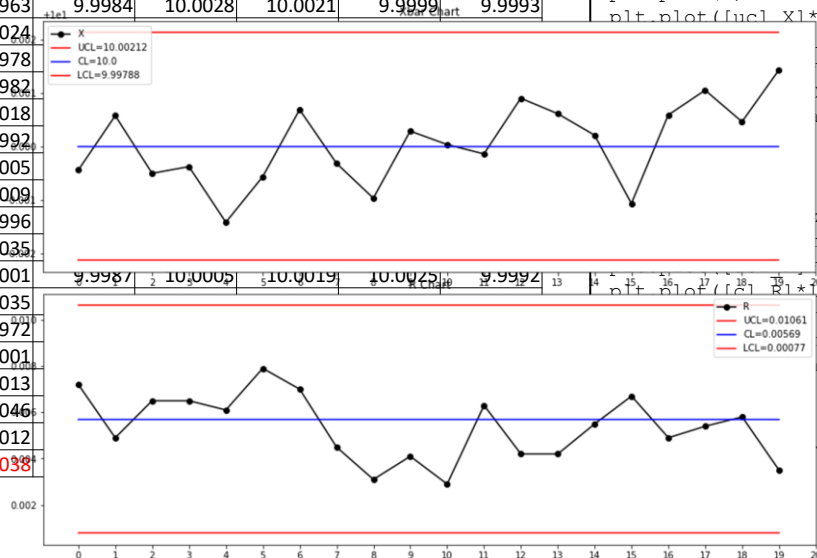


Python

Now redo with one more row of the added data.

Now we get an xbar outlier. This is out of control according to our SPC rules.

x1	x2	x3	x4	x5	x6	x7	x8
9.9976	10.0021	9.9961	9.9986	10.0033	9.9979	10.0026	9.9983
9.9998	10.0004	10.0006	9.9976	10.0017	10.0021	10.0025	9.9999
9.9998	9.9987	9.9972	9.9992	9.999	9.9996	9.9988	10.0037
10.0005	9.9977	9.9963	9.9984	10.0028	10.0021	9.9999	9.9993
9.9963	9.9986	10.0024	9.9992	9.999	9.999	9.999	9.999
10.0014	9.999	9.9978	9.9992	9.999	9.999	9.999	9.999
10.0009	10.0046	9.9982	9.9992	9.999	9.999	9.999	9.999
9.9994	10.0001	10.0018	9.9992	9.999	9.999	9.999	9.999
9.9981	9.9998	9.9992	9.9992	9.999	9.999	9.999	9.999
10.0024	10.0013	10.0005	9.9992	9.999	9.999	9.999	9.999
10.0018	10.0008	10.0009	9.9992	9.999	9.999	9.999	9.999
10.0027	10.0025	9.9996	9.9992	9.999	9.999	9.999	9.999
10.0029	10.0005	10.0035	9.9992	9.999	9.999	9.999	9.999
9.9991	10.0029	10.0001	9.9992	9.999	9.999	9.999	9.999
10.0008	10.002	10.0035	9.9992	9.999	9.999	9.999	9.999
9.9985	9.9983	9.9972	9.9992	9.999	9.999	9.999	9.999
10.001	10	10.0001	9.9992	9.999	9.999	9.999	9.999
9.9986	10.0036	10.0013	9.9992	9.999	9.999	9.999	9.999
9.9988	10.0009	10.0046	9.9992	9.999	9.999	9.999	9.999
10.001	10.0015	10.0012	9.9992	9.999	9.999	9.999	9.999
10.0019	10.0027	10.0038	9.9992	9.999	9.999	9.999	9.999



```
df = pd.read_excel('Lecture 12 SPC Data.xlsx')
data = df.to_numpy()
sample_size = len(data[0])
number_of_samples = len(data)
X = np.zeros((number_of_samples,1))
R = np.zeros((number_of_samples,1))
for i in range(number_of_samples):
    X[i] = data[i].mean()
    R[i] = data[i].max() - data[i].min()
```

```
Xmean = 10.0
Rmean = 0.002 * d2[sample_size]
ucl_X = Xmean + A2[sample_size]*Rmean
cl_X = Xmean
lcl_X = Xmean - A2[sample_size]*Rmean
```

```
ucl_R = D4[sample_size]*Rmean
cl_R = Rmean
lcl_R = D3[sample_size]*Rmean
```

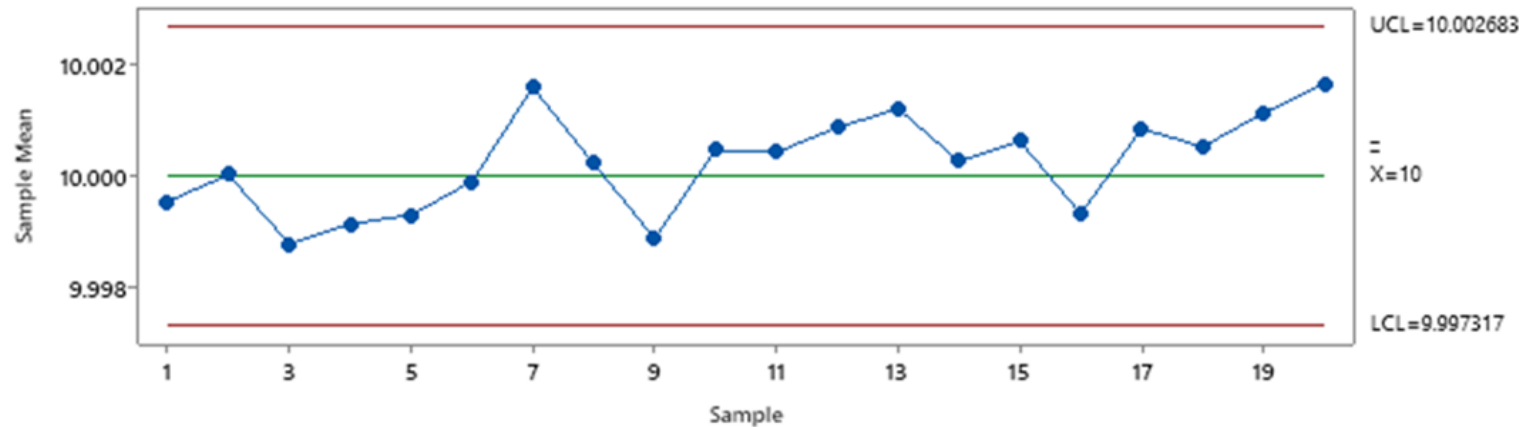
```
roundoffplaces = 5 # number of significant figures in legend
```

```
plt.figure(figsize=(15,5))
plt.plot(X,marker="o",color="k",label="X")
plt.plot([ucl_X]*len(X),color="r",label="UCL={}".format(round(ucl_X, roundoffplaces)))
plt.plot([cl_X]*len(X),color="b",label="CL={}".format(round(cl_X, roundoffplaces)))
plt.plot([lcl_X]*len(X),color="r",label="LCL={}".format(round(lcl_X, roundoffplaces)))
plt.legend()
plt.show()
```

```
plt.figure(figsize=(15,5))
plt.plot(R,marker="o",color="k",label="R")
plt.plot([ucl_R]*len(X),color="r",label="UCL={}".format(round(ucl_R, roundoffplaces)))
plt.plot([cl_R]*len(X),color="b",label="CL={}".format(round(cl_R, roundoffplaces)))
plt.plot([lcl_R]*len(X),color="r",label="LCL={}".format(round(lcl_R, roundoffplaces)))
plt.legend()
plt.show()
```

EXERCISE

A process is being statistically controlled with samples of 5 units. The historical nominal is again 10.000 and with a historical standard deviation of 0.002. The following Xbar chart has resulted.

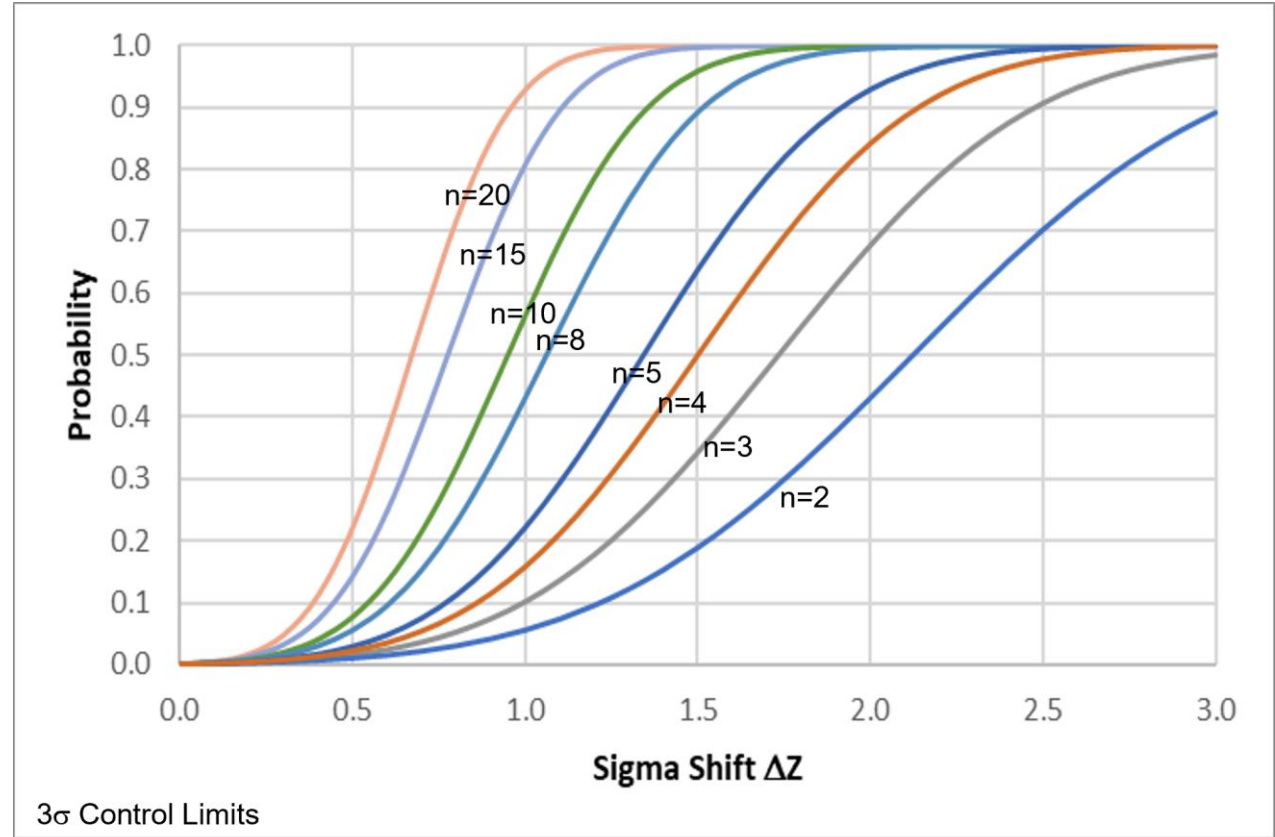


A new latest sample shows an xbar with a 1.5 sigma shift from nominal, $\bar{x}_{\text{new}} = 10.003$.

- What is probability this is a real shift in process mean?
- What is the probability this observation does not indicate a process mean shift and is due to simple random behaviour?

EXERCISE

The probability of detecting a mean shift is dependent on the sample size as discussed earlier. Here, a 1.5 sigma shift with a sample size of 5 shows a probability of detection of 65%. So there is a 35% chance this observed 1.5 sigma shift is due to simple random sampling error.



What Does “Out of Control” Mean?

A process that shows a non-random pattern is considered to be statistically *out of control*

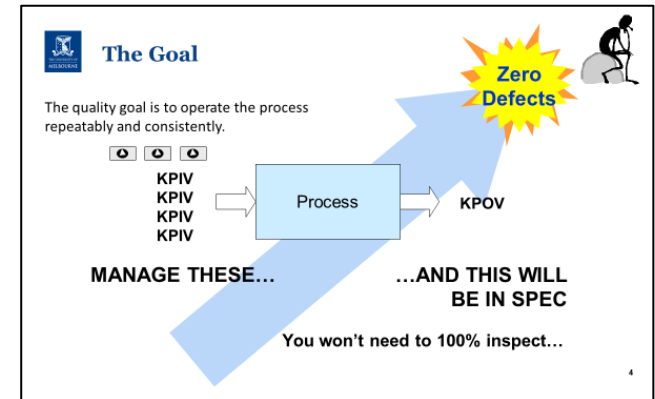
The non-random pattern means that there is strong statistical evidence that the process has changed.

Out of Control ≠ Out of Specification

The process may still be producing good parts.

Should you do something to return the process to where it was, even if it is not out of spec?

Do you want to be a six-sigma high quality manufacturer?





The OCAP

SPC charts are ***BAD NEWS*** charts. They tell you to stop producing (making money) and go fix something. People are often happier not fixing and would rather just keep working.

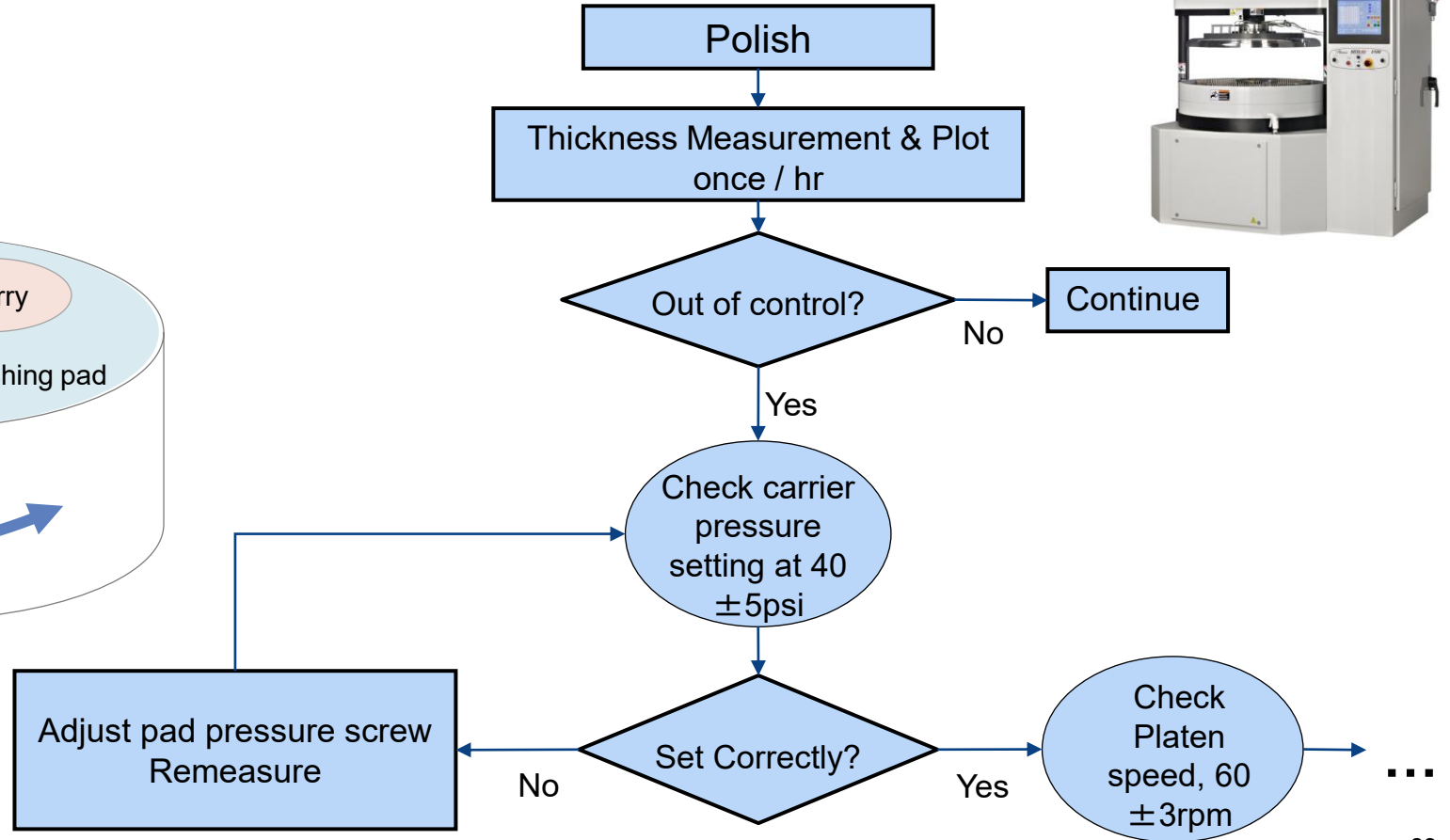
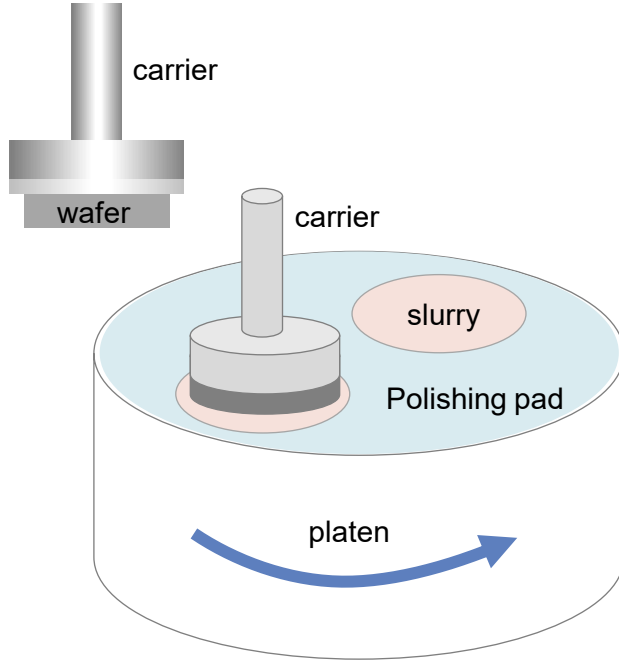
Over time, social problems can develop with control charts

- It can become a paper gathering exercise
- Out-of-control situations might be ignored
- There can be inconsistent or incorrect reactions

Therefore you need a uniform, documented plan of action to out-of-control conditions, the ***Out-of-Control Action Plan*** (OCAP)

- Each SPC chart should have a documented OCAP
- This is a subset of the entire Process Control Plan discussed earlier
 - For the few steps with an SPC implementation augment PCP with OCAP
 - The OCAP specifically states do when any special cause variation is detected

OCAP for a Polisher Machine





Where to Use SPC Charts

Use SPC when a mistake proofing solution is not feasible.

- The process step has high risk in the FMEA
- The process step has input variables that are critical
- Plant management has commitment

Use charts only where the input is critical to quality.

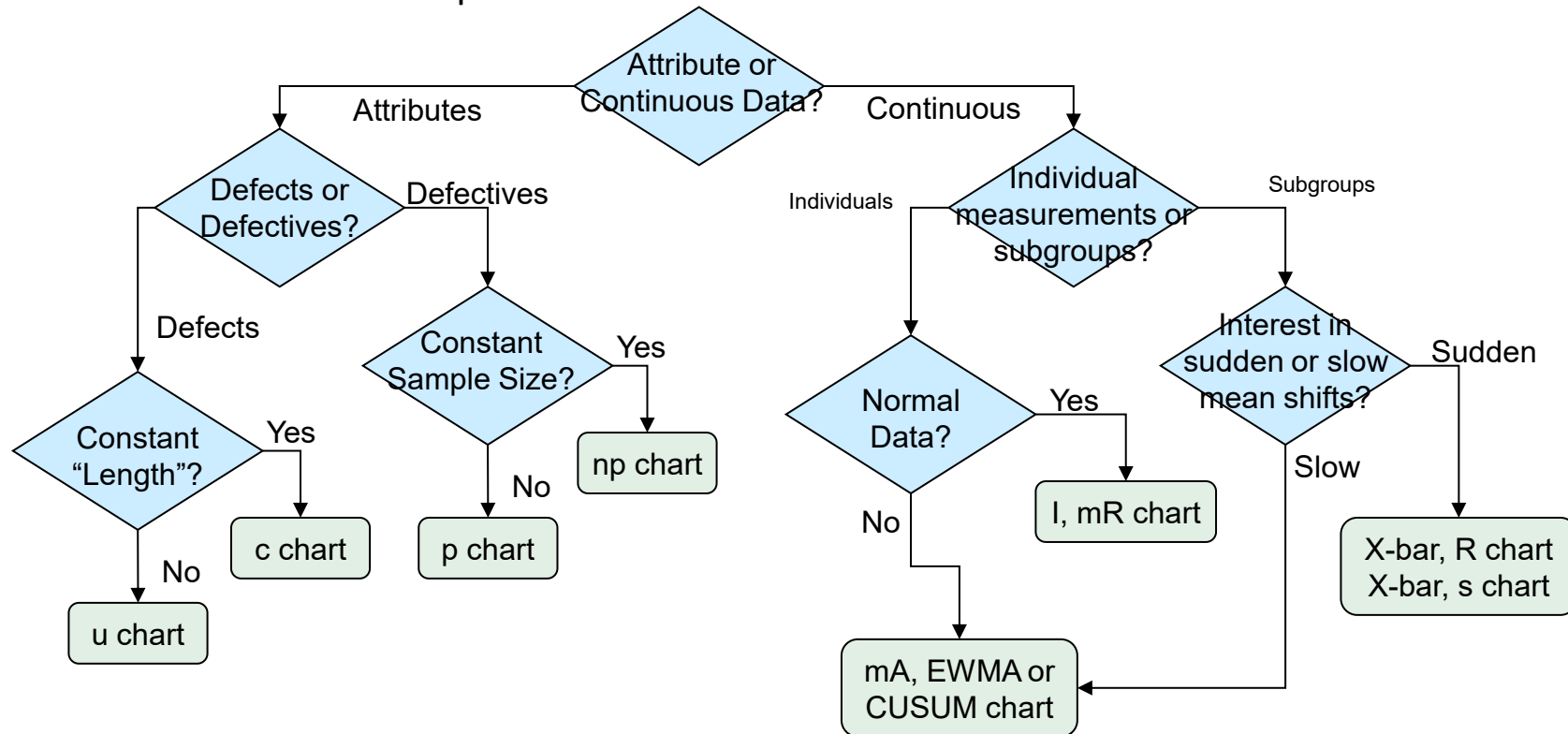
Remove charts that become unnecessary, when $C_{pk} \geq 1.5$

Goal: Monitor and control process inputs only, and, over time, ***eliminate the need entirely for SPC charts.***

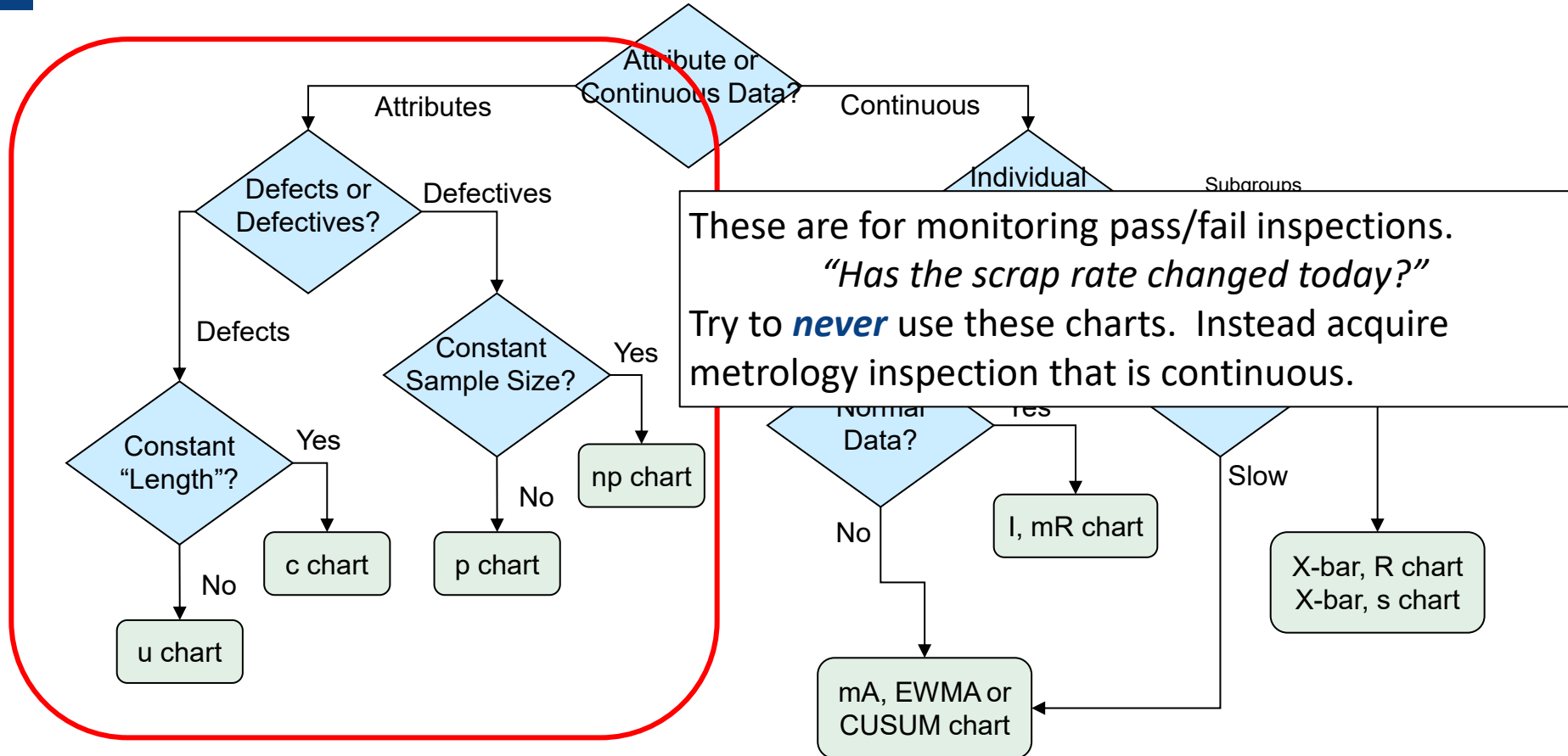
Other SPC Chart Types

SPC charts have been developed for all different situations

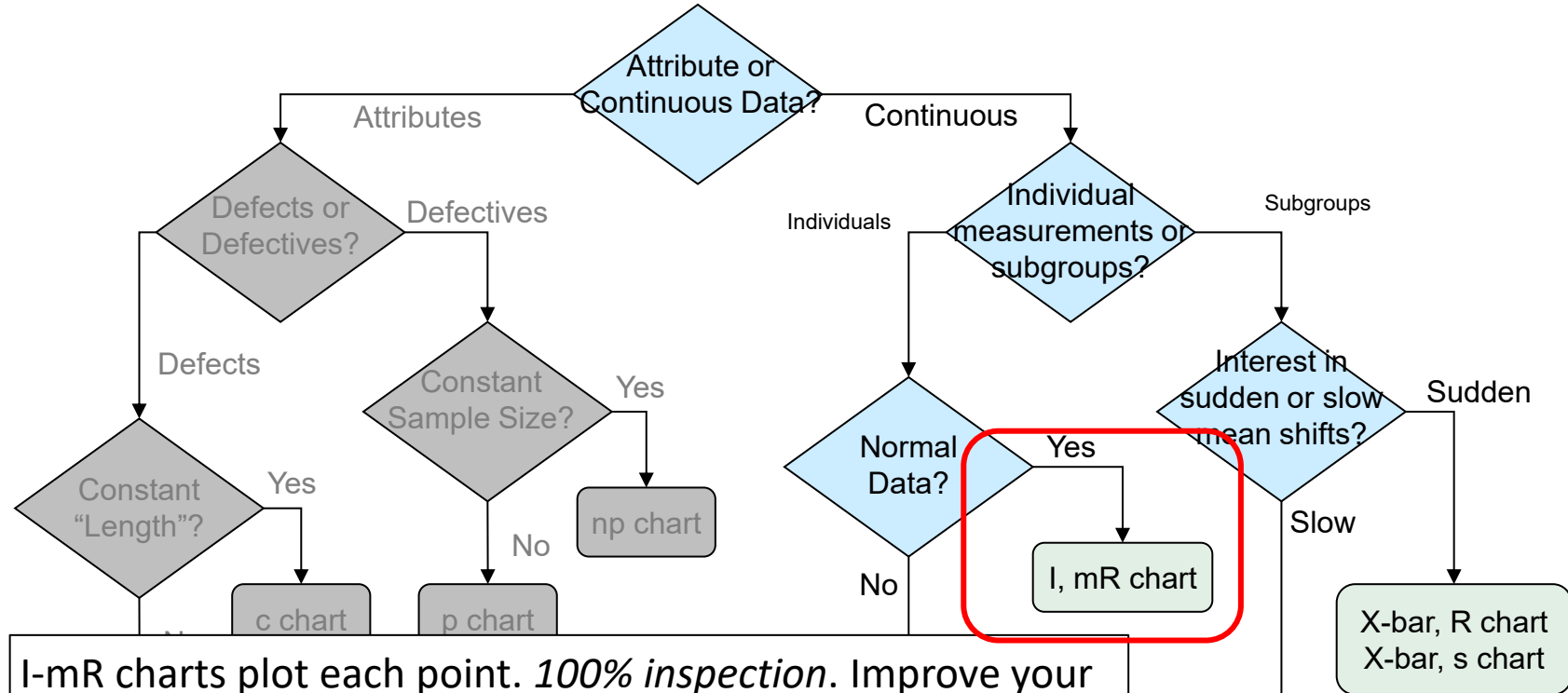
- Pass / Fail inspections. Counts of number of defects per product
- Hand calculated statistics. Computer calculated statistics



Pass/Fail Statistical Process Control



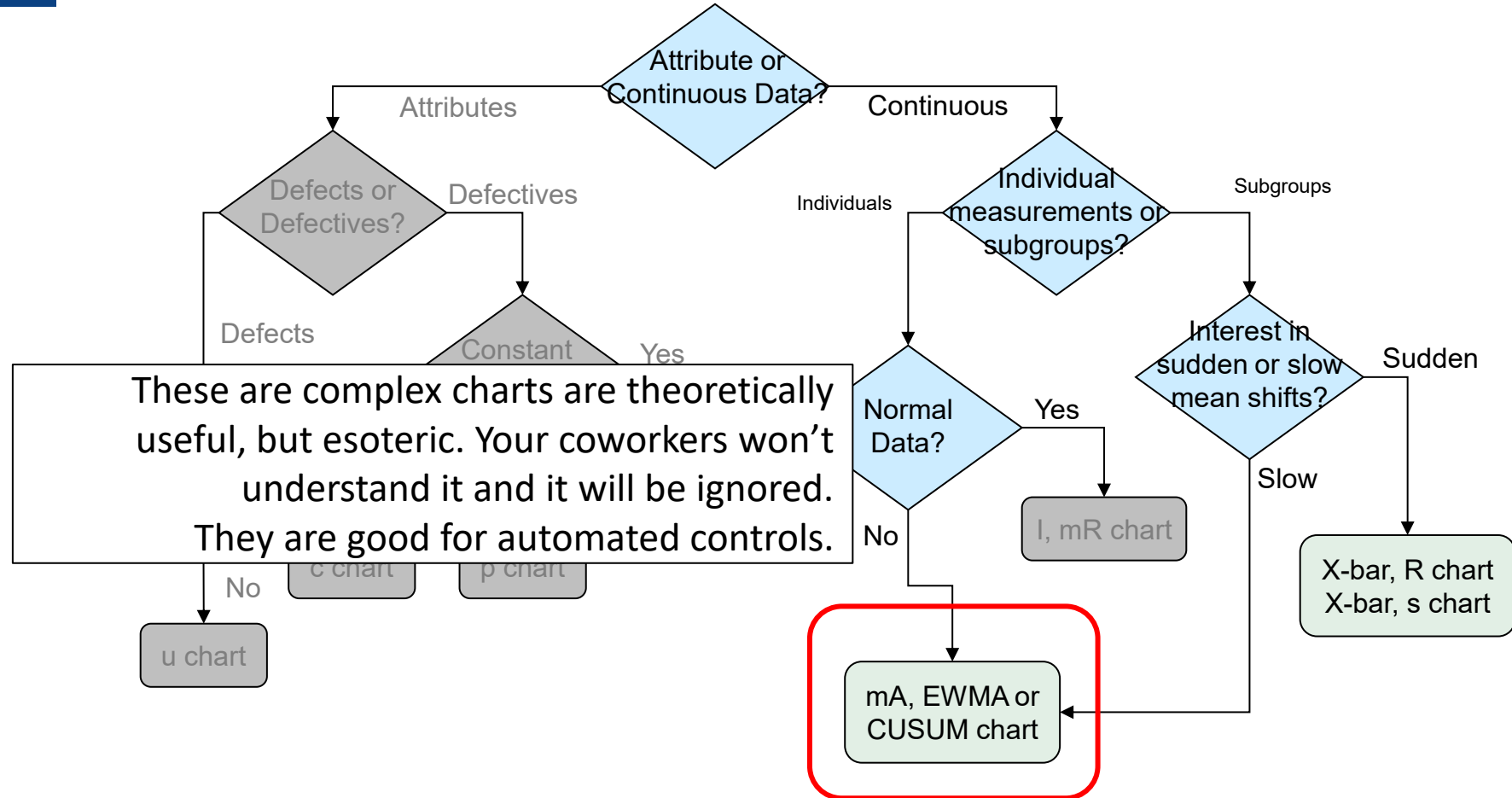
100% Inspection Statistical Process Control



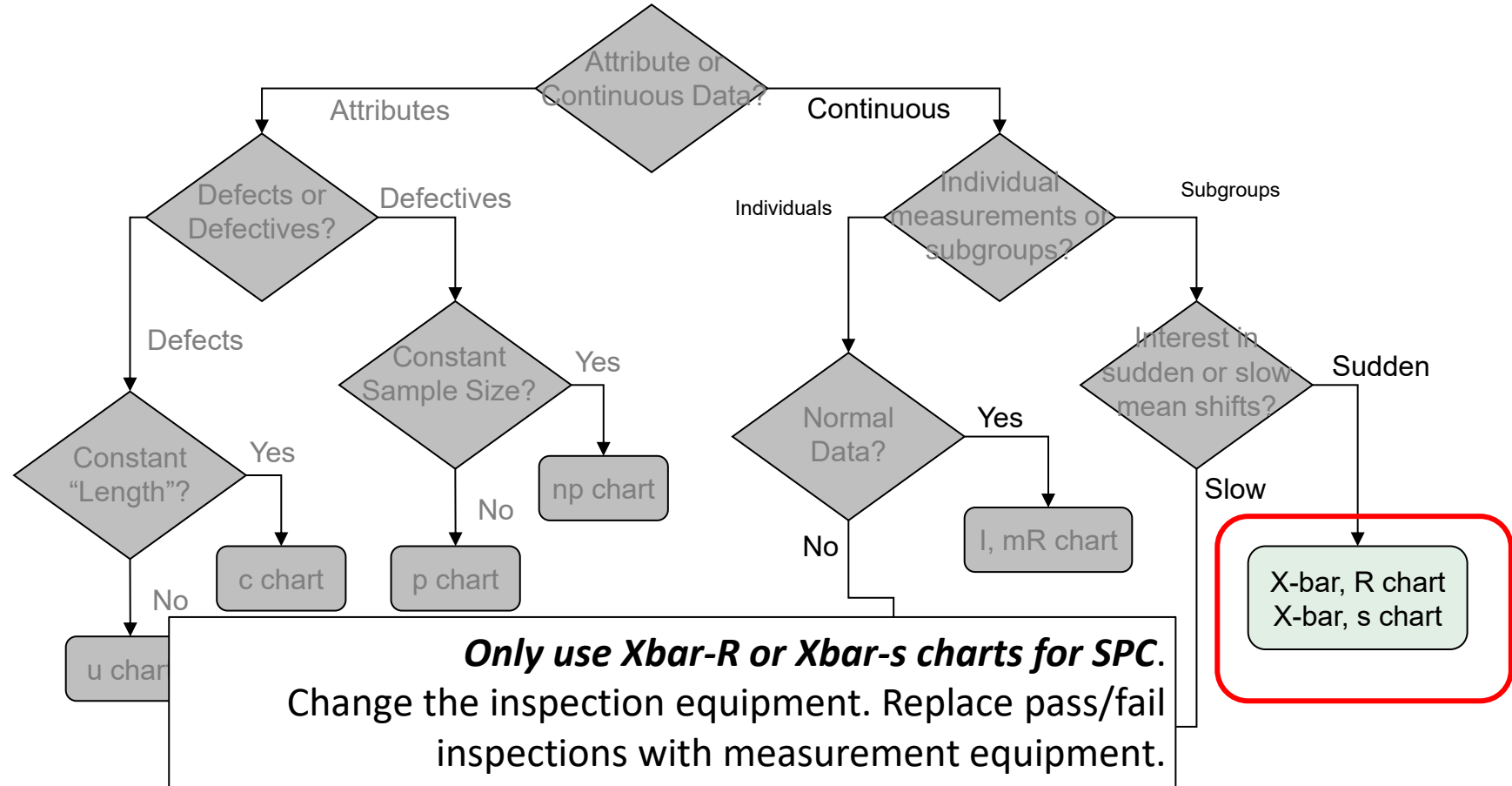
I-mR charts plot each point. *100% inspection*. Improve your process so you don't need 100% inspection.

I-MR charts are really only legitimate for low frequency data, such as calibration trends, and for 100% inspected safety critical production.

Other Statistical Process Control



Useful Statistical Process Control





SPC Chart Types

There are process monitoring SPC charts for pass / fail attributes: np, n, c, u charts.
Try to not use defect counting SPC – you accepted that you will make defects.
Replace pass / fail inspection with measured inspection.

To detect if the process has changed before going out of spec, use Xbar-R or Xbar-s charts.
SPC will have you make process corrections before making bad parts.
People won't do this unless you exert management discipline.

To detect if a process is in specification, use C_{pk} charts to compare data against specifications. Don't mix Cpk analysis with SPC.

NEXT: Incoming Inspection



Outline

Statistical Process Control

Xbar-R charts

Out of Control Action Plans

Incoming Inspection

Statistical Sampling Plans

Cpk Inspection Plans

PPAP



Supplier Quality Control

In today's modern global economy, you will become in charge of remote suppliers who must make parts or systems for you.

When do you accept their shipments, and when do you have evidence to legally reject them?





Incoming Inspection

A lot of material has arrived at your assembly plant's loading dock.

You need to inspect it, and sentence it.

How much inspection do you need to do?

Inspect each and every item?

5 of them?

5% of them?



100% Inspection

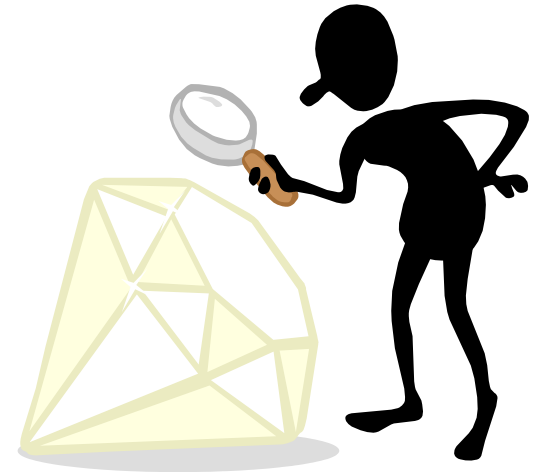
One approach to ensuring your supplier is to measure each and every incoming part, and pass/fail judge each one.

100% inspection.

100% inspection is typically required for safety critical parts.

Those whose failure can cause death or severe injury.

Yet, this is expensive. Is it always necessary?

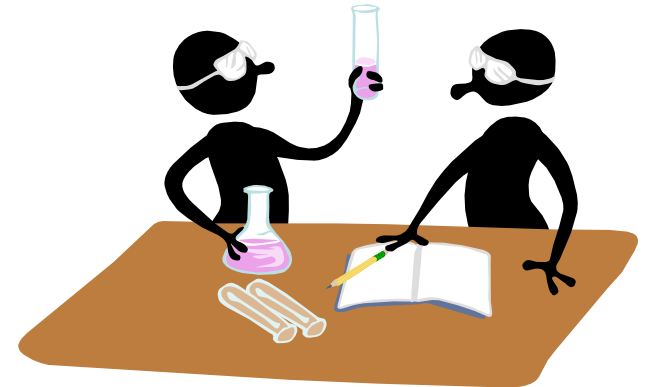


Sample Based Inspection

Instead of 100% inspection, we can inspect only a sample.

Acceptance sampling is useful when

- Testing is destructive.
- There is serious product risks if quality level changes.
- There is excellent quality, and reduction of 100% inspection is desired for cost savings.
- Inspection error increases with increased inspection.
- Large measurement error and 100% inspection can cause higher % of defective units.



Sample Inspection

For non-safety critical parts, you want to ensure the incoming materials are “the same” as you specify.

You expect an agreed-to quality level X.

Is this incoming lot of material “the same”?

Is this incoming lot of material “better than X”?



Sample Inspection

For non-safety critical parts, you want to ensure the incoming materials are “the same” as you specify.

You expect an agreed-to quality level X .

Is this incoming lot of material “the same”?

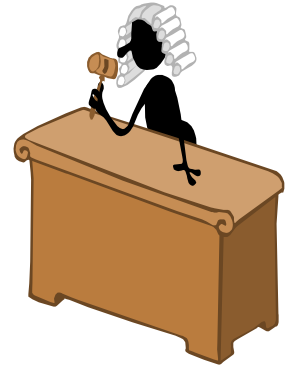
Is this incoming lot of material “better than X ”?

In this case, we are *sentencing* the lot.

We do not need to assess the quality level as $P\%$.

We only need to state whether or not it is better than $X\%$.

Sentencing needs less samples (testing) than assessing.

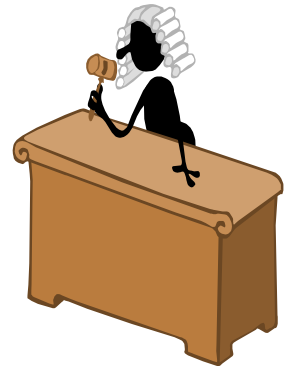


Acceptance Sampling

An inspection procedure for accepting or rejecting incoming material, using a sample of parts rather than 100% of the parts.

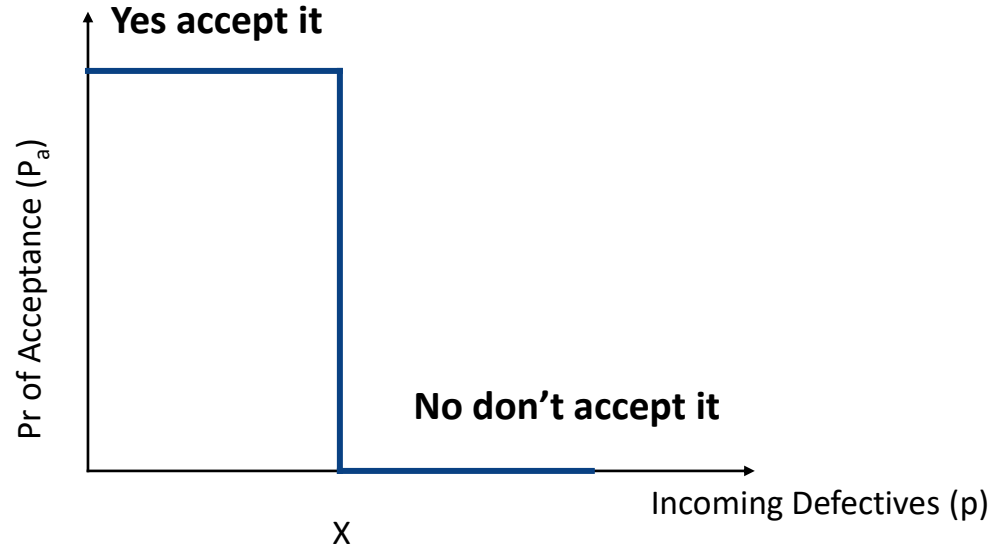
Used primarily for *Lot Sentencing*.

- Given a lot at incoming receiving, should it be accepted? Yes or No?



Ideal Lot Acceptance Test

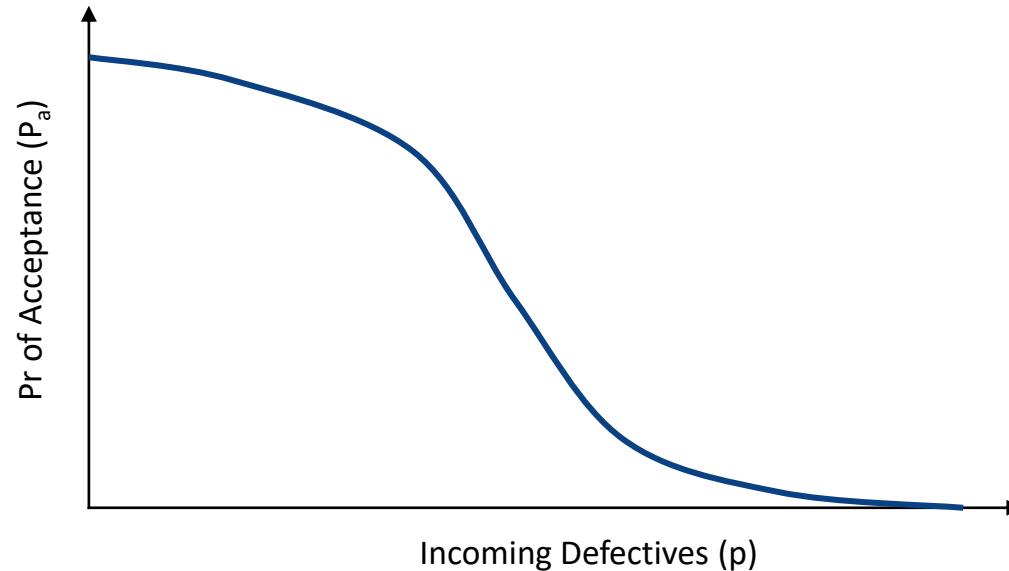
If we want the quality level $> X$, then the ideal acceptance test would disposition a lot perfectly against X .



What is the likelihood of accepting a lot, given it's quality level?

The **Operating Characteristic (OC) Curve** plots the probability of accepting a lot vs. the incoming lot % defective.

The OC curve shows the discriminatory power of a sampling plan.



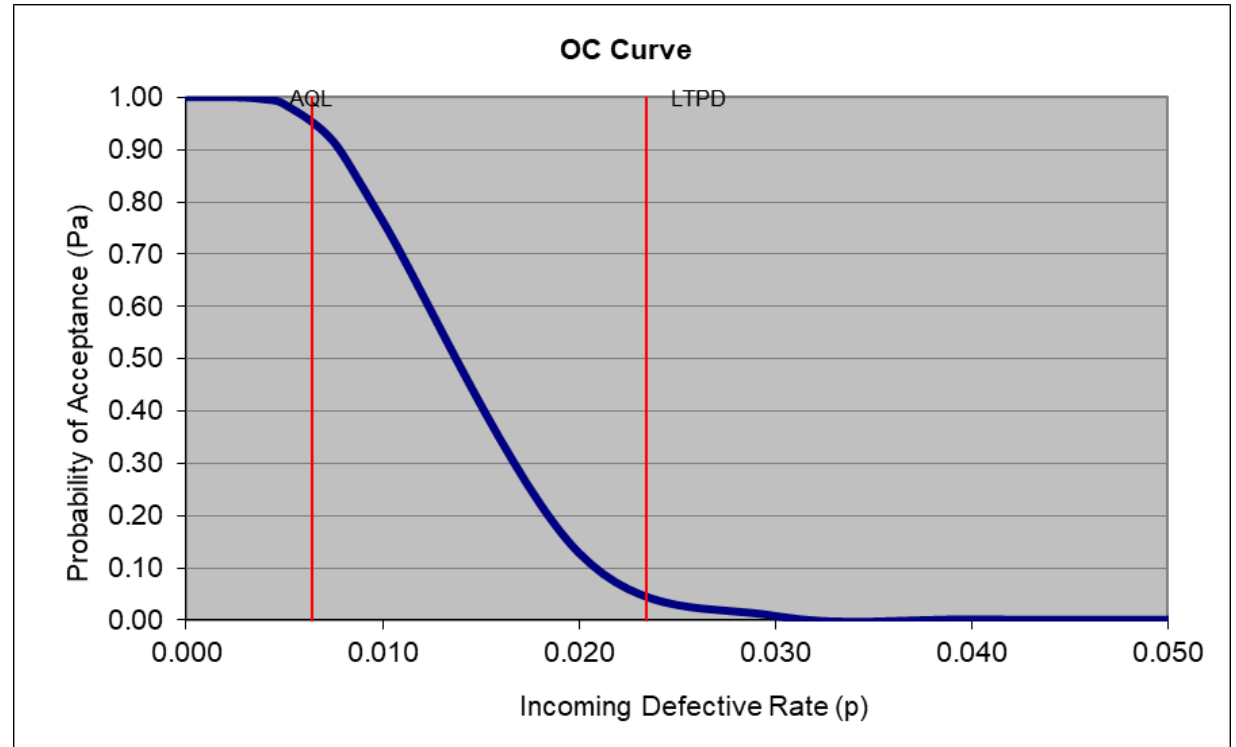
Example: OC Curve

Given an inspection sampling plan with the following OC curve, what is the probability of accepting a lot with:

0.5% defective rate?

1.0% defective rate?

2.0% defective rate?



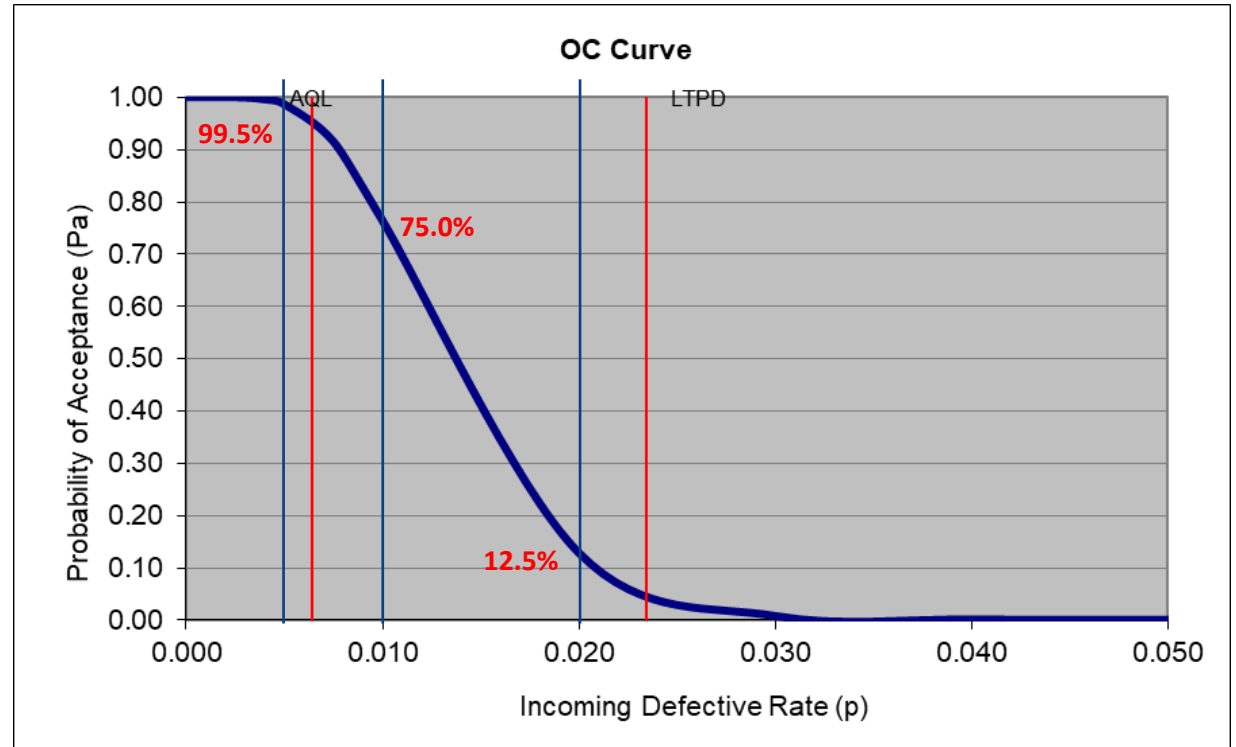
Example: OC Curve

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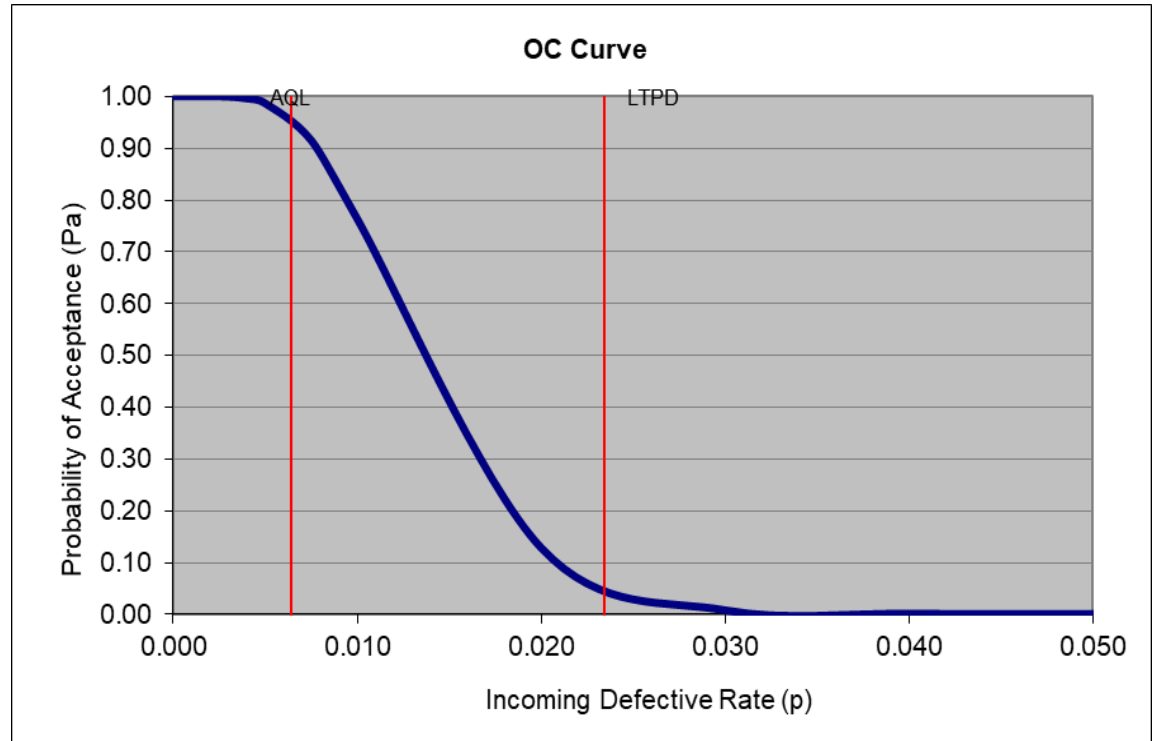
Exercise: OC Curve

Suppose a customer has the following OC curve.

Base on sampling,
your production is
running at 1% defective.

If you ship it, what do
you expect is the
probability of your
customer accepting it?

Is that a good thing?



Sample Plan Variables

Sample Size (n)

- The number of parts pulled from a lot.
- Not determined by the lot size, but rather the level of incoming quality and desired outgoing quality.

Acceptance Number (c)

- The maximum number of defectives allowed in a sample of n units.
- Again determined by the incoming and desired outgoing quality levels.

Practically, these two variables define the lot sentence.





OC Curve

Suppose the lot size N is very large (approaches ∞).

The distribution of the number of defectives (d) in a sample of n units follows a binomial distribution with parameters n and p , where p is the incoming percent defectives.

From the binomial distribution, the probability P of observing exactly d defectives is

$$P(d) = \frac{n!}{d! (n - d)!} p^d (1 - p)^{n-d}$$



OC Curve

The probability of accepting a lot P_a is then simply the probability that $d \leq c$ (where c is the acceptance number).

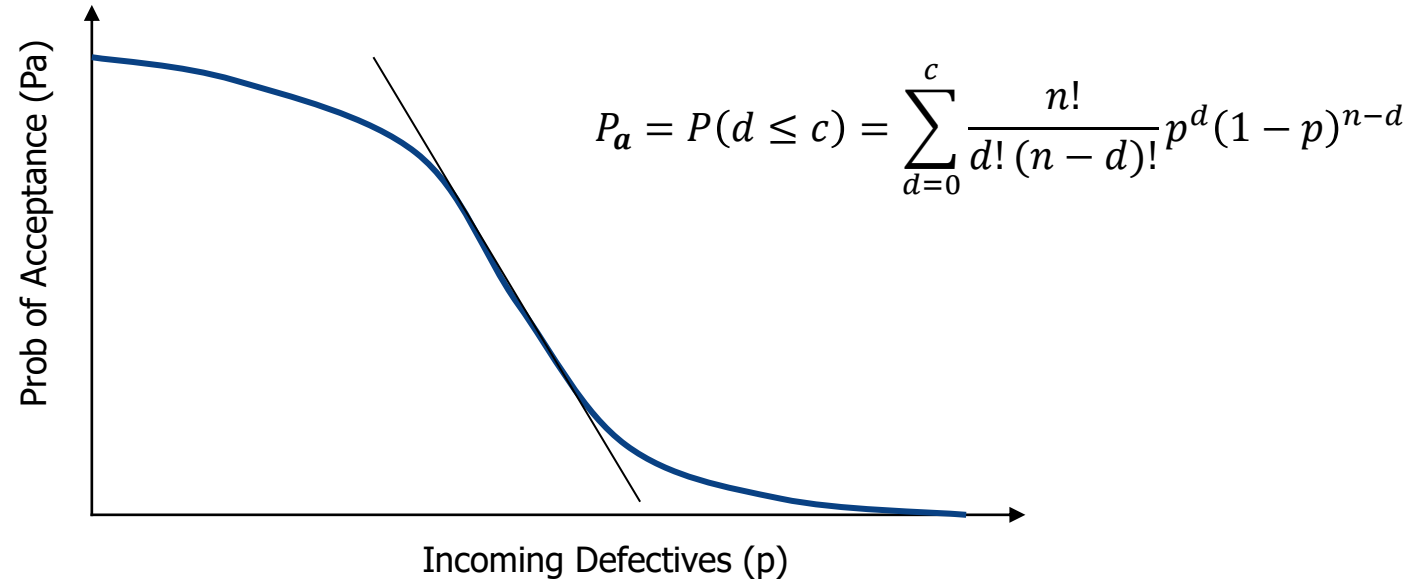
$$P_a = P(d \leq c) = \sum_{d=0}^c \frac{n!}{d! (n-d)!} p^d (1-p)^{n-d}$$

The **OC curve** is constructed by evaluating the above equation above for various values of p and graphing.

Effect of n and c on OC Curves

Given an incoming defective level (p), only the sample size (n) and acceptance number (c) affect the OC curve

- The steeper the OC curve, the better discriminatory power of the sampling plan.

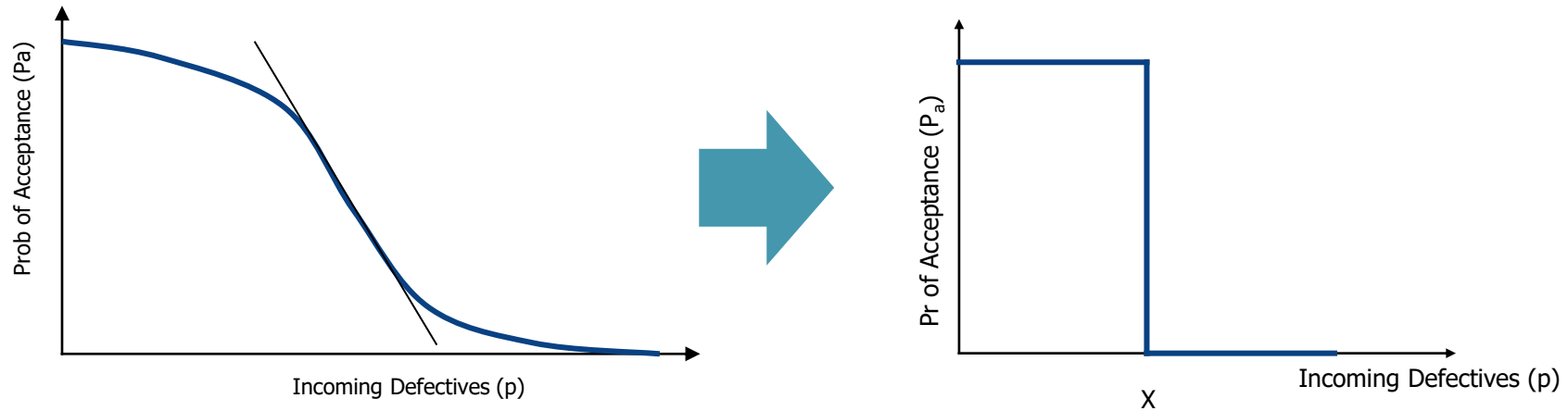


Effect of n and c on OC Curves

The perfect sampling plan can only be achieved with 100% inspection AND if inspection is free of error

The ideal OC curve can be approached by increasing the sample size

- Hence, the discriminatory power with which a sampling plan differentiates between good and bad lots increases with the sample size



The Binomial Distribution

The cumulative binomial distribution function

$$P_a = P(d \leq c) = \sum_{d=0}^c \frac{n!}{d! (n-d)!} p^d (1-p)^{n-d}$$

In Excel, this is simply

`=BINOM.DIST.RANGE( $n$ ,  $p$ , 0,  $c$ )`

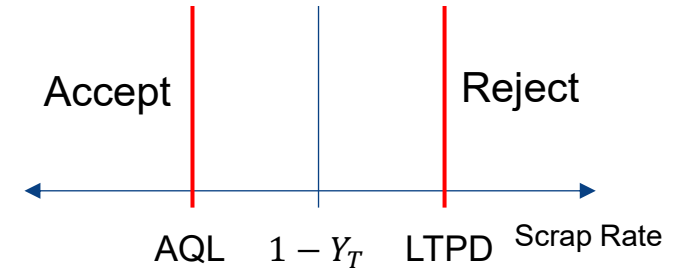
Defining Sample Plans

A practical sampling plan is designed by defining the **Acceptable Quality Level** (AQL) and the **Lot Tolerance Percent Defective** (LTPD)

The **AQL** is the scrap rate below which the lot will be accepted. A scrap rate above AQL *might* be accepted.

The **LTPD** is the scrap rate above which the lot will be rejected. A scrap rate below LTPD *might* be rejected.

Think of these as specification limits for a target scrap rate / yield.





Defining Sample Plans

Against the AQL and LTPD, we also need to define acceptable risk.

α is the risk level you are willing to take in rejecting a lot which should be accepted.

Producer's risk.

β is the risk level you are willing to take in accepting a lot which should be rejected.

Consumer's risk.

Acceptable Quality Level (AQL)

The **AQL** is the highest scrap rate that is still considered acceptable, with confidence $1-\alpha$. Equivalently, the lowest quality level that is still considered acceptable.

- Statistically, if the lot passes the sample test, one can state with $1-\alpha\%$ confidence that the quality level of the lot is $\geq \text{AQL}$.
- When $p = \text{AQL}$, then the probability of lot acceptance $P_a = 1 - \alpha$ (the producer's risk α)
- There is a high probability of acceptance (P_a) at the AQL

AQL states what the sampling plan will accept. The AQL is defined by the required (incoming) quality level of the vendor's process, NOT by the sampling plan.

Lot Tolerance Percent Defective (LTPD)

The **LTPD** is the lowest scrap rate that is still considered rejected, with power $1-\beta$. Equivalently, the highest quality level that is still considered rejected.

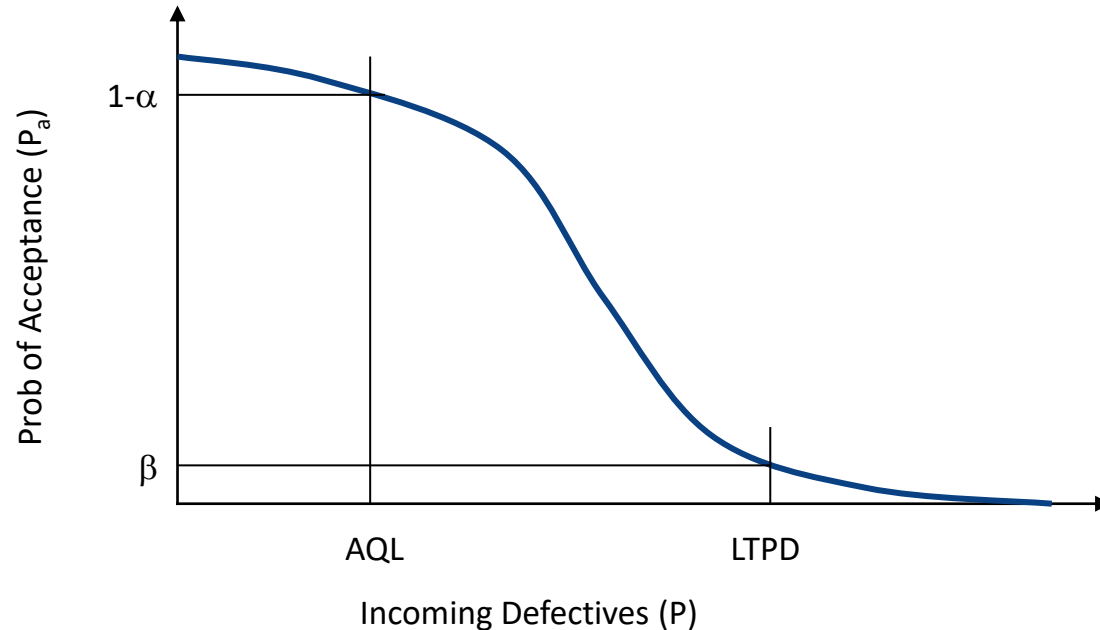
- Statistically, if the lot fails the sample test, one can state with $1-\beta\%$ power that the quality level of the lot is $\leq$ LTPD.
- When $p = \text{LTPD}$, then the probability of lot acceptance $P_a = \beta$ (the consumer's risk β)
- There is very low probability of acceptance at the LTPD

LTPD states what the sampling plan will reject.

LTPD is not a characteristic of the sampling plan, but a level of lot quality specified by the consumer. This protects the consumer from poor quality.

A Practical OC Curve

Combining these two points (4 variables):

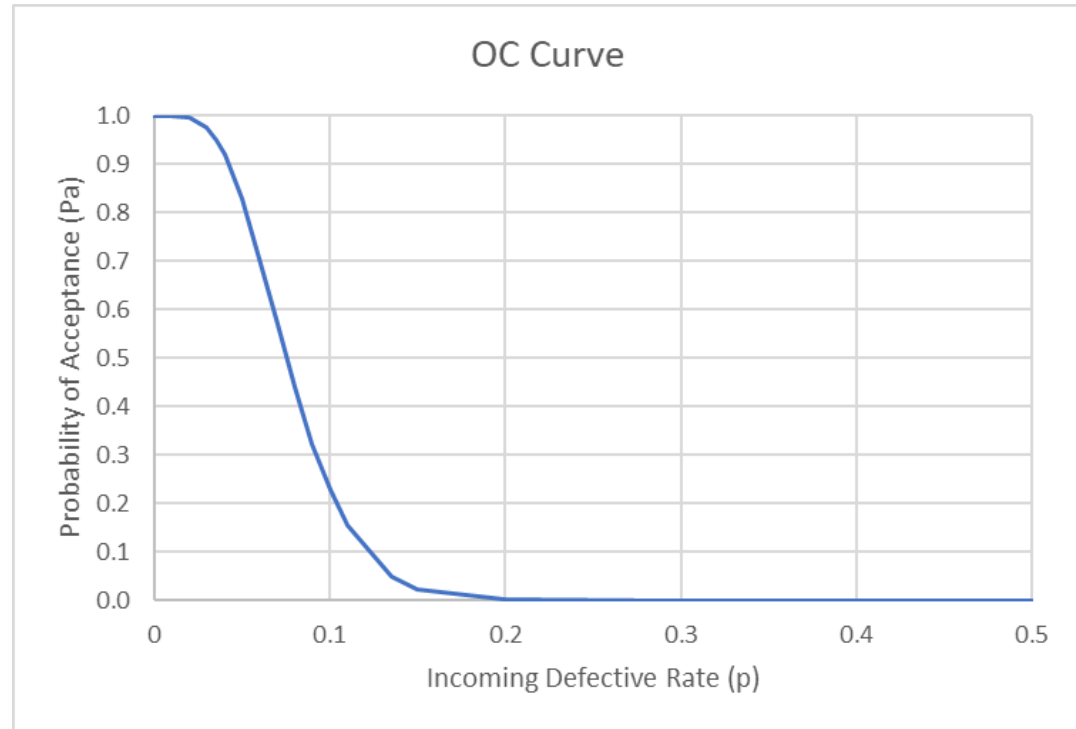


To define a sampling plan, you need to define the 4 variables.

Example: OC Curve

Given a sampling plan of 75 samples from a lot with 5 allowed defective. This has the following OC curve.

With $\alpha = \beta = 0.05$, what is the AQL and LTPD?



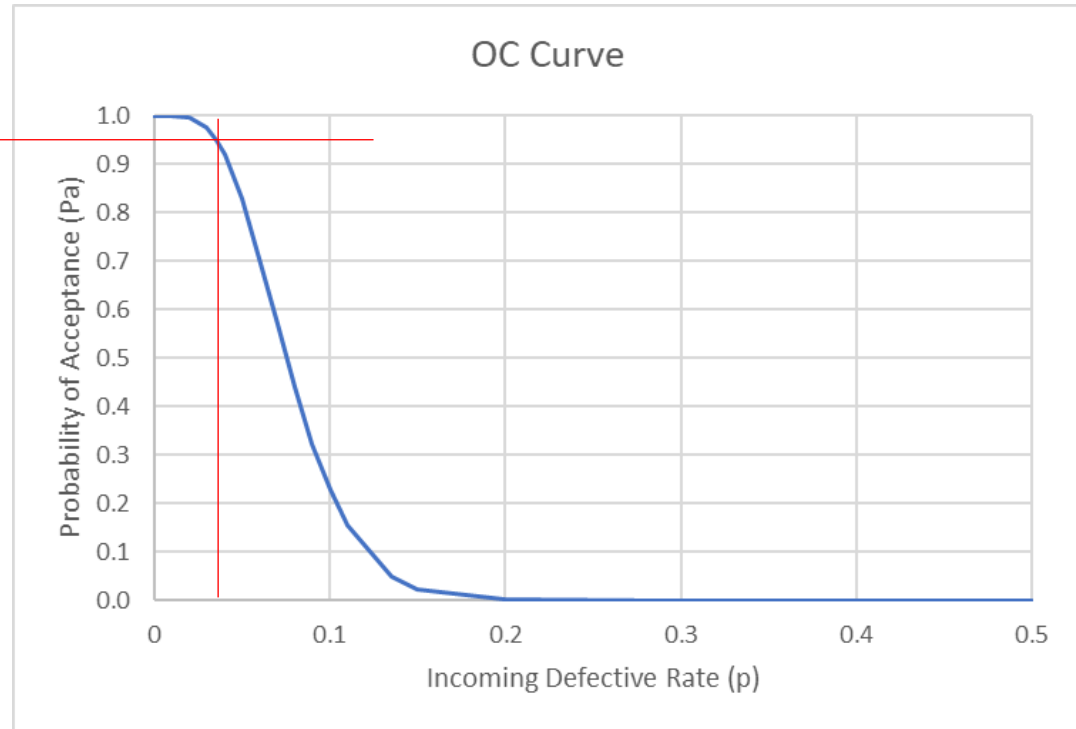
Example: OC Curve

Given a sampling plan of 75 samples from a lot with 5 allowed defective. This has the following OC curve.

With $\alpha = \beta = 0.05$, what is the AQL and LTPD?

$1 - \alpha = 95\%$

AQL = minimum
defective rate
rejected at
5% risk = 3.5%
(producers risk)

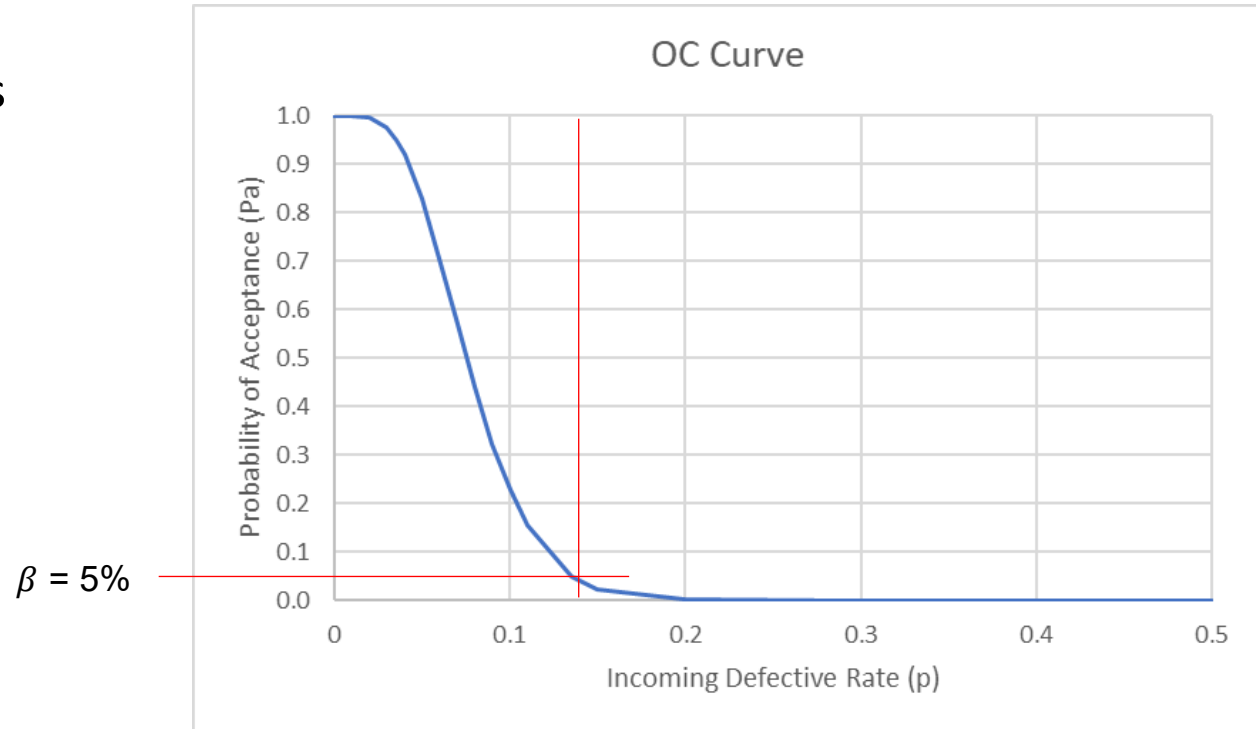


Example: OC Curve

Given a sampling plan of 75 samples from a lot with 5 allowed defective. This has the following OC curve.

With $\alpha = \beta = 0.05$, what is the AQL and LTPD?

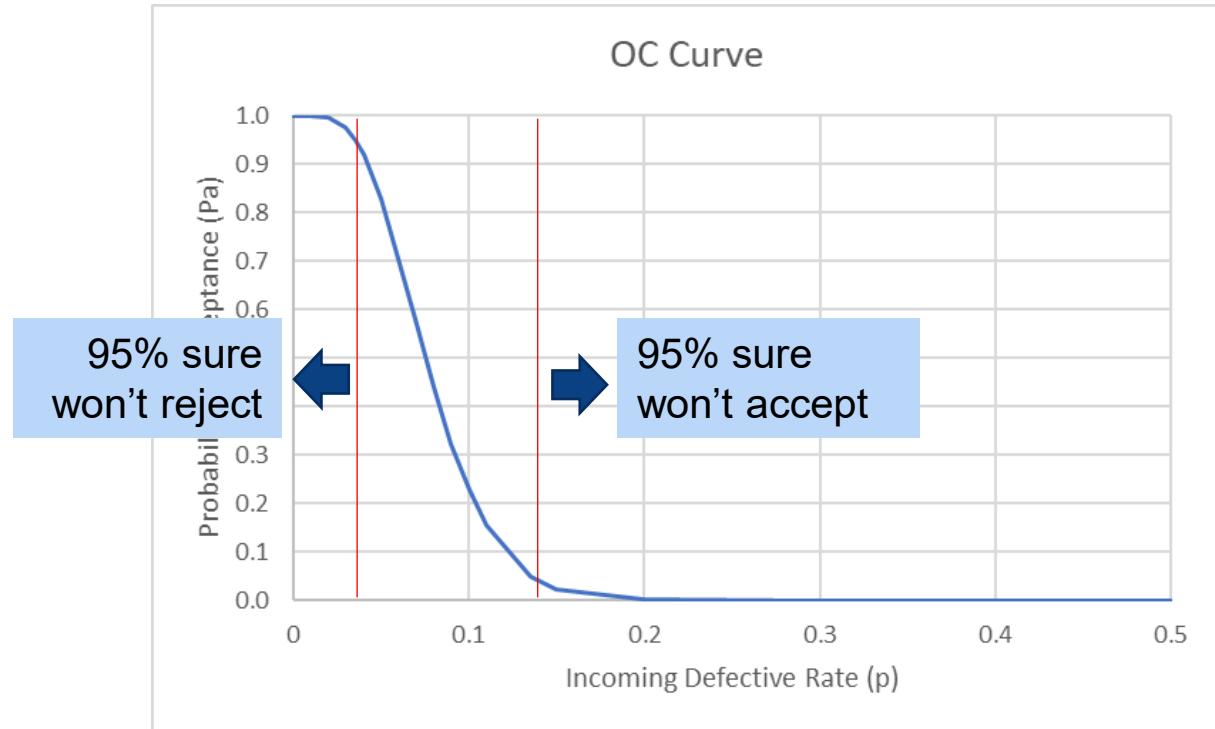
LTPD = maximum
defective rate
accepted
at 5% risk = 13.5%
(consumers risk)



Example: OC Curve

Given a sampling plan of 75 samples from a lot with 5 allowed defective. This has the following OC curve.

To confidence and power of 95%, the range of the lot defective rate is from 2.5 – 13.5%.





Outline

Statistical Process Control

Xbar-R charts

Out of Control Action Plans

Incoming Inspection

Statistical Sampling Plans

Cpk Inspection Plans

PPAP

Single Sampling Plan Design

A **sampling plan** (n,c) can be computed as the OC curve passes through these two designated points.

As equations, the sampling plan (n,c) can be computed from two simultaneous equations:

$$1 - \alpha = \sum_{d=0}^c \frac{n!}{d! (n-d)!} AQL^d (1 - AQL)^{n-d} \quad \text{Equation 1}$$

$$\beta = \sum_{d=0}^c \frac{n!}{d! (n-d)!} LTPD^d (1 - LTPD)^{n-d} \quad \text{Equation 2}$$



Single Sampling Plan Design

To obtain the 2 unknowns (n, c), we need to solve the equations simultaneously.

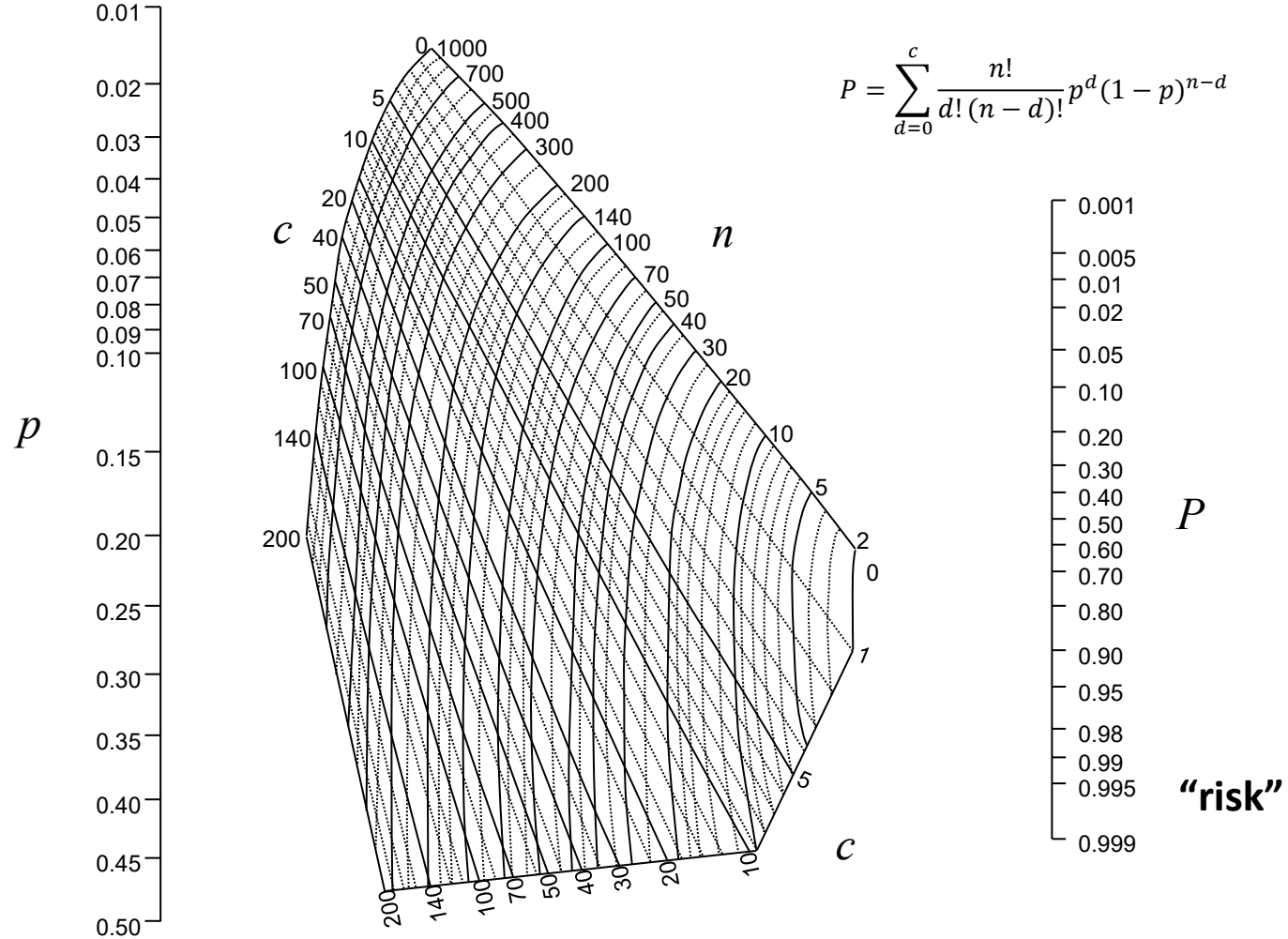
Note that the equations are non-linear and therefore iterative numerical methods must be used to compute a solution.

A graphical method can be used, the *binomial nomograph* chart

- Given (AQL, LTPD, α , β), it can be used to compute required n and c .

Binomial Nomograph

$$P = \sum_{d=0}^c \frac{n!}{d!(n-d)!} p^d (1-p)^{n-d}$$



“defective rate”

“risk”



Single Sampling Plan Design: Nomograph

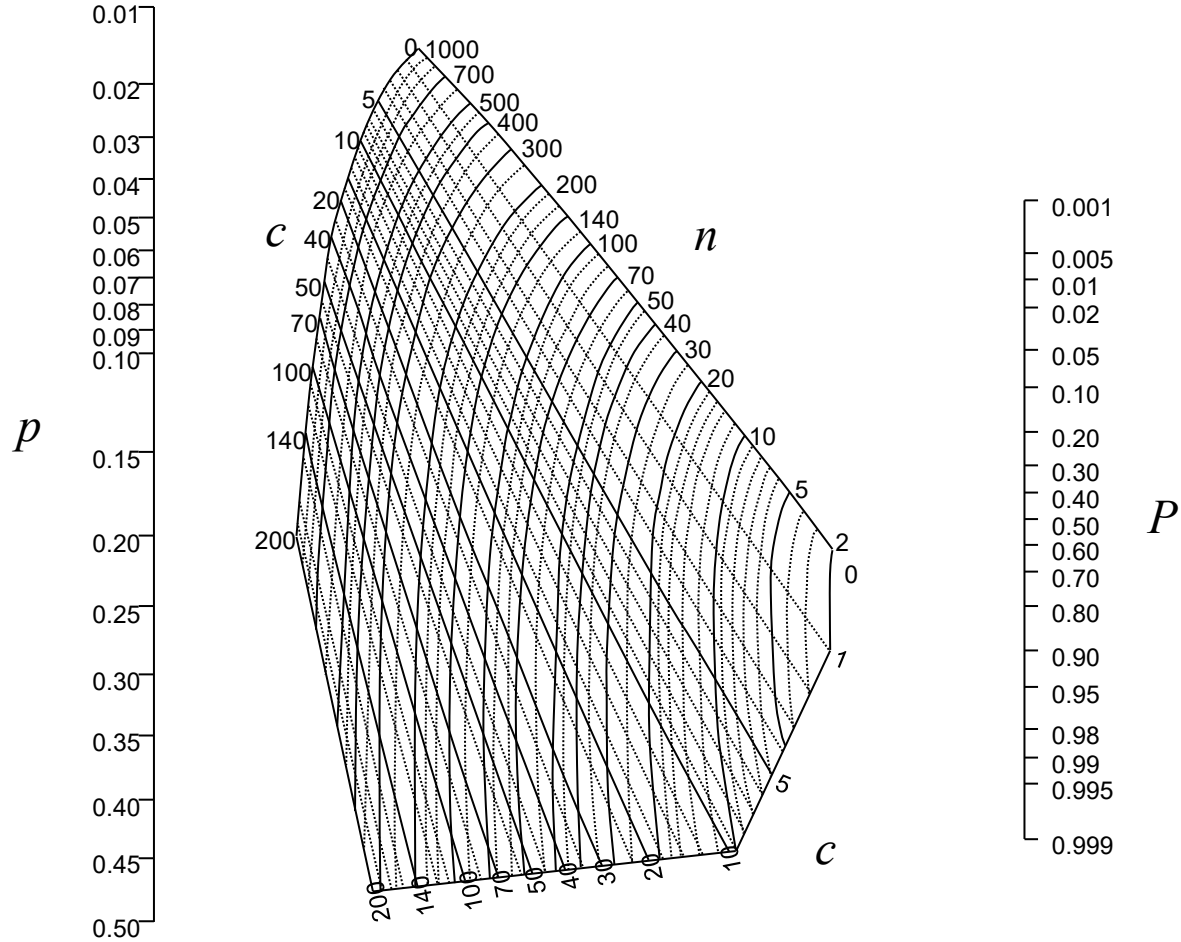
Consider

- $\alpha = 5\%$: a willingness for 5% risk in rejecting a good lot
- $\beta = 10\%$: a willingness for 10% risk in accepting a bad lot
- AQL = 1%: a good lot means less than 1% defective
- LTPD = 8%: a bad lot means more than 8% defective

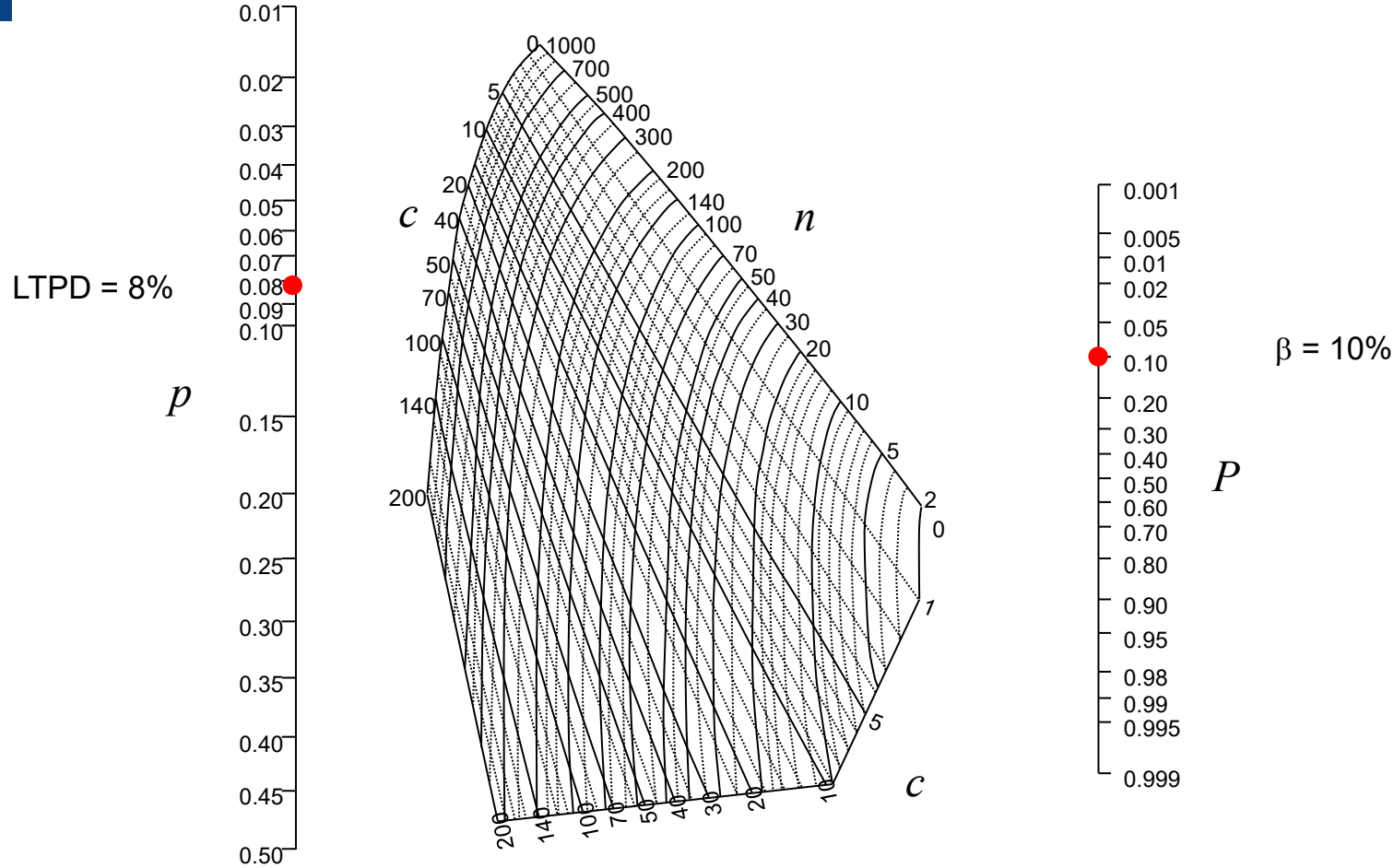
What is a good sampling plan?

- $n = ?$
- $c = ?$

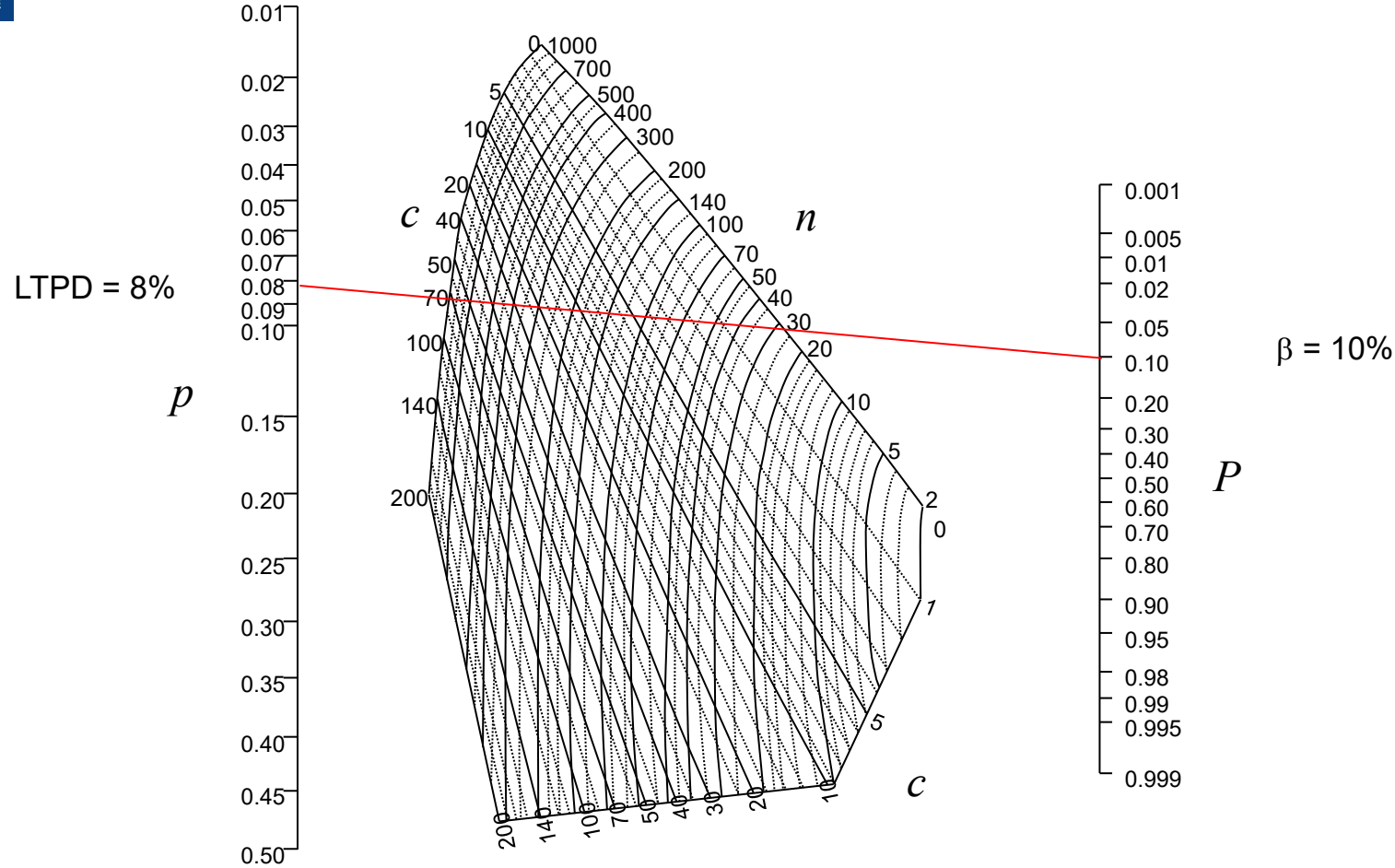
Binomial Nomograph



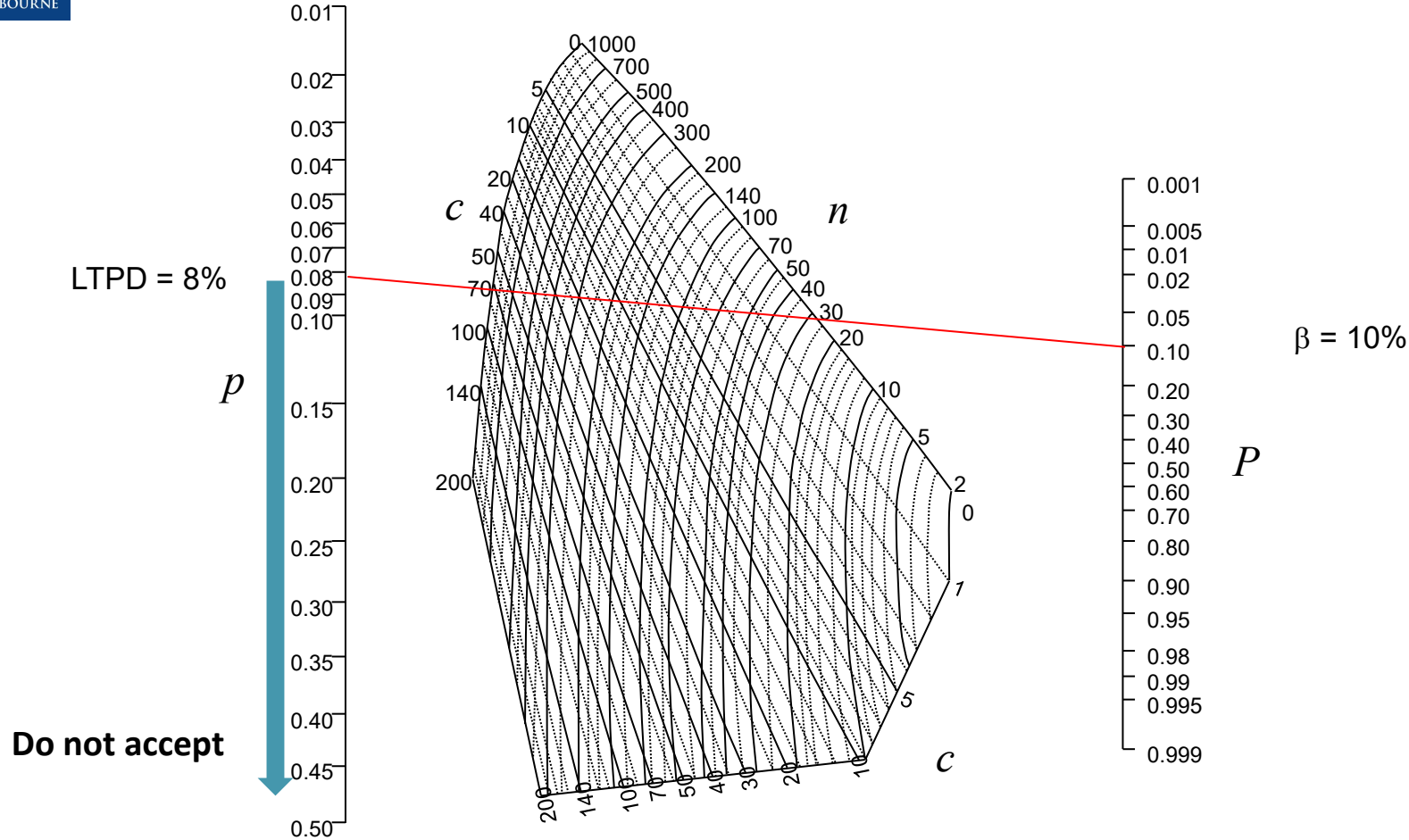
Binomial Nomograph



Binomial Nomograph



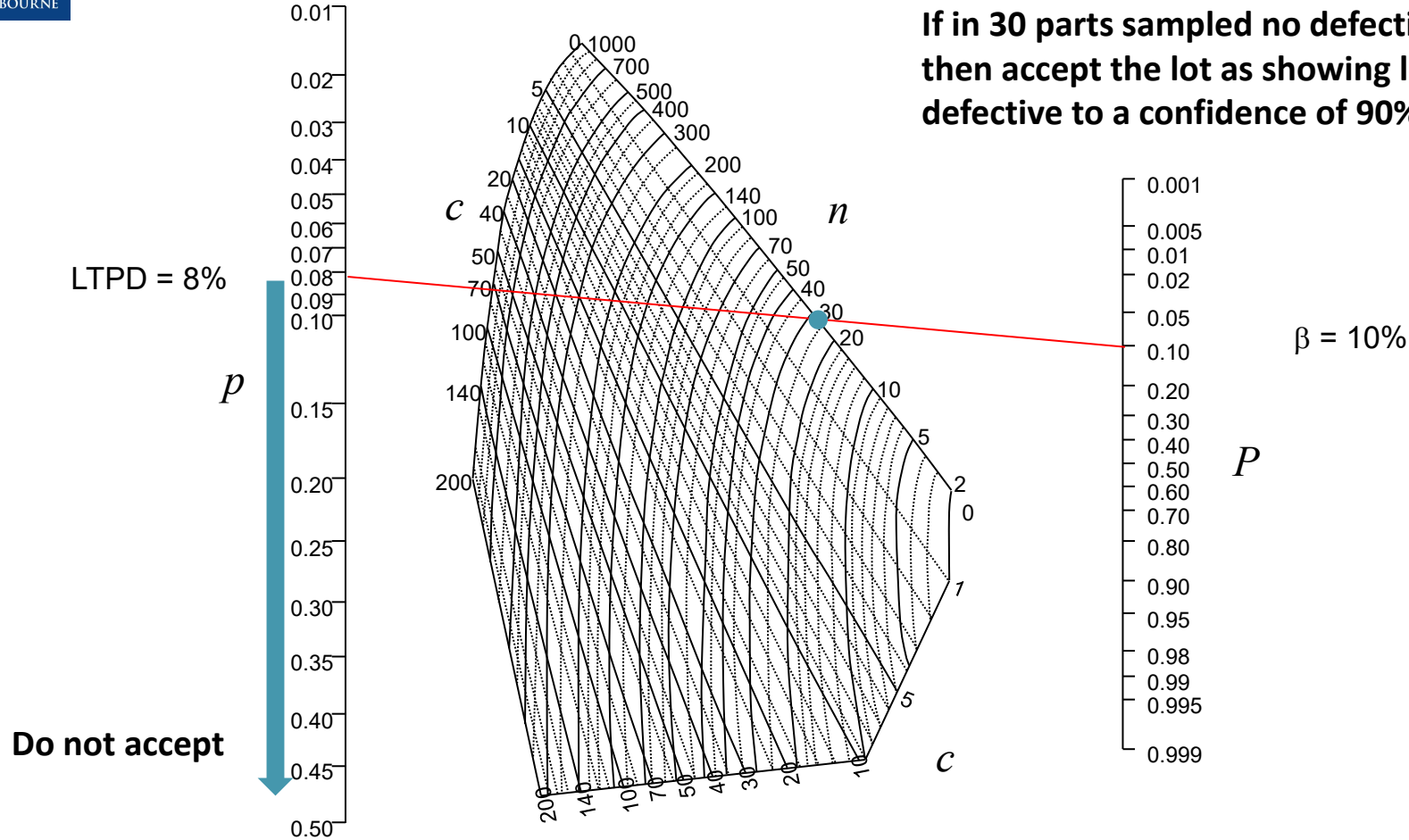
Binomial Nomograph



Binomial Nomograph

For $c=0$, $n=30$.

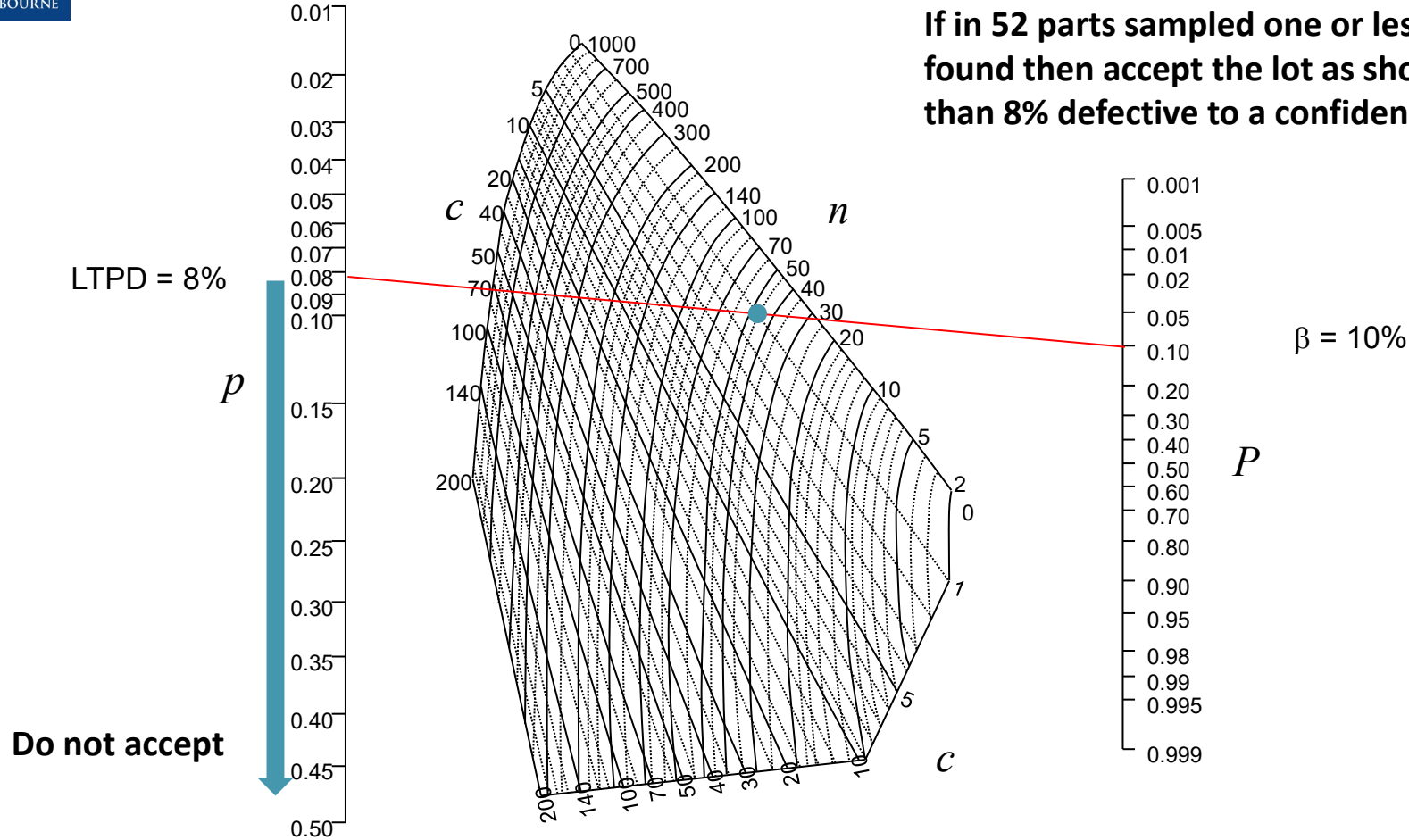
If in 30 parts sampled no defective found
then accept the lot as showing less than 8%
defective to a confidence of 90%.



Binomial Nomograph

For $c=1$, $n=52$.

If in 52 parts sampled one or less defective found then accept the lot as showing less than 8% defective to a confidence of 90%.



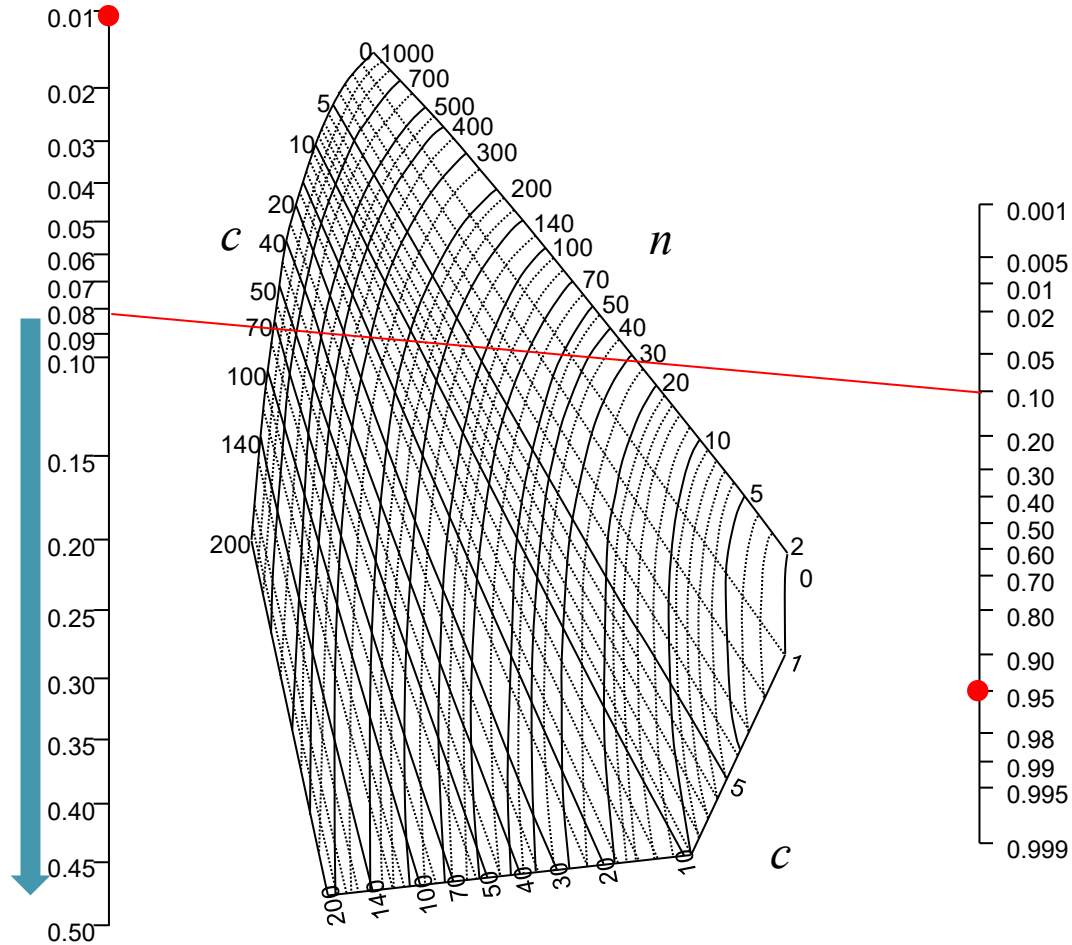
Binomial Nomograph

AQL = 1%

LTPD = 8%

p

Do not accept



$\beta = 10\%$

P

$1 - \alpha = 95\%$

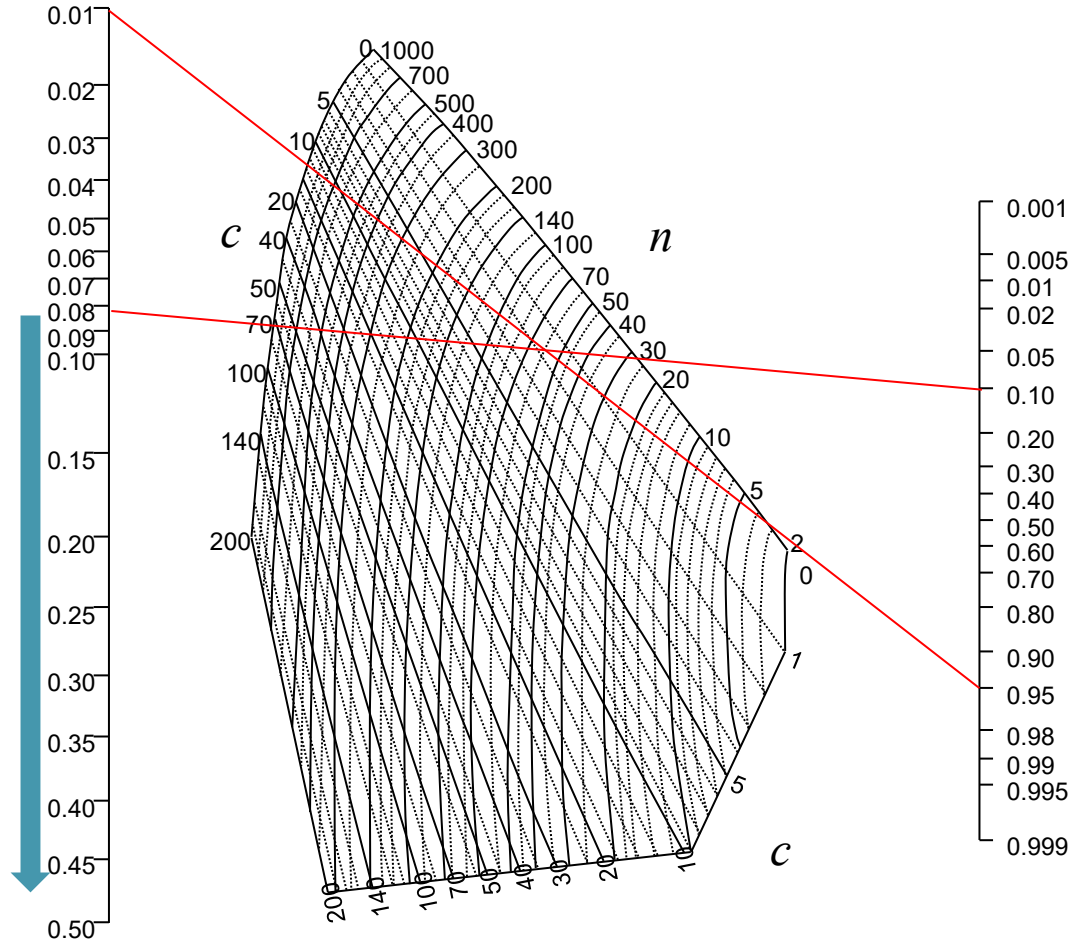
Binomial Nomograph

AQL = 1%

LTPD = 8%

p

Do not accept

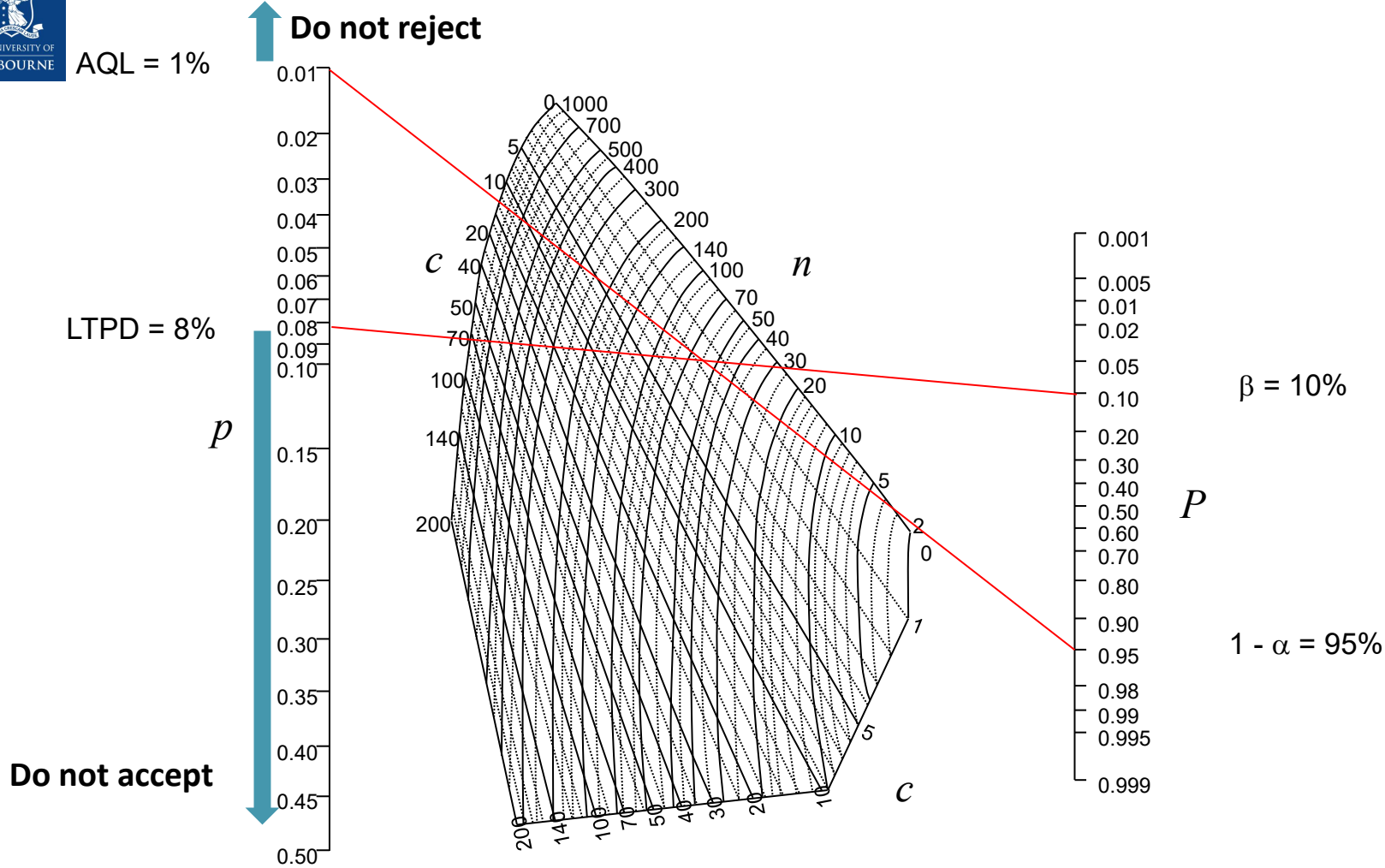


$\beta = 10\%$

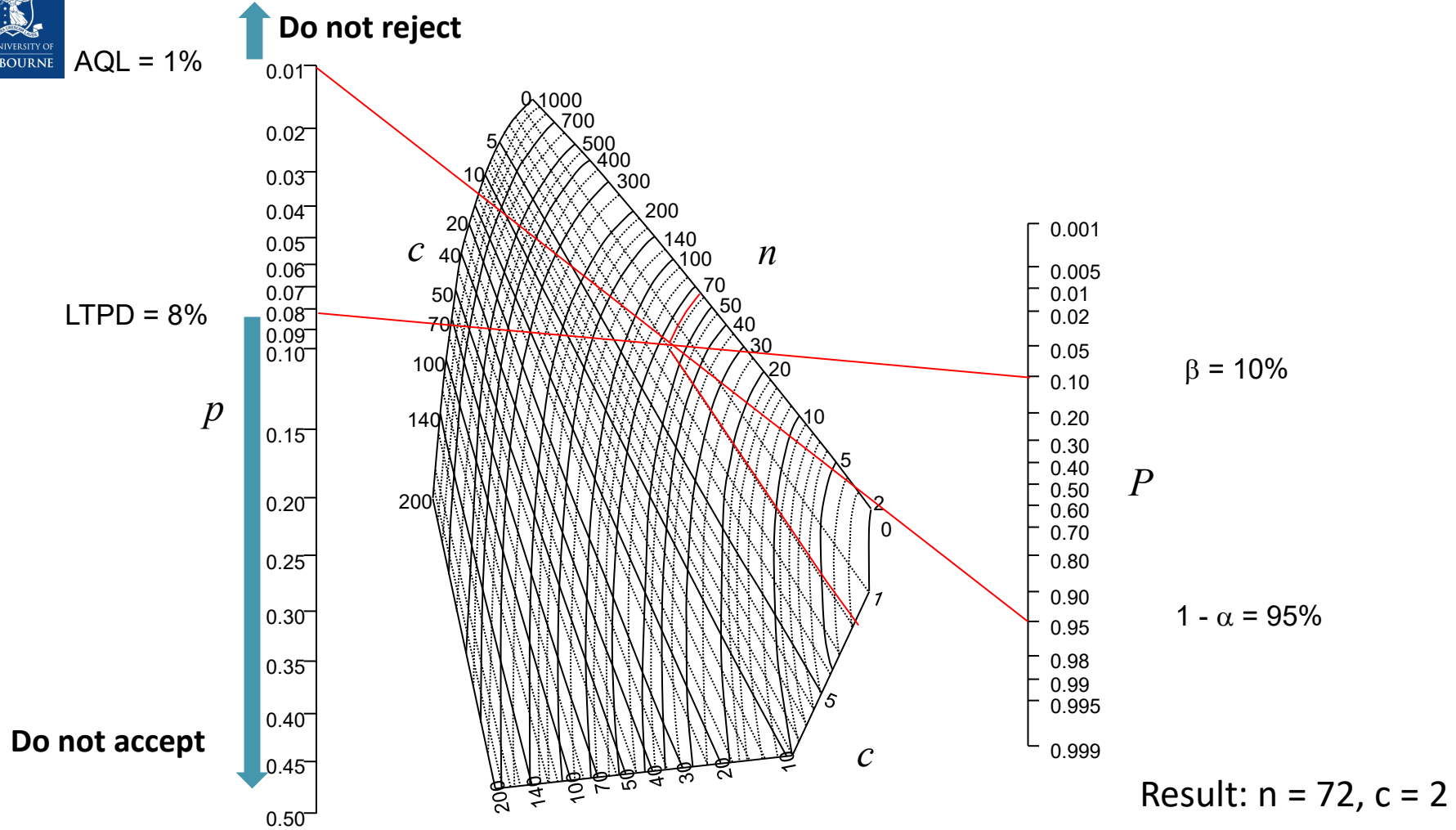
P

$1 - \alpha = 95\%$

Binomial Nomograph



Binomial Nomograph





Incoming Lot Reject Rate

Given a sampling plan (n, c) , and the expected incoming quality level p , you can compute the expected reject rate.

$$P_a = \sum_{d=0}^c \frac{n!}{d! (n-d)!} p^d (1-p)^{n-d}$$

Expected Rejection Rate

With the previous lot sample inspection plan, how many lots will generally be rejected?

With a typical incoming quality level of 8% defective

$$P_a = \sum_{d=0}^c \frac{n!}{d! (n-d)!} p^d (1-p)^{n-d}$$

$$P_a = \sum_{d=0}^2 \frac{72!}{d! (72-d)!} 0.08^d (1-0.08)^{72-d}$$

$$P_a = 6.6\%$$

Notice:

- The sample plan (n,c) is independent of the incoming quality level p.
- The probability of accepting a typical lot (P_a) depends on the sample plan inputs (AQL and LTPD) versus the incoming quality level p.

Binomial Nomograph

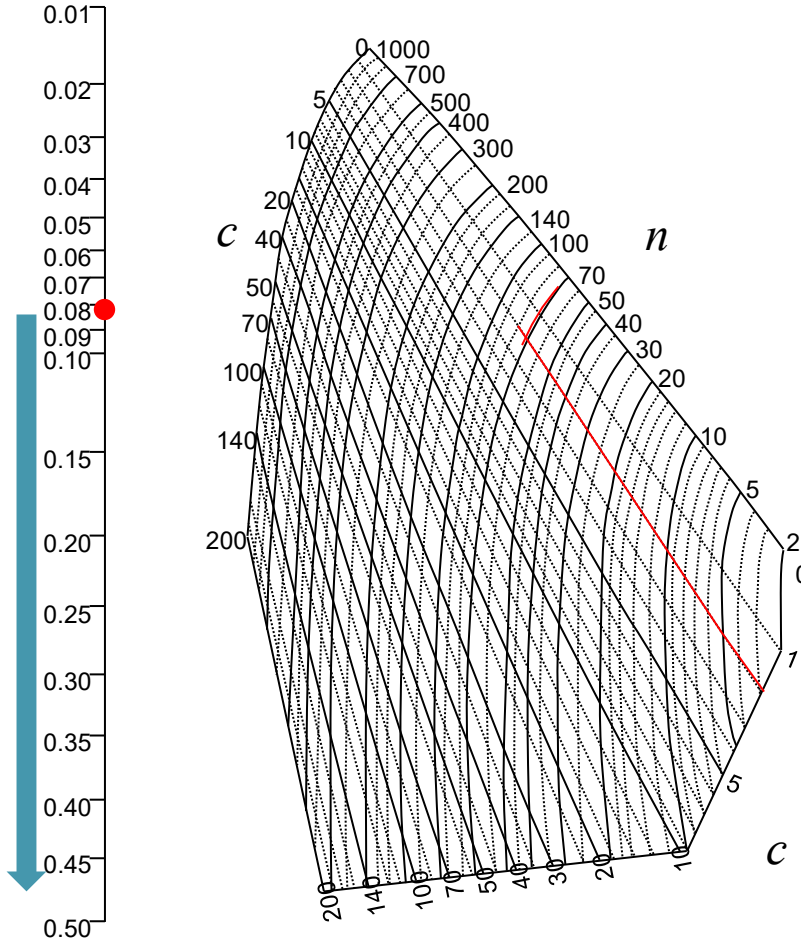
↑ Do not reject

AQL = 1%

LTPD = 8%

p

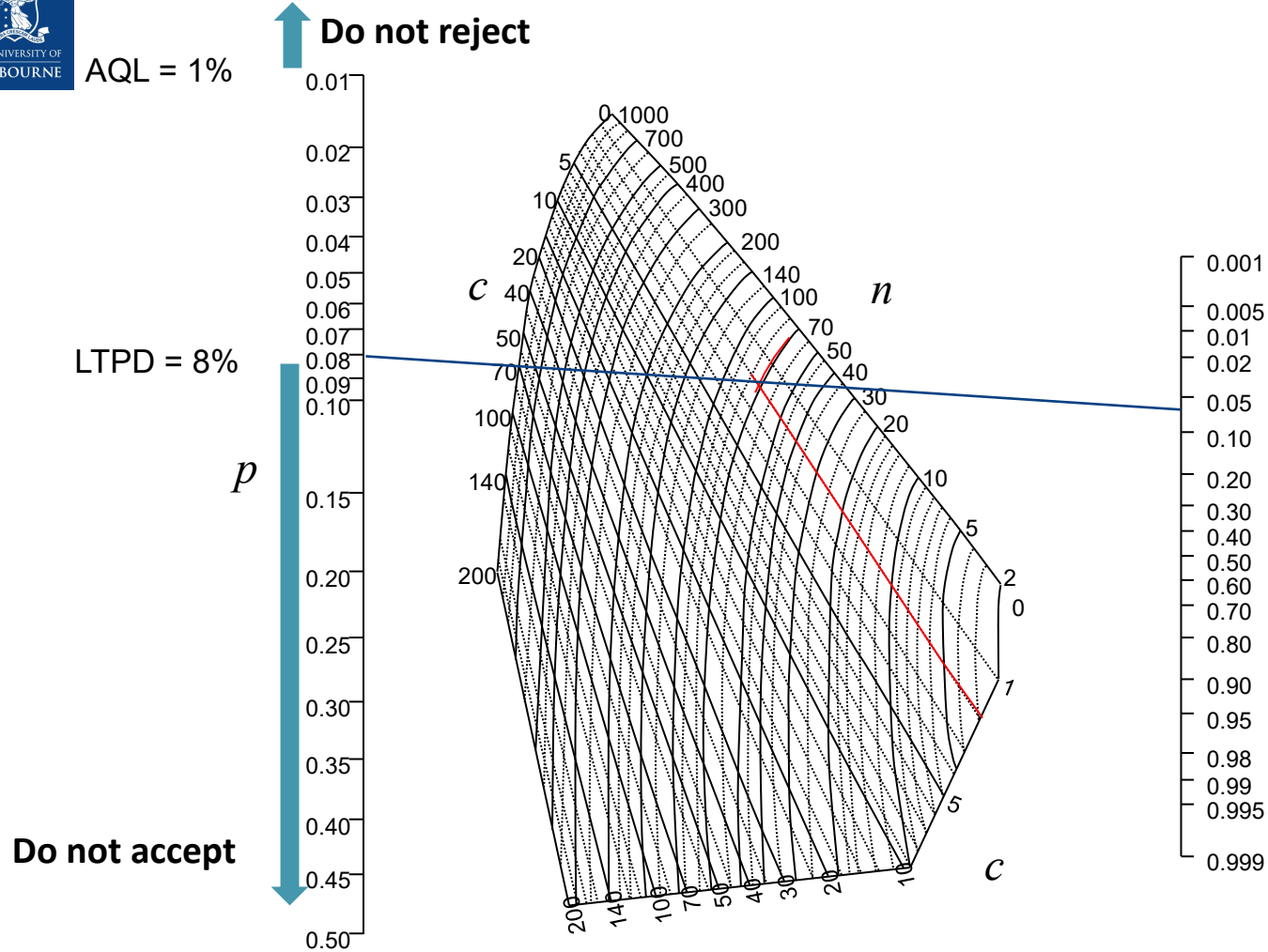
Do not accept



0.001
0.005
0.01
0.02
0.05
0.10
0.20
0.30
0.40
0.50
0.60
0.70
0.80
0.90
0.95
0.98
0.99
0.995
0.999

P

Binomial Nomograph



If a typical 8% defective rate, then an 6.6% chance of acceptance.

The Binomial Equation

The binomial equation is the basis for lot sampling

$$P = \sum_{d=0}^c \frac{n!}{d! (n-d)!} p^d (1-p)^{n-d}$$

However, often we stop sampling if we find even one defect.

That is, $c = 0$.

We then find a sample size n with acceptable $(\alpha, \beta, AQL, LTPD)$.

The Binomial Equation

$$P = \sum_{d=0}^c \frac{n!}{d! (n-d)!} p^d (1-p)^{n-d}$$

Notice for $c = 0$, the binomial equation reduces to

$$P = (1-p)^n \quad \text{and so} \quad p = 1 - P^{\frac{1}{n}} \quad \text{and} \quad n = \frac{\log P}{\log(1-p)}$$

Notice for $c = 1$, the binomial equation reduces to

$$P = np(1-p)^{n-1} + (1-p)^n$$

The Binomial Equation in Excel

For the $c=0$ and $c=1$ single and double sampling plans, one can solve in Excel easily.

Closed form equation $n = \frac{\log(1-Conf)}{\log(1-LTPD)}$.

90%		Confidence											
1.00%		LTPD											
c	n	P	Error							Formulas for P			
0	230	90%	0.000895							=1-(1-B13)^D16			
1	388	90%	3.793731	<= goal seek this to zero						=1-(D17*B13*(1-B13)^(D17-1)+(1-B13)^D17)			

No closed form equation. Instead 'Goal Seek' n using $1 - Conf = np(1 - p)^{n-1} + (1 - p)^n$



Practical Sampling Plans

Given AQL, LTDP and willingness to single, double or otherwise sample lots, there are alternative sampling plans

- Recipes for how many samples to take and levels of defectives

Historically, there are tables with sampling plans.

These are simplifications from the 2 equation solution shown.

Instead, only one binomial equation is used.

This leaves the other as a dependent variable.



Example Sampling Plan

Here is the LTPD single and double Sampling Plan table at a 90% confidence level.

Sampling Plan for		90% Confidence																
Maximum Percent Defective		50	30	20	15	10	7	5	3	2	1.5	1	0.7	0.5	0.3	0.2	0.15	0.1
c	0	4	7	11	15	22	32	45	76	114	153	230	328	460	767	1151	1534	2302
	1	6	11	18	24	37	54	76	128	193	258	388	554	776	1295	1943	2592	3888



Example Sampling Plan

Here is the LTPD double Sampling Plan table at a 90% confidence level.

Desired confidence level

Sampling Plan for		90% Confidence																	
Maximum Percent Defective		50	30	20	15	10	7	5	3	2	1.5	1	0.7	0.5	0.3	0.2	0.15	0.1	
	c	0	4	7	11	15	22	32	45	76	114	153	230	328	460	767	1151	1534	2302
		1	6	11	18	24	37	54	76	128	193	258	388	554	776	1295	1943	2592	3888

Example Sampling Plan

Here is the LTPD double Sampling Plan table at a 90% confidence level.

Desired confidence level

Desired maximum percent defectives

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Maximum Percent Defective		50	30	20	15	10	7	5	3	2	1.5	1	0.7	0.5	0.3	0.2	0.15	0.1
c	0	4	7	11	15	22	32	45	76	114	153	230	328	460	767	1151	1534	2302
	1	6	11	18	24	37	54	76	128	193	258	388	554	776	1295	1943	2592	3888

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	1	6	11	18	24	37	54	76	128	193	258	388	554	776	1295	1943	2592	3888

↑
**Allowed
defectives in
the sample**

Example Sampling Plan

Here is the LTPD double Sampling Plan table at a 90% confidence level.

Desired confidence level

Desired maximum percent defectives

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	1	6	11	18	24	37	54	76	128	193	258	388	554	776	1295	1943	2592	3888

↑
Allowed defectives in the sample

Required sample size



Example Sampling Plan

Suppose you want less than 1% defective at 90% confidence.
How many samples are needed to be inspected?

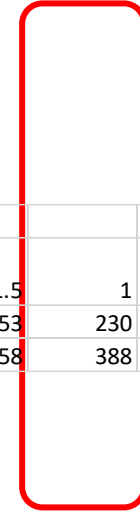
Sampling Plan for		90% Confidence																
Maximum Percent Defective		50	30	20	15	10	7	5	3	2	1.5	1	0.7	0.5	0.3	0.2	0.15	0.1
c	0	4	7	11	15	22	32	45	76	114	153	230	328	460	767	1151	1534	2302
	1	6	11	18	24	37	54	76	128	193	258	388	554	776	1295	1943	2592	3888

Example Sampling Plan

Sampling Plan for		90% Confidence																
Maximum Percent Defective		50	30	20	15	10	7	5	3	2	1.5	1	0.7	0.5	0.3	0.2	0.15	0.1
c	0	4	7	11	15	22	32	45	76	114	153	230	328	460	767	1151	1534	2302
	1	6	11	18	24	37	54	76	128	193	258	388	554	776	1295	1943	2592	3888

The 1% maximum defectives column

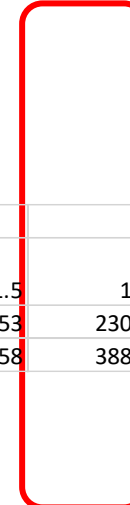
Example Sampling Plan



Sampling Plan for		90% Confidence																
Maximum Percent Defective		50	30	20	15	10	7	5	3	2	1.5	1	0.7	0.5	0.3	0.2	0.15	0.1
c	0	4	7	11	15	22	32	45	76	114	153	230	328	460	767	1151	1534	2302
	1	6	11	18	24	37	54	76	128	193	258	388	554	776	1295	1943	2592	3888

If no defectives found in the first 230 parts inspected, accept the lot.

Example Sampling Plan



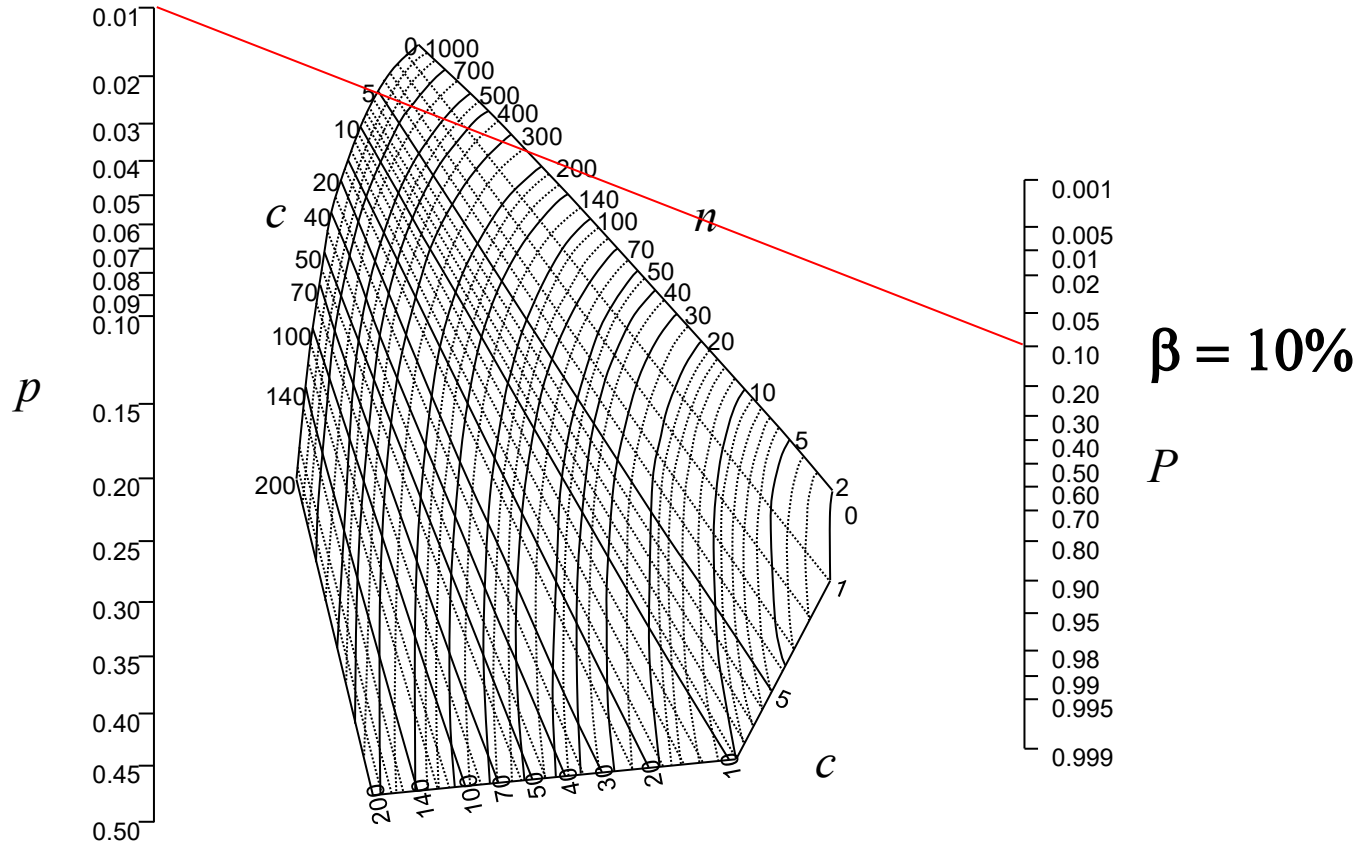
Sampling Plan for		90% Confidence																
Maximum Percent Defective		50	30	20	15	10	7	5	3	2	1.5	1	0.7	0.5	0.3	0.2	0.15	0.1
c	0	4	7	11	15	22	32	45	76	114	153	230	328	460	767	1151	1534	2302
	1	6	11	18	24	37	54	76	128	193	258	388	554	776	1295	1943	2592	3888

If no defectives found in the first 230 parts inspected, accept the lot.

If you find 1 defective, keep going to 388 parts. If no more found, then accept the lot. If you find 2 or more in the first 388, reject the lot.

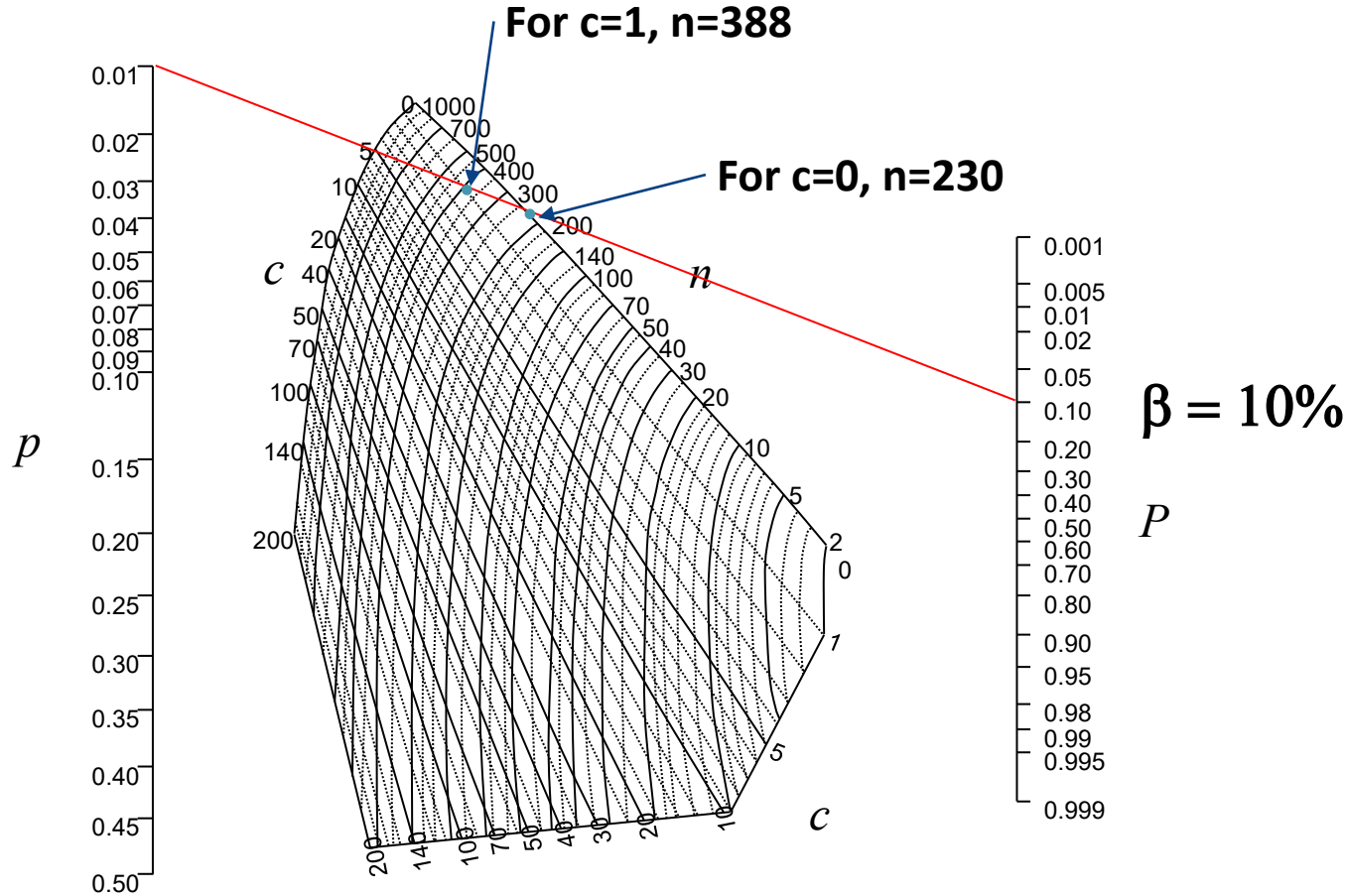
On the Binomial Nomograph

LTPD = 1%



On the Binomial Nomograph

LTPD = 1%



Using the Binomial Equation

$$\beta = \sum_{d=0}^c \frac{n!}{d! (n-d)!} LTPD^d (1 - LTPD)^{n-d}$$

For $\beta = 0.1$, $LTPD = 0.01$, solve with $c = 0$:

$$0.1 = \frac{n!}{0!(n-0)!} 0.01^0 (1 - 0.01)^n \text{ or } n=230$$

And with $c = 1$:

$$0.1 = \frac{n!}{0!(n-0)!} 0.01^0 (1 - 0.01)^n + \frac{n!}{1!(n-1)!} 0.01^1 (1 - 0.01)^{n-1} \text{ or } n = 388$$



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Adaptive Sampling Requirements

Typically, the level of inspection necessary is defined by the process capability C_{pk} . Poor quality needs more inspection.

- If C_{pk} is low, 100% inspection is necessary. If C_{pk} is high, sample inspection suffices. Is this lot different?

Incoming Quality Level	Part Severity	Sampling
$C_{pk} \geq 1.67$	Safety Critical	100%
$C_{pk} \geq 1.67$	Major Critical	SPC and 5 samples
$C_{pk} \geq 1.67$	Minor Critical	SPC and 5 samples
$C_{pk} \geq 1.33$	Safety Critical	100%
$C_{pk} \geq 1.33$	Major Critical	0.5% AQL & LTPD
$C_{pk} \geq 1.33$	Minor Critical	2.5% AQL & LTPD
$C_{pk} \leq 1.33$	Safety Critical	Not allowed
$C_{pk} \leq 1.33$	Major Critical	100%
$C_{pk} \leq 1.33$	Minor Critical	100%

Inspection Plans for Acceptable C_{pk}

The traditional OC Curve inspection plans are pass-fail based.

A better approach would be to sentence the lot based on whether the lot's C_{pk} is adequate.

- Assurance of quality as C_{pk}

Incoming Quality Level	Part Severity	Sampling
$C_{pk} \geq 1.67$	Safety Critical	100%
$C_{pk} \geq 1.67$	Major Critical	SPC and 5 samples
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$C_{pk} \geq 1.33$	Safety Critical	100%
$C_{pk} \geq 1.33$	Major Critical	0.5% AQL & LTPD
$C_{pk} \geq 1.33$	Minor Critical	2.5% AQL & LTPD
$C_{pk} \leq 1.33$	Safety Critical	Not allowed
$C_{pk} \leq 1.33$	Major Critical	100%
$C_{pk} \leq 1.33$	Minor Critical	100%

Inspection Plans for Acceptable C_{pk}

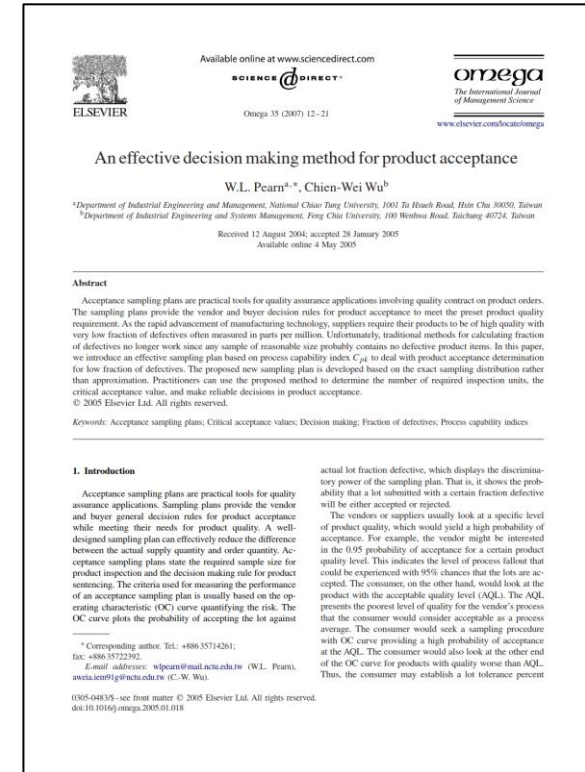
The traditional OC Curve inspection plans are pass-fail based.

A better approach would be to sentence the lot based on whether the lot's C_{pk} is adequate.

- Assurance of quality as C_{pk}

Pearn and Wu created this. (Omega 2007)

This sentences lots on C_{pk} , not on the binary pass/fail conformance to specifications.



Inspection Plans for Acceptable C_{pk}

Define α and β risk levels.

Then also define

C_{AQL} : The maximum rejectable C_{pk} value with α -risk

C_{LTPD} : The minimum acceptable C_{pk} value with β -risk

With this, one can solve for the sampling plan (C_0, n):

n : the size of the random sample drawn from the large lot

C_0 : the minimum level of C_{pk} to accept the lot



Two Simultaneous Equations

The math relating α , β , C_{AQL} and C_{LTPD} to two variables C_0 and n :

$$1 - \alpha = \int_0^{b_1\sqrt{n}} G\left(\frac{(n-1)(b_1\sqrt{n}-t)^2}{9nc_0^2}\right) (\phi(t + \xi\sqrt{n}) + \phi(t - \xi\sqrt{n})) dt$$

$$\beta = \int_0^{b_2\sqrt{n}} G\left(\frac{(n-1)(b_2\sqrt{n}-t)^2}{9nc_0^2}\right) (\phi(t + \xi\sqrt{n}) + \phi(t - \xi\sqrt{n})) dt$$

where the sample size n and the cut-off capability level C_0 are the two variables being solved for, and where $b_1 = 3C_{AQL} + |\xi|$, $b_2 = 3C_{LTPD} + |\xi|$, $\xi = \frac{\mu-M}{\sigma}$, G is the cumulative chi-squared distribution function with degree of freedom $n-1$, and ϕ is the standard normal distribution density function.

Inspection Plans for Acceptable C_{pk}

α	β	$C_{AQL} = 1.33$ $C_{LTPD} = 1.00$		$C_{AQL} = 1.50$ $C_{LTPD} = 1.33$		$C_{AQL} = 1.67$ $C_{LTPD} = 1.33$		$C_{AQL} = 2.00$ $C_{LTPD} = 1.67$	
		n	c_0	n	c_0	n	c_0	n	c_0
0.01	0.01	158	1.1645	834	1.4149	232	1.4994	358	1.8345
	0.025	132	1.1509	704	1.4077	195	1.4853	301	1.8207
	0.05	112	1.1372	600	1.4005	166	1.4712	256	1.8069
	0.075	100	1.1271	537	1.3952	148	1.4607	229	1.7967
	0.10	91	1.1186	491	1.3907	135	1.4519	209	1.7881
0.025	0.01	136	1.1790	713	1.4222	200	1.5144	307	1.8489
	0.025	113	1.1655	593	1.4151	165	1.5033	254	1.8352
	0.05	94	1.1517	498	1.4078	139	1.4860	213	1.8212
	0.075	83	1.1414	441	1.4024	122	1.4754	188	1.8108
	0.10	75	1.1327	399	1.3977	110	1.4663	170	1.8019
0.05	0.01	119	1.1937	616	1.4297	174	1.5294	266	1.8635
	0.025	97	1.1805	505	1.4227	142	1.5157	218	1.8501
	0.05	80	1.1669	418	1.4154	117	1.5016	180	1.8362
	0.075	70	1.1566	366	1.4099	102	1.4908	157	1.8257
	0.10	62	1.1477	328	1.4052	91	1.4816	141	1.8167
0.075	0.01	108	1.2047	557	1.4352	158	1.5407	241	1.8743
	0.025	87	1.1919	451	1.4284	128	1.5273	195	1.8613
	0.05	71	1.1785	369	1.4123	104	1.5134	159	1.8476
	0.075	62	1.1683	320	1.4158	90	1.5028	138	1.8372
	0.10	55	1.1595	285	1.4110	80	1.4937	123	1.8282
0.10	0.01	100	1.2140	513	1.4399	146	1.5502	223	1.8836
	0.025	80	1.2016	412	1.4333	117	1.5373	179	1.8709
	0.05	65	1.1886	334	1.4263	95	1.5238	145	1.8576
	0.075	56	1.1787	288	1.4209	81	1.5134	124	1.8473
	0.10	49	1.1700	254	1.4161	72	1.5043	110	1.8384

Inspection Plans for Acceptable C_{pk}

α	β	$C_{AQL} = 1.33$ $C_{LTPD} = 1.00$		$C_{AQL} = 1.50$ $C_{LTPD} = 1.33$		$C_{AQL} = 1.67$ $C_{LTPD} = 1.33$		$C_{AQL} = 2.00$ $C_{LTPD} = 1.67$	
		n	c_0	n	c_0	n	c_0	n	c_0
0.01	0.01	158	1.1645	834	1.4149	232	1.4994	358	1.8345
	0.025	133	1.1509	704	1.4077	195	1.4853	301	1.8207
	0.05	113	1.1372	600	1.4005	166	1.4712	256	1.8069
	0.075	100	1.1271	537	1.3952	148	1.4607	229	1.7967
	0.10	94	1.1186	491	1.3907	135	1.4519	209	1.7881
0.025	0.01	136	1.1790	713	1.4222	200	1.5144	307	1.8489
	0.025	113	1.1655	593	1.4151	165	1.5033	254	1.8352
	0.05	94	1.1517	498	1.4078	139	1.4860	213	1.8212
	0.075	83	1.1414	441	1.4024	122	1.4754	188	1.8108
	0.10	75	1.1327	399	1.3977	110	1.4663	170	1.8019
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	0.05	65	1.1886	334	1.4263	95	1.5238	145	1.8576
	0.075	56	1.1787	288	1.4209	81	1.5134	124	1.8473
	0.10	49	1.1700	254	1.4161	72	1.5043	110	1.8384

Reject no lot with $C_{pk} > C_{AQL}$
Accept no lot with $C_{pk} < C_{LTPD}$

Inspection Plans for Acceptable C_{pk}

**Risk of rejecting a lot
that actually has
adequate C_{pk}**

α	β	$C_{AQL} = 1.33$ $C_{LTPD} = 1.00$		$C_{AQL} = 1.50$ $C_{LTPD} = 1.33$		$C_{AQL} = 1.67$ $C_{LTPD} = 1.33$		$C_{AQL} = 2.00$ $C_{LTPD} = 1.67$	
		n	c_0	n	c_0	n	c_0	n	c_0
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Inspection Plans for Acceptable C_{pk}

**Risk of accepting a lot
that actually has
inadequate C_{pk}**

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Inspection Plans for Acceptable C_{pk}

Sample sizes n and
allowed minimum
 C_{pk} of the sample.

$$C_{pk} \geq c_0$$

α	β	$C_{AQL} = 1.33$ $C_{LTPD} = 1.00$		$C_{AQL} = 1.50$ $C_{LTPD} = 1.33$		$C_{AQL} = 1.67$ $C_{LTPD} = 1.33$		$C_{AQL} = 2.00$ $C_{LTPD} = 1.67$	
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Incoming Inspection Summary

The purpose is to sentence lots, not to estimate lot defective rate.

- To decide whether the defective rate is $< X$.
- Not to determine the defective rate.
- Not to inspect the quality level up to X by filtering out bad parts.

Acceptance Sampling does not provide quality control of the lot.

- It does not even estimate the actual lot defective rate.
- It does not assure the supplier process is in control, which ought be done with SPC.

Acceptance Sample does not “inspect up” the quality level.

- Only 100% inspection can remove defectives and “inspect up” quality.
- Acceptance sampling is simply an audit to statistically ensure the lot conforms to requirements, with an associated level of risk.

Start using C_{pk} sampling plans instead of old binomial pass-fail inspection plans.

- For high quality processes, pass/fail inspection does not work!
- I guarantee nobody at your company will know what you are talking about.
It is an opportunity for quality improvement!

Example: Single Sampling Plan

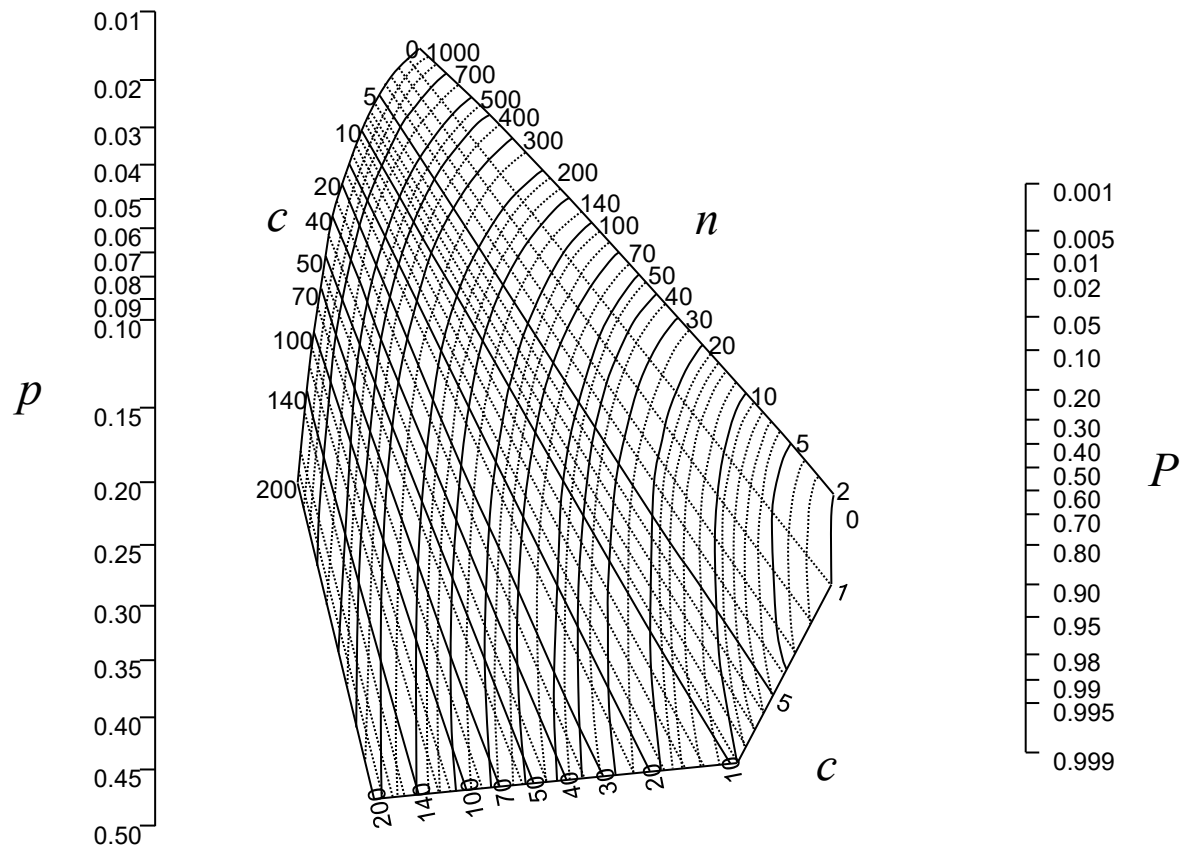
Design a single acceptance sampling plan

- A vendor is running at a typical 8% defective rate.
- The worst tolerable lot defective level is 20% to meet requirements.
- The factory says they can accept average level of defectives to be 5%, since lower levels have excessive price and rejected material costs.
- A lot comprises of 10,000 parts.
- Acceptable producer and consumer risks are 5% and 10% respectively

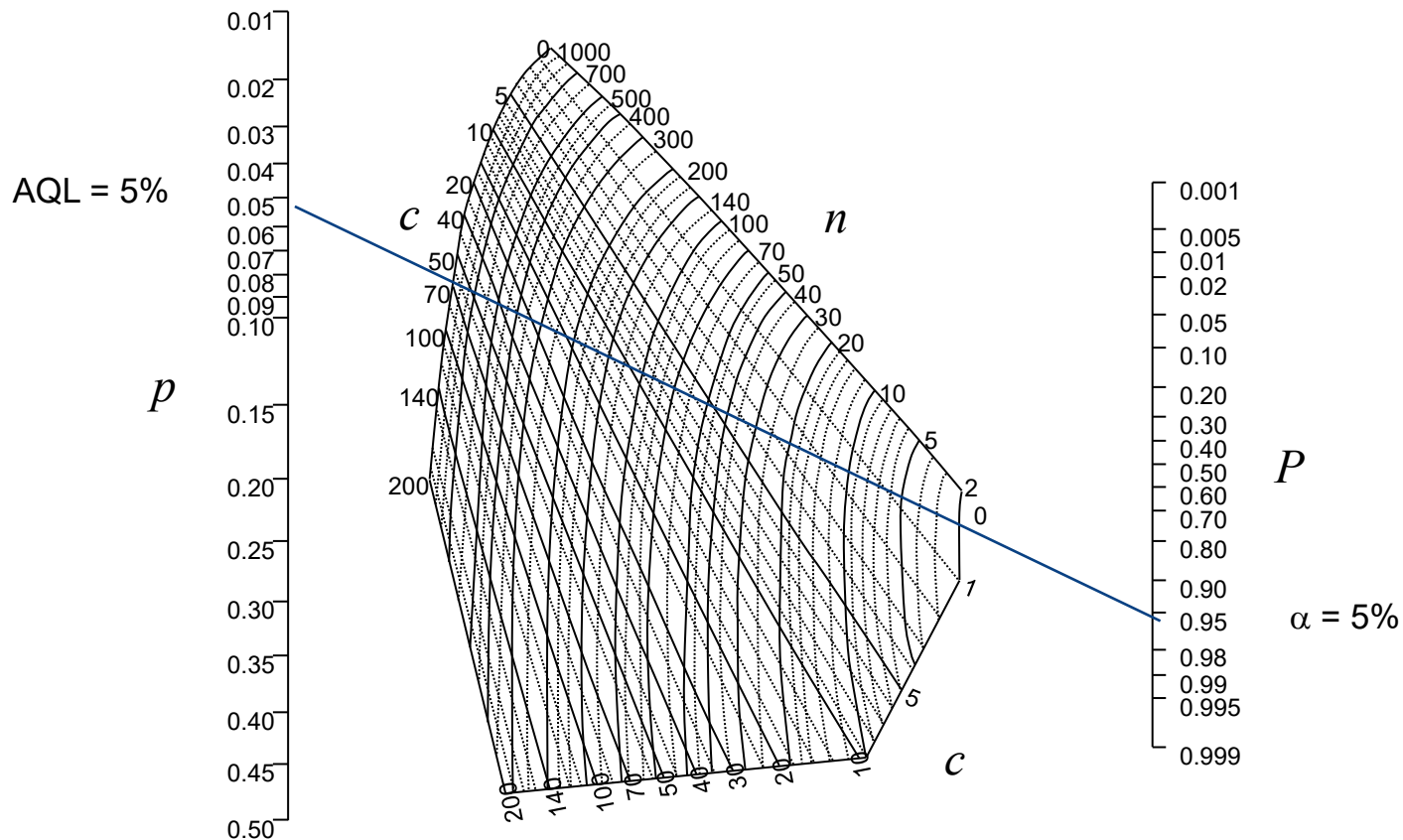
Design a Single Sampling Plan and propose the following:

- The sample size (n) required / lot
- The acceptance number (c)
- What is the probability of lot acceptance (P_a) for a typical lot?

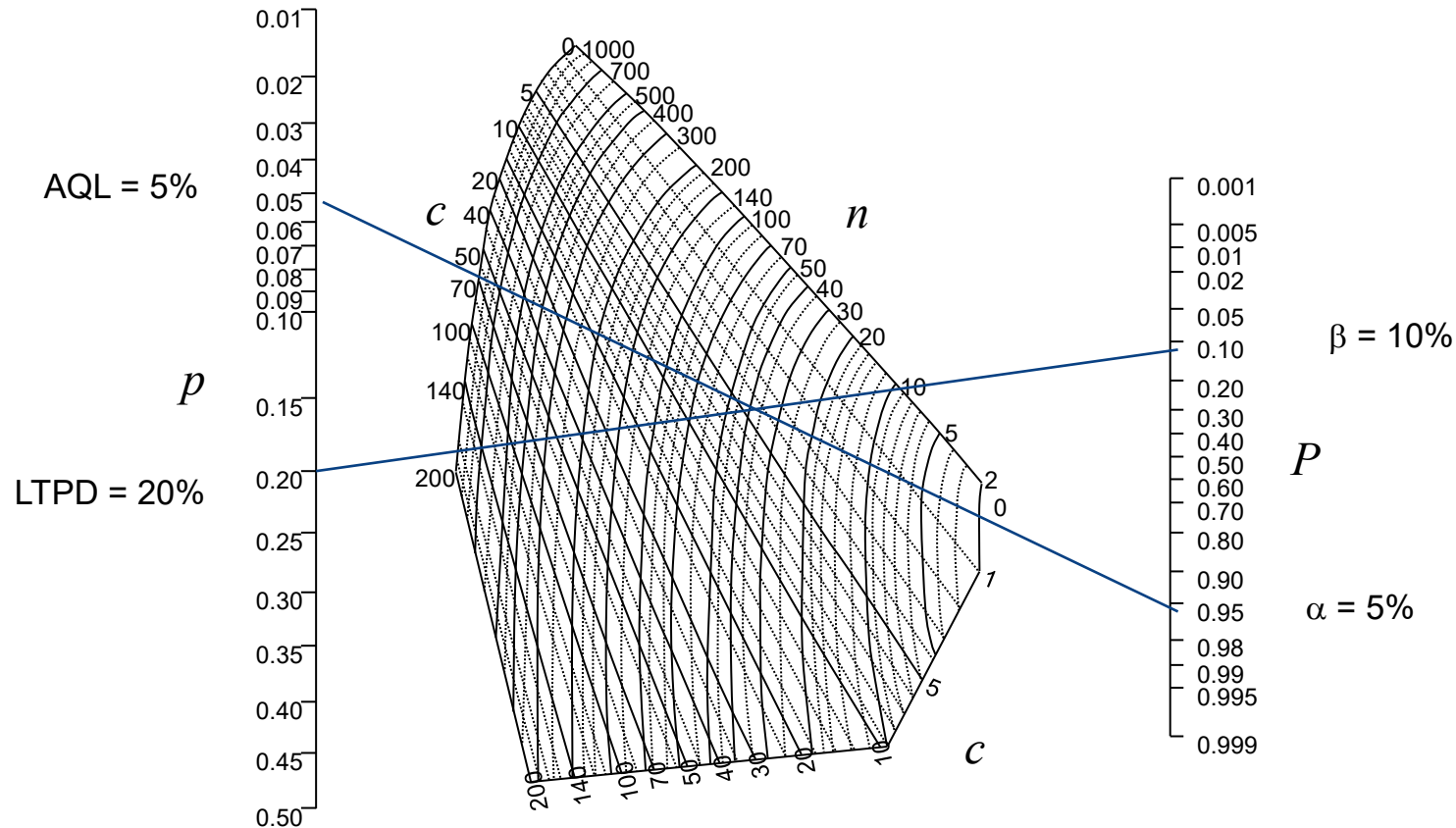
Exercise 1 Solution



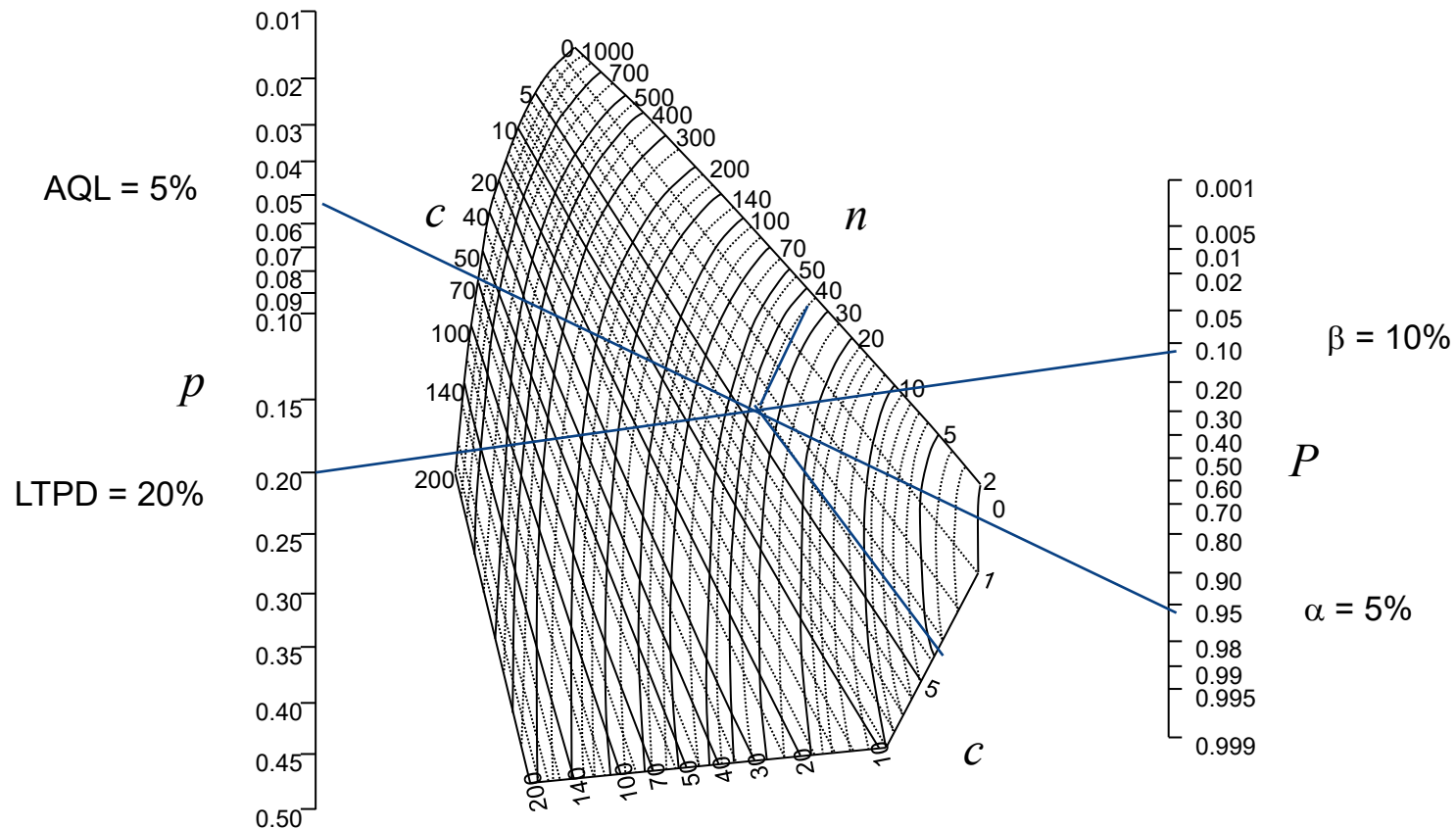
Exercise 1 Solution



Exercise 1 Solution



Exercise 1 Solution





Exercise 1 Solution

As equations, solve simultaneously for (n,c)

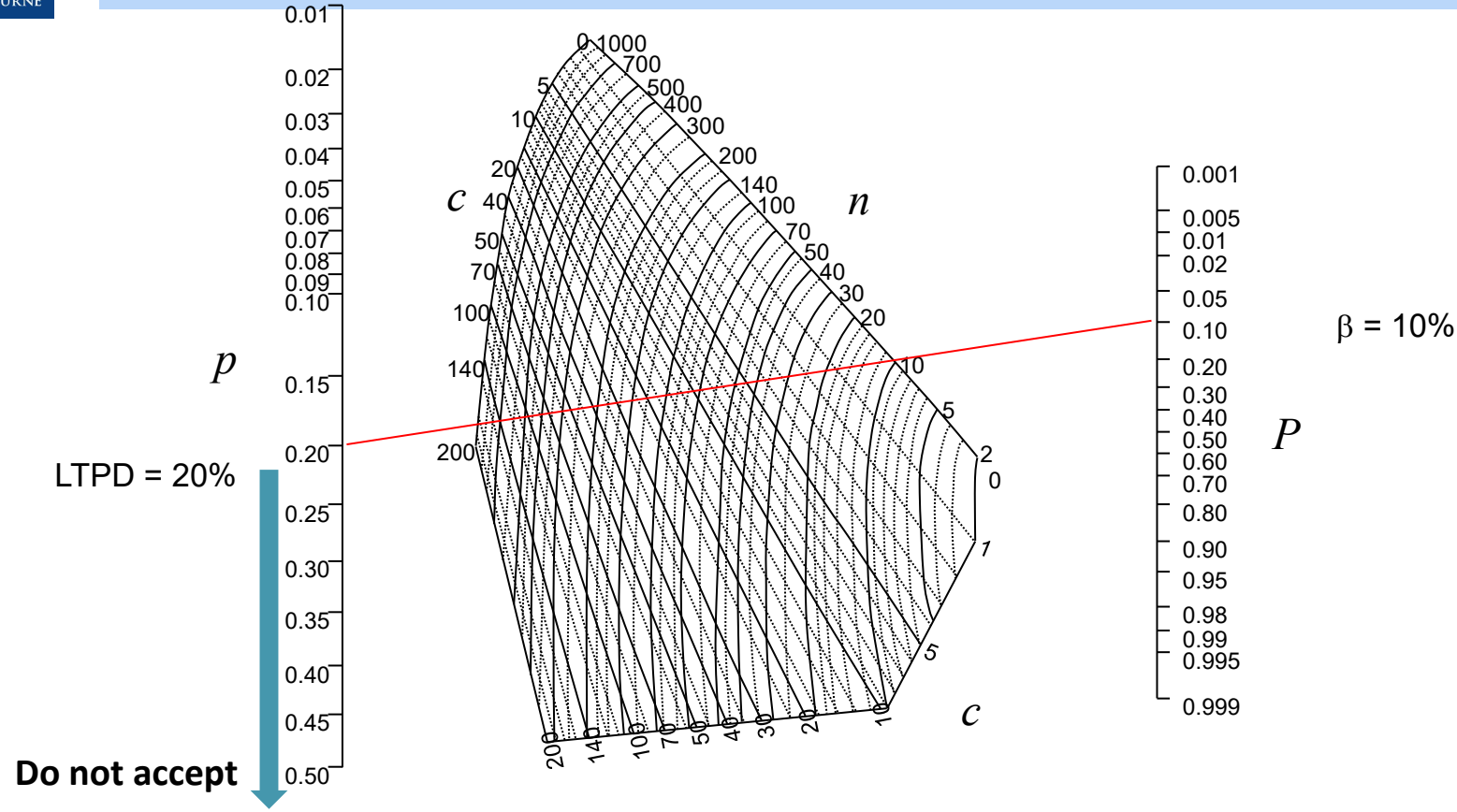
$$1 - 0.05 = \sum_{d=0}^c \frac{n!}{d! (n-d)!} 0.05^d (1 - 0.05)^{n-d}$$
$$0.10 = \sum_{d=0}^c \frac{n!}{d! (n-d)!} 0.20^d (1 - 0.20)^{n-d}$$

Which is n=38, c=4.

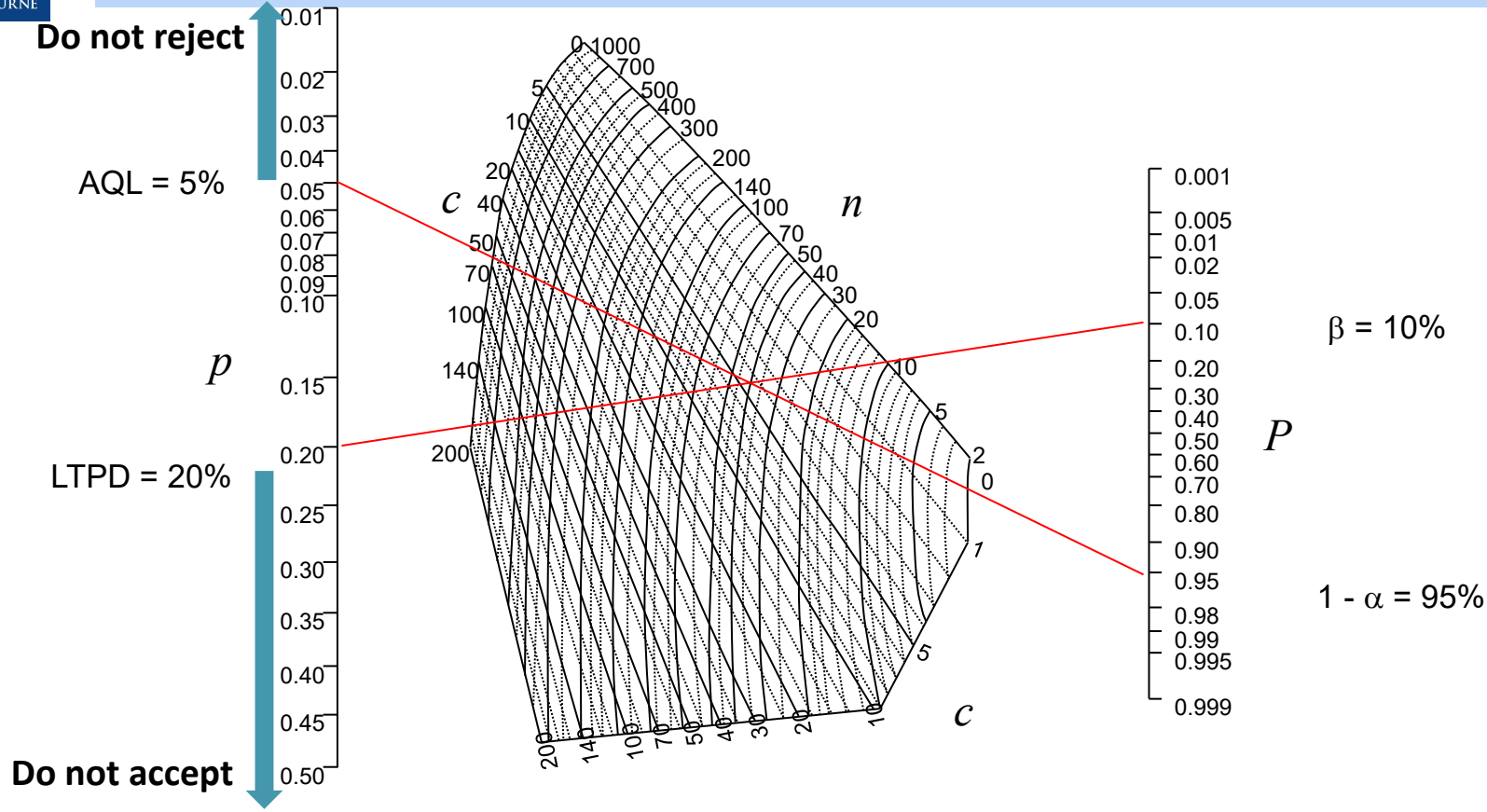
Then, with a typical 8% defective rate,

$$P_a = \sum_{d=0}^4 \frac{38!}{d! (38-d)!} 0.08^d (1 - 0.08)^{38-d} = 82\%$$

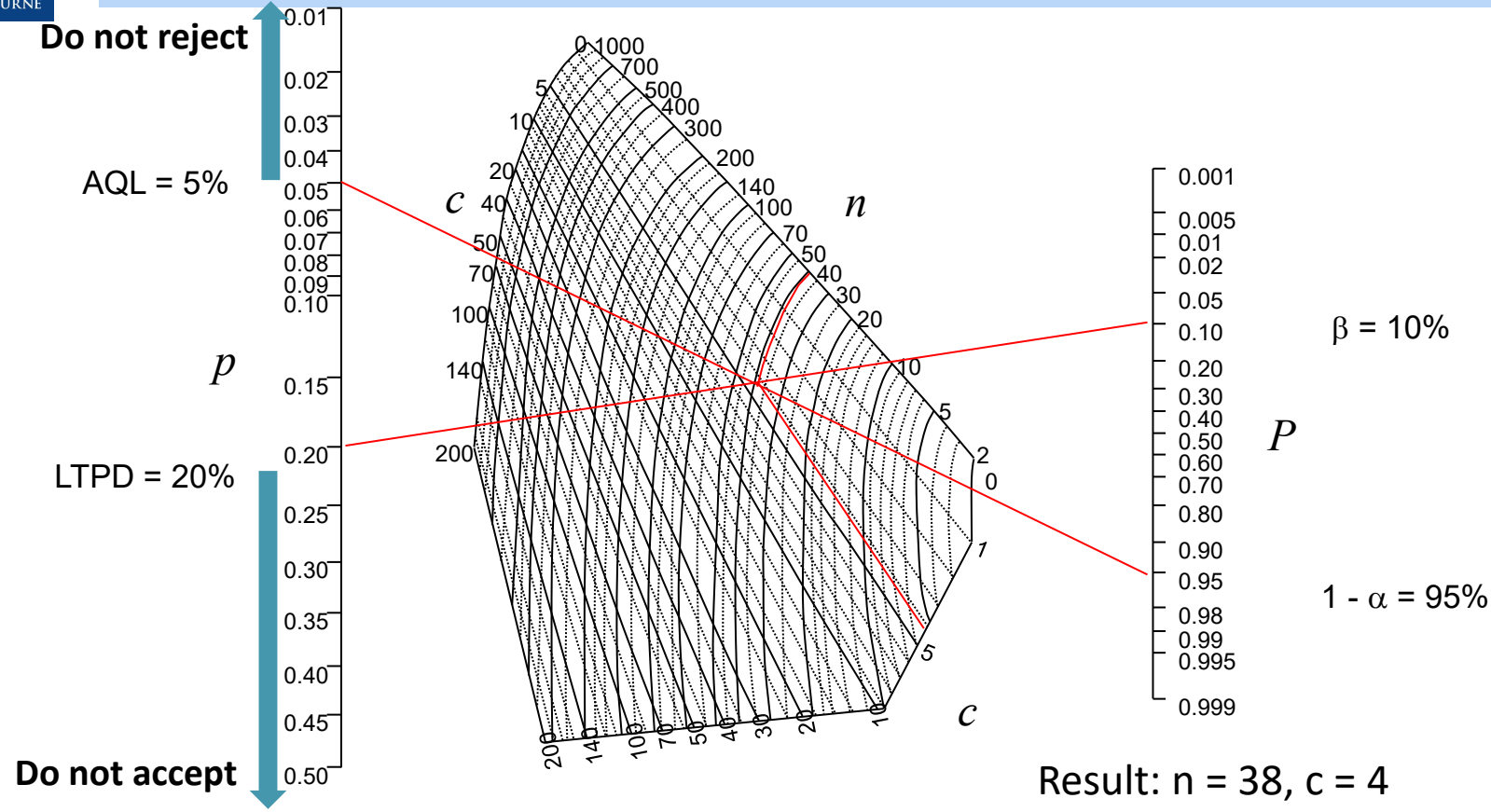
Exercise 1 Solution



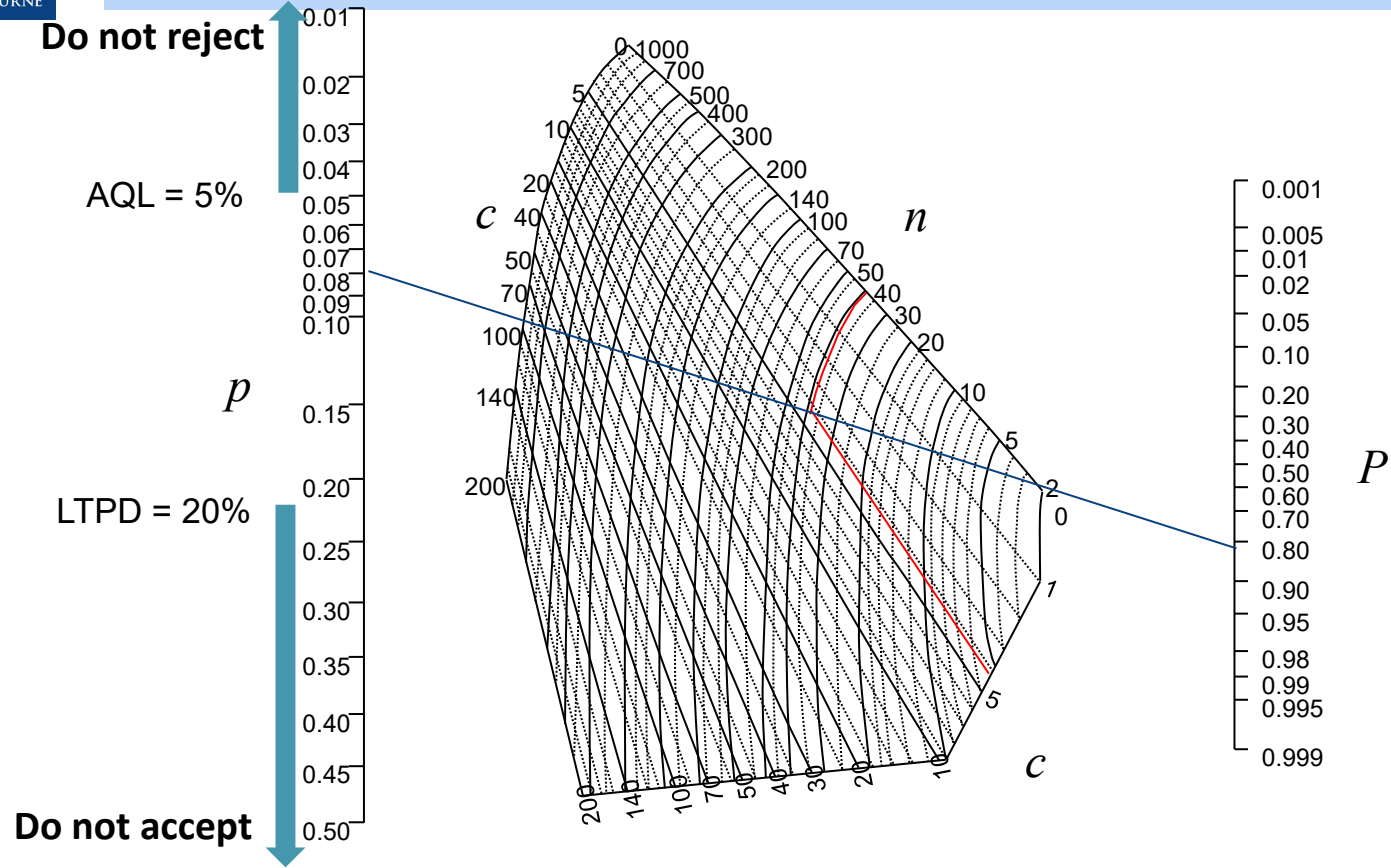
Exercise 1 Solution



Exercise 1 Solution



Exercise 1 Solution



If a typical 8% reject, then an 82% chance of acceptance.

EXERCISE

Suppose your customer declares they are using this sampling plan: sampling 45 parts and no defects allowed.

Sampling Plan Table for 90% Confidence.

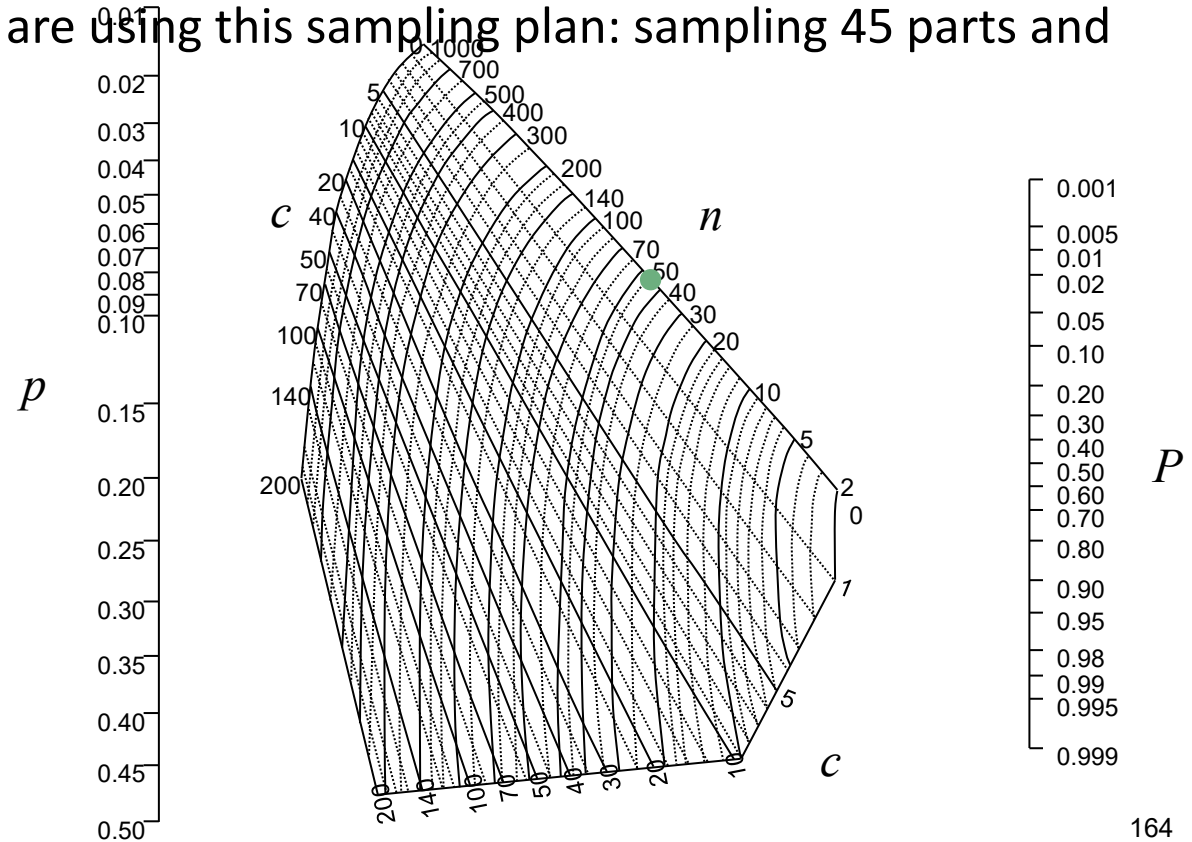
Maximum Percent Defective		50	30	20	15	10	7	5	3	2	1.5	1	0.7	0.5	0.3	0.2	0.15	0.1
		Minimum Sample Size (n)																
c	0	4	7	11	15	22	32	45	76	114	153	230	328	460	767	1160	1540	2350
	1	7	12	18	25	38	55	77	129	194	258	388	555	777	1300	1950	2600	3900

EXERCISE

Suppose your customer declares they are using this sampling plan: sampling 45 parts and no defects allowed.

Your process was running at 1.5% defective.

What is probability your customer will accept the lot?



EXERCISE

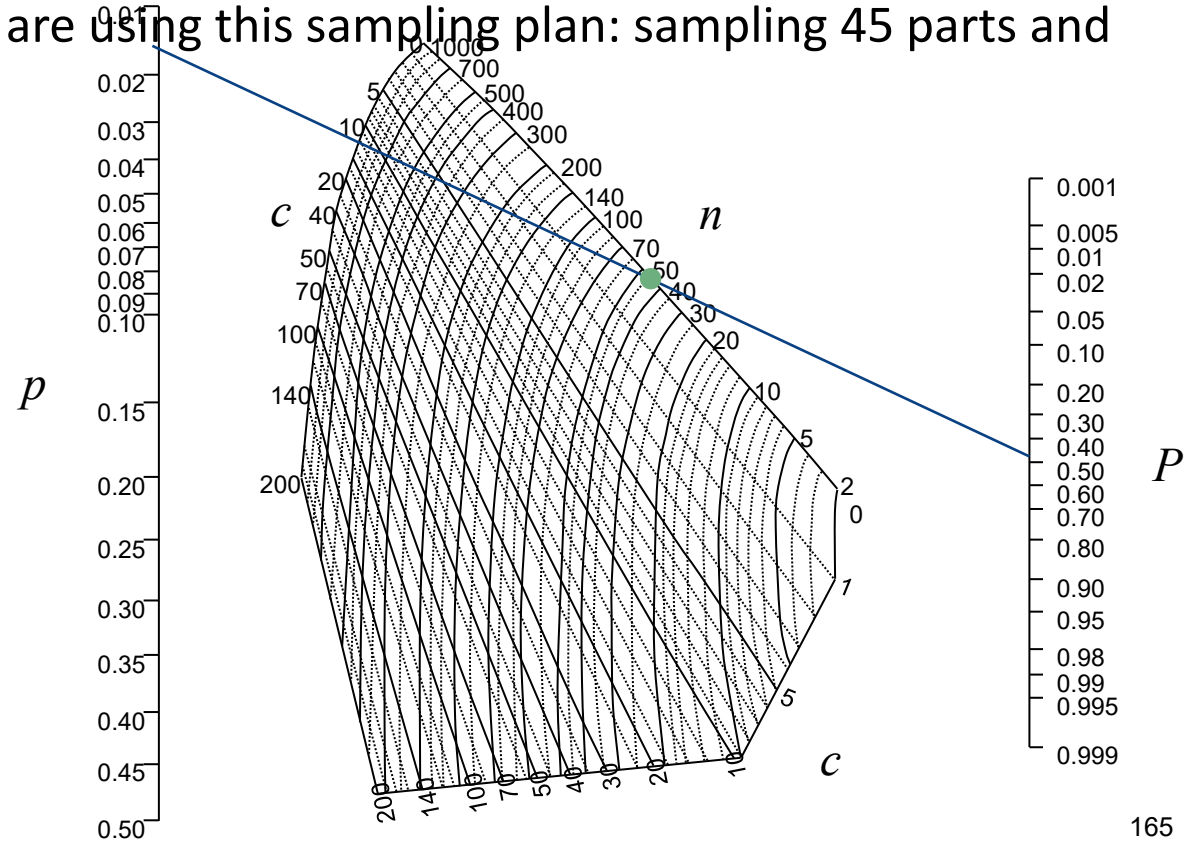
Suppose your customer declares they are using this sampling plan: sampling 45 parts and no defects allowed.

Your process was running at 1.5% defective.

What is probability your customer will accept the lot?

About 50-50.

Note to have 90% confidence in a customer accepting your lot, your defective rate must be 10X better than their acceptance rate.



Example

Suppose you have a supplier for which you want a $C_{pk} = 1.5$.

You risk levels are $\alpha = \beta = 5\%$.

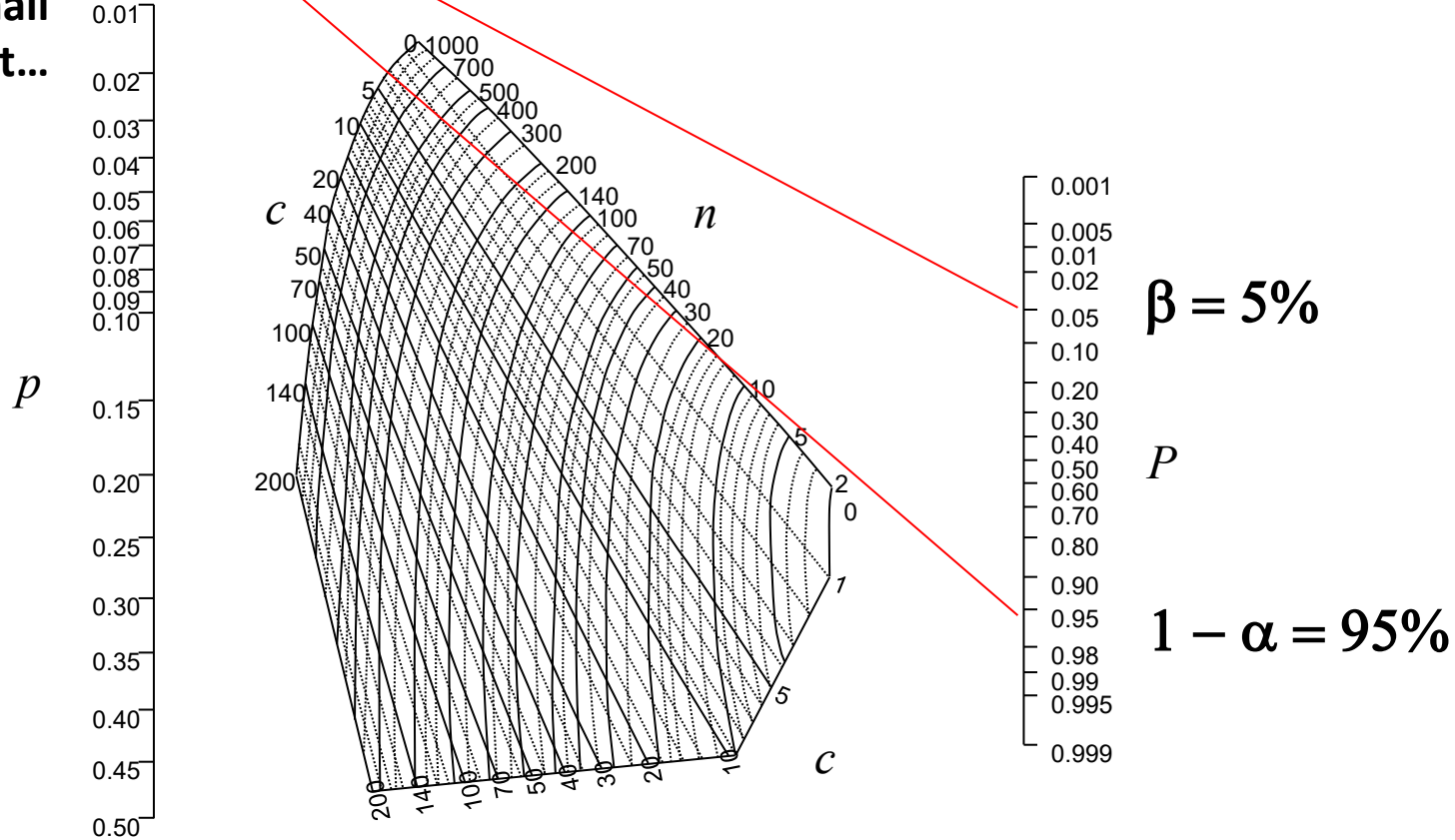
To maintain $1.33 \leq C_{pk} \leq 1.67$, the associated defective rates are AQL = $0.7/10^6$ ($C_{AQL} = 1.67$) and LTPD = $67/10^6$ ($C_{LTPD} = 1.33$).

Questions:

1. What is the traditional AQL sampling plan size (n) for $c=0$?
2. What is the traditional LTPD sampling plan size (n) for $c=0$?
3. What would be your C_{pk} sampling plan size (n) using the C_{pk} inspection method of Pearn and Wu?

On the Binomial Nomograph

LTPD and AQL
values are too small
and off the chart...



Exercise Solution

The AQL equation with $c=0$:

$$1 - \alpha = (1 - AQL)^n$$
$$1 - \alpha = (1 - AQL)^n \text{ or } n = 77,000$$

At this sampling plan (77000,0), the risk of escapes β is

$$\beta = (1 - 0.000067)^{77000} = 0.6\%. \text{ 99.4\% confidence in no escapes.}$$

Exercise Solution

The AQL equation with $c=0$: $n = 77,000$ $\beta = 0.6\%$.

The LTPD equation with $c=0$:

$$\beta = (1 - LTPD)^n$$

$$0.05 = (1 - 0.000067)^n \text{ or } n = 45,000$$

At this sampling plan (45000,0), the risk of a rejecting good lot is
 $1 - \alpha = (1 - 0.0000007)^{45000} = 97\%$. 3% risk in rejecting a good lot.

Exercise Solution

The AQL equation ($C_{AQL} = 1.67$) with $c=0$: $n = 77,000$ $\beta = 0.6\%$.

The LTPD equation ($C_{LTPD} = 1.33$) with $c=0$: $n = 45,000$ $\alpha = 3\%$.

To find a sampling plan that satisfies both AQL and LTPD, we must relax the $c=0$ constraint and solve for n and c simultaneously. However, as c increases so will n . n is already too large.

As C_{pk} increases, pass/fail inspection sampling plans become impractical.

Exercise Solution

The C_{pk} sampling plan with $\alpha = \beta = 0.05$, $C_{AQL} = 1.67$ and $C_{LTPD} = 1.33$:

α	β	$C_{AQL} = 1.33$ $C_{LTPD} = 1.00$		$C_{AQL} = 1.50$ $C_{LTPD} = 1.33$		$C_{AQL} = 1.67$ $C_{LTPD} = 1.33$		$C_{AQL} = 2.00$ $C_{LTPD} = 1.67$	
		n	c_0	n	c_0	n	c_0	n	c_0
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	0.05	80	1.1669	418	1.4154	117	1.5016	180	1.8362
	0.075	70	1.1566	366	1.4099	102	1.4908	157	1.8257
	0.10	62	1.1477	328	1.4052	91	1.4816	141	1.8167

Use 120 samples and a minimum acceptance level: sample $C_{pk} \geq c_0 = 1.5$.

If the sample $C_{pk} \geq 1.5$, then the population has $1.33 \leq C_{pk} \leq 1.67$.



Outline

Statistical Process Control

Xbar-R charts

Out of Control Action Plans

Incoming Inspection

Statistical Sampling Plans

Cpk Inspection Plans

Production Part Assurance Process

Supplier Certification

Suppose you have hired a new supplier to provide parts.

You have provided them your design documentation

- Drawings, Specifications, Instructions

What do you ask the supplier to show to assure you that they will produce high quality parts?

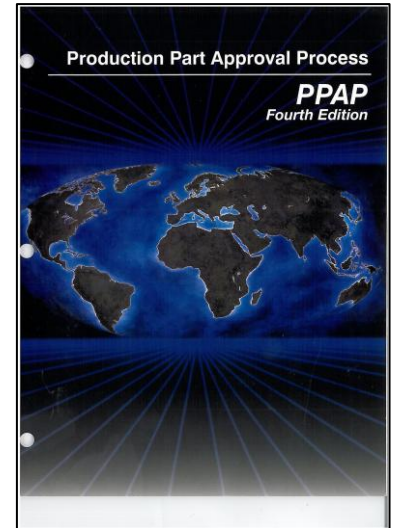
When do you feel confident to hire them to produce high quality?



A supplier ***Production Part Assurance Process (PPAP)*** is a specification to define requirements that a supplier must demonstrate to become quality-certified to produce parts. It was first created by the Automotive Industry Action Group.

- Provide evidence the design, record and specification quality requirements are understood and fulfilled.
- Provide evidence the established manufacturing process has the potential to produce product that consistently meets all requirements at the intended production rate.

The ***PPAP package*** is a legal document under revision control, providing documented evidence of *quality* compliance.





PPAP vs Typical Part Approval

Other standards for suppliers exist

- Declaration of conformity: signed document by a supplier saying they can produce to specification
- Typically includes a sample of parts and their inspection reports

PPAP includes these contracting elements, but also quality elements:

- DFMEA, Process Map, PFMEA, Control Plan, As Built Drawings, MSA, C_{pk} data

These are necessary to assure quality.

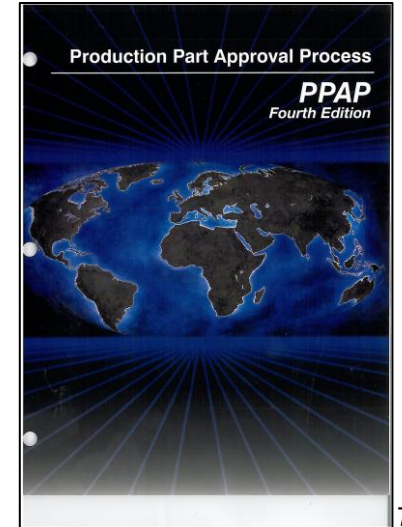
While developed in the auto industry, PPAP requirements are spreading to all industries.



PPAP Elements

1. Released production drawings
2. Supplementary Inspection sheets
3. Production Purchase Order
4. Design FMEA
5. Process Flow
6. Process FMEA
7. Process Control Plan
8. Process Readiness Study
9. Initial Process Capability Studies
10. Measurement Systems Analysis
11. Dimensional Report
12. Production Verification Testing
13. Material Certification Documentation
14. Raw Material Approval
15. Part Marking Approval
16. Packaging Approval
17. Review and Signoff

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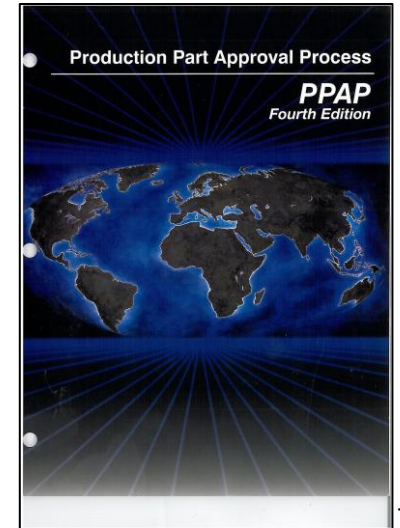
PPAP Elements

1. Released production drawings
2. Supplementary Inspection sheets
3. Production Purchase Order
4. Design FMEA
5. Process Flowchart
6. Process FMEA
7. Process Control Plan
8. Process Readiness Study
9. Initial Process Capability Studies

**Everything we have
discussed in this subject.**

10. Measurement Systems Analysis
11. Dimensional Report
12. Production Verification Testing
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PPAP vs ISIR

In Germany, the Verband der Automobilindustrie (VDA) subsequently created the *Initial Sampling Inspection Report*

- Same as PPAP





PPAP Elements

1. Released production drawings

- Latest updated drawings showing as-produced dimensions, under revision control, with design responsible party approval.

2. Supplementary Inspection sheets

- Beyond steady state production inspection, supplementary inspections are often requested at startup. A copy of these copy of measurements and statistics of the capability studies.

3. Production Purchase Order

- A copy of the production purchase order is typically kept in the PPAP file.



PPAP Elements

4. Design FMEA

- A copy of the design failure modes and effects analysis (DFMEA), even for parts not designed by the supplier, detailing what can go wrong with a part and its effect.

5. Process Flowchart

- A flowchart of the process and supporting text describing the process steps and standard work, including rework.

6. Process FMEA

- A copy of the process failure modes and effects analysis (PFEMA) detailing what can go wrong with a process and its effect.

7. Process Control Plan

- A copy of the work instructions for detecting all process failure modes and correcting them, including sampling frequency plans

8. Process Readiness Study

- A copy of the pre-production process readiness study, including minimally the checklist reviews of
 - Tools, fixtures, equipment and gages, Standard work instructions, Process control methods, Maintenance program, Foreign object prevention, detection and removal.

9. Initial Process Capability Studies

- A copy of measurements and statistics of the capability studies on all key product characteristics, key process outputs and key process inputs using a statistically significant sample against all measurement errors.
- Demonstration that all key measurements have $C_{pk} \geq 1.33$.



PPAP Elements

10. Measurement Systems Analysis

- For all key measurements, a copy of the measurement systems analysis including instrument calibration and gage repeatability and reproducibility.

11. Dimensional Report

- For all design drawing key characteristics, a copy of the dimensional capability. This includes the first article inspection reports.

12. Production Verification Testing

- For all part performance requirements, production part test results with responsible design party certifications.



PPAP Elements

13. Material Certification Documentation

- A copy of documented evidence all incoming materials are purchased from approved sources.

14. Raw Material Approval

- A copy of report in approved format demonstrating compliance of materials used by supplier.

15. Part Marking Approval

- Copy of approval for compliance marking of parts. Digital photograph of compliance approval marking on part.

16. Packaging Approval

- Copy of approval for compliance marking of packaging.

17. Review and Signoff

- Copy of submission records and disposition approval, at all levels.



PPAP Process

Typically, these elements are provided in phases aligned with development and contracting

- Level 1 – Part Submission Warrant (PSW) declaration only
- Level 2 – PSW with product samples and limited supporting data
- Level 3 – PSW with product samples, FMEA, and short run C_{pk} data
- Level 4 – PSW and all process controls, MSA, data.
- Level 5 – PSW with product samples and complete documentation.

The complete documentation is the PPAP Package, available for review at the supplier's manufacturing location.



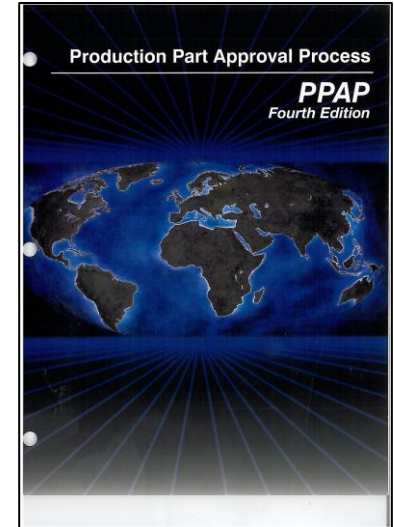
PPAP Contracting

When hiring a supplier, have them complete all PPAP elements and submit for review.

Phase it in alignment with the development process.

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Summary

To assure your suppliers will provide high quality, you need more than ISO 9000 certification. That is an entry point. ISO9000 documents the quality processes, good or bad.

To assure you will actually get high quality parts, you need

- Measured evidence on samples. What is their Cpk?
- Understanding of what can go wrong in their processes, FMEA.
- Understanding of what they will do when things go wrong. Control Plan.
- Understanding of their measurement capability. MSA studies.
- Understanding of how things going wrong compare to spec's. Cpk data.

PPAP is designed to minimally document this.